



FITTER-EU

DELIVERABLE D2.3

Embedding social justice in twin transition research: Concepts, definitions and methodological approach (3)



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Embedding Social Justice in Twin Transition Research: Concepts, Definitions and Methodological Approach (3)

Summary

D2.3 “Embedding Social Justice in Twin Transition Research: Concepts, Definitions and Methodological Approach” refines the project’s methodological approach to ensure that fairness and inclusivity are at the heart of FITTER-EU R&I activities.

Building on earlier work, FITTER-EU now focuses on targeted policy objectives in three key sectors of analysis: Energy, Transport and Housing. Specifically, the focus is on renewable energy, electric mobility, and climate-resilient housing. These themes, selected for their relevance and policy infrastructure, are explored through case studies in six European countries. The project’s overarching aim is to understand and address the roots of inequality that may arise as Europe moves toward a more sustainable future, providing policymakers with tools and insights to anticipate challenges and design policy solutions.

Central to the FITTER-EU methodology are three interlinked approaches. First, systems inquiry helps researchers and policymakers see beyond surface-level problems, using tools like causal loop diagrams and leverage points to identify the deeper structures that drive unsustainable outcomes and social disparities. Second, risk assessment is used to evaluate the likelihood and severity of unintended consequences, with an emphasis on supplementing traditional analysis with systems thinking to better address structural inequalities. Third, impact assessment considers the broader effects of policies, paying special attention to how changes might affect different groups, especially in terms of gender and intersectional inequalities.

FITTER-EU’s approach is firmly rooted in a rights-based perspective, drawing on international frameworks such as the International Bill of Human Rights and the European Social Charter. This ensures that policies are not only effective but also just, upholding principles of universality, non-discrimination, participation, and accountability. The project’s commitment to social justice means that the benefits of the twin transition should be shared equitably, and its burdens should not fall disproportionately on disadvantaged groups.

Each sector presents unique challenges. In transport, FITTER-EU highlights issues like induced car demand and underfunded public transit, advocating for transformative policies such as reallocating road space and integrating mobility systems. In energy, the focus is on rapid decarbonisation and equitable access to renewables, with attention to energy poverty and fair distribution of costs and benefits. Housing analysis centres on the right to adequate shelter, the barriers to retrofitting homes, and the impact of financialisation, emphasising the need for climate-resilient, affordable housing for all.

To support these goals, the FITTER platform offers a modular, scalable, and transparent digital environment for policy appraisal. It integrates predictive modelling, risk assessment, systems thinking, and stakeholder engagement, helping users visualise hazards, identify disadvantaged groups, and plan mitigation strategies. The platform acknowledges its limitations, such as data quality and predictive uncertainty, and stresses the importance of qualitative insights alongside quantitative analysis.

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1. Introduction

1.1. Overview of FITTER-EU

FITTER-EU aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem that includes a Digital Platform. Through a co-creation methodological approach, this ecosystem aims to enable policymakers to understand the structures and systems through which social groups risk being adversely affected by twin transition policies. This understanding is supported through the co-creation of different future scenarios, and the development of mitigation plans with the support of Better Practice Guides.

1.2. Purpose and scope of the Deliverable

Building on the conceptual background developed in D2.1 and D2.2 this deliverable further develops the FITTER approach and methodology.

1.3. Structure of the Deliverable

Building on the eco-social policy development approach outlined in [D2.1](#) and [D2.2](#), Chapter 2 specifies the human rights framework and planetary limits that policy makers must account for. An outline of the function of systems inquiry, risk assessment and impact assessment is provided in Chapter 3. Chapter 4 provides an overview of how inequalities are reproduced in the energy, building and transport sectors. Using transport as an example, Chapter 5 details how the FITTER platform will support policy makers in identifying leverage points for transformative change, assess systemic risks, and design interventions that address both environmental and social inequalities within mobility systems

2. A rights-based approach to policy design and implementation

As outlined in [D2.1](#) and [D2.2](#), FITTER does not anticipate that a just transition to a carbon neutral economy is possible without a fundamental restructuring of economic models, land stewardship, energy sources, and mindsets. We situate the ‘eco-social-growth’ trilemma as a ‘systems trilemma’ with complex relationships between ‘reproduction, production and consumption systems’ (economy), ‘planetary ecosystems’ (environment) and ‘political structures and norms’ (social) framed in terms of the reality defined by the climate emergency. We situate ‘justice’ at the core of the wider triangle.

Law and policy, guided by justice concerns, should effectively intervene to (re)structure systems so that we remain within planetary boundaries and enable liveable lives for all.

To achieve this, FITTER will take a human rights-based approach (HRBA), with due regard for climate justice, through a two-stage process - a **risk assessment** supplemented by systems inquiry.

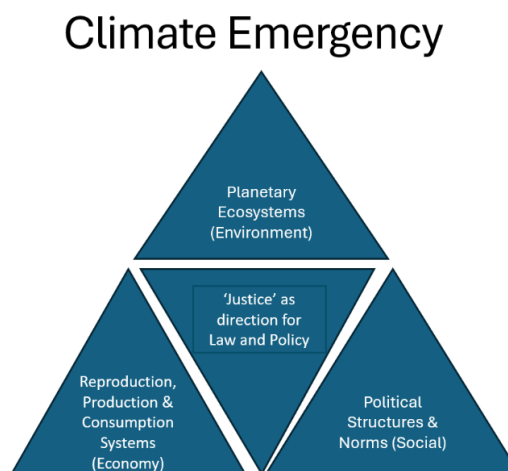


Figure 1: The Systems Trilemma (See also [Galgóczi, 2022](#) and [Heffron and McCauley, 2017](#))

As outlined in [D2.2](#), the scope of FITTER has been refined. Project findings demonstrate that labour market impacts of the twin transition are very difficult to predict. However, the consumers disadvantaged by twin transition measures are low-income population cohorts and/or people at risk of poverty in every sector. A low level of digital literacy is also one of the strongest indicators of vulnerability to structural harm, both for consumers and workers (see [D3.1](#)).

To address these findings, four areas of focus were chosen - renewable energy sources, electric mobility, housing climate insulation, and the digitalisation of agriculture ([D2.2](#)).

The agriculture case study was subsequently deemed unfeasible for two reasons. Most significantly, research found that the digitalisation of agriculture is at an early stage of development. As of early 2025 there was no agriculture digitalisation policy infrastructure in the case study countries - Ireland and Italy – and the policy infrastructure at EU level is under development. As such, the case study teams had no relevant policies to assess. Second, although the root causes of structural disadvantage in agriculture may well be exacerbated by technological developments, they cannot be addressed with digital solutions. They require transformation of underlying systems and structures – laws, norms, policy priorities.

The focus on consumers across energy, housing and transport interconnect well and there is overlapping policy infrastructure. The focus on producers in agriculture, while much needed, was deemed to be unfeasible given the lack of policy infrastructure, the complexities of the sector, and time and resource constraints in FITTER.

The scope of FITTER is now further refined, with Ireland and Italy switching from agriculture to housing and transport respectively.

- The transition to renewable energy sources – all case study countries
- The transition to electric mobility – Spain, Hungary, Italy
- The transition to climate resilient housing – Germany, Portugal, Ireland

The focus of the case study teams remains on an assessment of the extent to which green and digital transition policies interconnect and function to address eco-social justice concerns within each area of focus (renewable energy sources, electric mobility, climate resilient housing).

As outlined in [D2.1](#) and [D2.2](#), twin transition policies need to be meaningfully guided by justice concerns while enabling reproduction, production and consumption systems to remain within planetary boundaries. This chapter provides a summary of the human rights framework, justice concerns, and planetary limits that policy makers must account for.

2.1. Human Rights Framework

Under international law, the State is the main duty bearer when it comes to defending human rights. People employed by the State are responsible for respecting, protecting and fulfilling these rights. A human rights-based approach to policy design and implementation should strengthen the capacity of duty bearers to meet their obligations and strengthen the capacity of rights holders to realise their rights and claim their entitlements.

2.1.1 A Human Rights-Based Approach

A human rights-based approach (HRBA) requires that human rights principles guide policy design and implementation. There are two main rationales for taking a human rights-based approach:

- Intrinsic rationale – recognises that a rights-based approach is morally and legally right
- Instrumental rationale – recognises that a rights-based approach leads to more sustainable outcomes

In practice, the motivation for taking a human rights-based approach often combines both intrinsic and instrumental rationales ([UN Women, 2024](#)).

FITTER has committed to supporting a just transition. This requires the FITTER platform to enable policy makers to uphold human rights while staying within planetary boundaries ([Kortetmäki et al. 2025](#))

A human rights-based approach requires that:

1. Policy design and implementation aim to realise and advance human rights
2. Design and implementation are guided by human rights principles:
 - Universality and Inalienability – All people everywhere are free and equal in dignity and rights.
 - Indivisibility – All human rights have equal status; there is no hierarchy of rights.
 - Interdependence and Interrelatedness: The fulfilment of one right often depends on the fulfilment of other rights. Rights are interdependent and interrelated.
 - Equality and Non-discrimination: All people are equal and no one should suffer discrimination on the basis of grounds of discrimination as established by human rights standards.
 - Participation: Everyone has the right to participate in and access information with regard to decision-making processes that affect their lives. Rights-based approaches require a high degree of participation by all members of affected communities.
 - Accountability and Rule of Law: States and other duty-bearers are answerable for the observance of human rights.
3. Policy design and implementation should focus on developing the capacities of duty-bearers (to meet obligations) and rights-holders (to claim rights) ([UN Women, 2024](#))

This connects with the goals of climate justice wherein there is a shift to thinking of **adaptation as transformative** - in line with a revised eco-social contract. As outlined in [D2.1](#), justice movements need to

build social systems that enable the sustainability and protection of socio-ecological systems, ensure governmental accountability to all those impacted by climate change and foreground a notion of security based on meeting basic human needs ([Schlosberg and Collins, 2014: 370](#)).

In practical terms, taking a human rights approach in the context of climate justice goals requires:

- understanding how inequalities are reproduced,
- supporting/amplifying affected communities to participate in planning and decision-making,
- transforming systems and structures such that they produce both *ecologically sustainable and socially equitable outcomes* by design.

2.1.2 International Bill of Human Rights

The goal of the just transition to 'leave no-one behind' can be partly achieved through the realisation of the right to an adequate standard of living to enable health and wellbeing.

[The Universal Declaration of Human Rights \(1948\)](#), the [International Covenant on Civil and Political Rights \(1966\)](#), and the [International Covenant on Economic, Social and Cultural Rights \(1966\)](#) make up the International Bill of Human Rights.

The right to an adequate standard of living to enable health and wellbeing is outlined in Article 25 (1) of the Universal Declaration of Human Rights, and Article 11 of the International Covenant on Economic Social and Cultural Rights (ICESCR).

The ICESCR (1966) also includes

- Right of self-determination
- Principle of non-discrimination
- Right to work
- Right to just conditions of work
- Right to form and join trade unions
- Right to strike
- Right to social security
- Protection and assistance accorded to the family
- Maternity protection and supports
- Protection from economic and social exploitation for children and young people
- Right to adequate standard of living including adequate food, clothing and housing
- Right to physical and mental health
- Right to education
- Right to take part in cultural life
- Right to enjoy the benefits of scientific progress

International Covenant on Civil and Political Rights (1966)

Right to adequate housing

Article 11 of the ICESCR (1996) states that:

The States Parties to the present Covenant recognize the right of everyone to an adequate standard of living for himself and his family, including adequate food, clothing and housing, and to the continuous improvement of living conditions.

The UN Committee on Economic Social and Cultural Rights General Comment No. 4 (1991) establishes 7 characteristics of the right to adequate housing.

- a) *Legal security of tenure*: Notwithstanding the type of tenure, all persons should possess a degree of security of tenure which guarantees legal protection against forced eviction, harassment and other threats. States parties should take measures aimed at conferring legal security of tenure.

- b) *Availability of services, materials, facilities, and infrastructure*: An adequate house must contain facilities essential for health, security, comfort and nutrition. Occupants should have sustainable access to natural and common resources, safe drinking water, **energy for cooking, heating and lighting**, sanitation and washing facilities, means of food storage, refuse disposal, site drainage and emergency services.
- c) *Affordability*: Financial costs associated with housing should be at such a level that the satisfaction of other basic needs is not threatened or compromised. States parties should ensure that the percentage of housing-related costs is, in general, commensurate with income levels. States parties should establish housing subsidies for those unable to obtain affordable housing, ensure forms and levels of housing finance which adequately reflect housing needs, and ensure tenants are protected against unreasonable rent levels or rent increases. In societies where natural materials constitute the chief sources of building materials for housing, steps should be taken by States parties to ensure the availability of such materials.
- d) *Habitability*: Adequate housing must provide occupants with adequate space and protection from cold, damp, heat, rain, wind or other threats to health, structural hazards, and disease vectors. The physical safety of occupants must be guaranteed. The Committee encourages States parties to comprehensively apply the Health Principles of Housing prepared by WHO;
- e) *Accessibility*: Disadvantaged groups must be accorded full and sustainable access to adequate housing resources. Housing law and policy should fully account for the special housing needs of disadvantaged groups. Within many States parties increasing access to land by landless or impoverished segments of the society should constitute a central policy goal. Discernible governmental obligations need to be developed aiming to substantiate the right of all to a secure place to live in peace and dignity, including access to land as an entitlement.
- f) *Location*: Adequate housing must be in a location which allows access to employment options, health-care services, schools, childcare centres and other social facilities. This is true both in large cities and in rural areas. Housing should not be built on polluted sites nor in immediate proximity to pollution sources that threaten the right to health of the inhabitants.
- g) *Cultural adequacy*: The way housing is constructed, the building materials used and the policies supporting these must appropriately enable the expression of cultural identity and diversity of housing. Activities geared towards development or modernization in the housing sphere should ensure that the cultural dimensions of housing are not sacrificed, and that, inter alia, modern technological facilities, as appropriate are also ensured.

Mobility rights

Public authorities increasingly recognise that the realisation of human rights, such as access to **employment, health, education and social amenities** are dependent on rights-based urban and transport planning. Without sufficient proximate services and access to transport, disadvantaged communities remain disconnected from social amenities, employment, education, and health services.

In relation to *employment*, Article 6 of the ICESCR states that:

1. The States Parties to the present Covenant recognize the right to work, which includes the right of everyone to the opportunity to gain his living by work which he freely chooses or accepts, and will take appropriate steps to safeguard this right.

In relation to *health*, Article 12 of the ICESCR states that

2. The States Parties to the present Covenant recognize the right of everyone to the enjoyment of the highest attainable standard of physical and mental health.

In relation to *education*, Article 13 states that

3. The States Parties to the present Covenant recognize the right of everyone to education. They agree that education shall be directed to the full development of the human personality and the sense of its dignity and shall strengthen the respect for human rights and fundamental freedoms.

In relation to *cultural life*, Article 15 states that

1. The States Parties to the present Covenant recognize the right of everyone:
 - a) To take part in cultural life.
 - b) To enjoy the benefits of scientific progress and its applications.

Each of the FITTER case study countries have ratified the ICESCR with no declarations or reservations in relation to Articles 6, 11, 12, 13 or 15.

2.1.3 The European Social Charter System

[The European Social Charter \(ESC\) \(1961\)](#) focused on labour rights. It was updated in 1996 with the [Revised European Social Charter \(RESC\)](#). This contains a broader range of social rights.

To ratify the ESC/RESC a State must accept a minimum number and range of provisions in Part II of the ESC/RESC but it can then decide what other provisions of the ESC/RESC it wishes to be bound by.

Right to adequate housing

Article 16 of the RESC states that:

With a view to ensuring the necessary conditions for the full development of the family, which is a fundamental unit of society, the Parties undertake to promote the economic, legal and social protection of family life by such means as social and family benefits, fiscal arrangements, **provision of family housing**, benefits for the newly married and other appropriate means.

As part of Article 23 - The right of elderly persons to social protection – the RESC states that States must:

enable elderly persons to choose their lifestyle freely and to lead independent lives in their familiar surroundings for as long as they wish and are able, by means of:

- a) **provision of housing** suited to their needs and their state of health or of adequate support for adapting their housing;
- b) the health care and the services necessitated by their state;

Article 31 of the RESC states that:

With a view to ensuring the effective exercise of the right to housing, the Parties undertake to take measures designed:

1. to promote access to housing of an adequate standard;
2. to prevent and reduce homelessness with a view to its gradual elimination;
3. to make the price of housing accessible to those without adequate resources.

Mobility rights

As outlined above public authorities increasingly recognise that the realisation of rights to employment, health and education are dependent on rights-based and integrated land-use and transportation planning.

The RESC refers to mobility rights under Article 15 - The right of persons with disabilities to independence, social integration and participation in the life of the community. Therein, in relation to people with disabilities, Parties undertake:

to promote their full social integration and participation in the life of the community in particular through measures, including technical aids, aiming to overcome barriers to communication and mobility and enabling access to transport, housing, cultural activities and leisure.

Integrated land-use and transportation planning is arguably also central to the fulfilment of Article 30 of the RESC – The right to protection against poverty and social exclusion. The RESC states that:

With a view to ensuring the effective exercise of the right to protection against poverty and social exclusion, the Parties ensure:

- a) to take measures within the framework of an overall and co-ordinated approach to promote the effective access of persons who live or risk living in a situation of social exclusion or poverty, as well as their families, to, in particular, employment, housing, training, education, culture and social and medical assistance;
- b) to review these measures with a view to their adaptation if necessary.

Not all of the FITTER case study countries have agreed to be bound by the provisions detailed above ([Council of Europe, 2024](#)).

Table 1 - Ratification Status of Selected Social Rights Provisions under the Revised European Social Charter in FITTER Case Study Countries

Article	Provision	Spain	Italy	Portugal	Ireland	Germany	Hungary
15	The right of persons with disabilities to independence, social integration and participation in the life of the community	✓	✓	✓	✓	✓	✓
16	The right of the family to social, legal and economic protection	✓	✓	✓	✓	✓	✓
23	The right of elderly persons to social protection	✓	✓	✓	✓	✓	✗
30	The right to protection against poverty and social exclusion	✓	✓	✓	✓	✗	✗
31	The right to housing	✓	✓	✓	✗	✗	✗

[Council of Europe, 2025](#)

2.1.4 EU Framework – Entitlements to Energy Services

In contrast to the former agreements/charter, where energy was just amongst the topic of housing, the European Pillar of Social Rights (2017) and the UN Sustainable Development Goals (2015) include energy as an essential service which everyone is entitled to access.

The concept of energy poverty was introduced by Directive [2009/72/EC](#) of the European Parliament and of the Council. The [Recommendation \(EU\) 2020/1563](#) and the accompanying staff working document provides guidance on energy poverty as well as on the definition of what constitutes a household affected by energy poverty. It provides a set of 13 energy poverty indicators from which Member States can choose those available and relevant to their context. Member States' chosen indicators need to reflect different facets of energy poverty and use supplemental data sets to reflect local realities.

As outlined in Commission Recommendation (EU) 2023/2407

Energy poverty is a multidimensional phenomenon. In many cases, this situation is driven primarily by three underlying root causes, linked to **high energy expenditure in proportion to the household budget, low levels of income, and low energy performance of buildings and appliances**. The situation of a household can be

further influenced by geographic and climate factors, household characteristics, gender, health, and specific household energy and transportation needs. As such, households with higher energy needs, which include families with children, persons with disabilities and older persons, are also more susceptible to energy poverty and to its effects. Women, and in particular those who are single parents and older women, are also particularly affected by energy poverty due to structural inequalities in income distribution, socioeconomic status and the gender care gap ([EC, 2023:1](#))

The principles of the [European Pillar of Social Rights Action Plan](#) set the vision for the ‘social rulebook’. Principles 19 and 20 cover energy, housing and transport:

19. Housing and assistance for the homeless

- a. Access to social housing or housing assistance of good quality shall be provided for those in need.
- b. Vulnerable people have the right to appropriate assistance and protection against forced eviction.
- c. Adequate shelter and services shall be provided to the homeless in order to promote their social inclusion.

20. Access to essential services

Everyone has the right to access essential services of good quality, including water, sanitation, energy, transport, financial services and digital communications. Support for access to such services shall be available for those in need.

2.1.5. The realisation of rights

Core principles which apply to the realisation of social, economic and cultural rights include the principles that:

- States should respect, protect and fulfil rights
- States have a continuing obligation to take appropriate steps to realise rights as quickly and effectively as possible
- States are obliged to guarantee rights without discrimination of any kind.
- States must realise rights to the maximum available resources
- States should not deprive people of rights they previously enjoyed
- States are obliged to provide for the ‘minimum essential’ levels of rights
- States must enable transparency, accountability and participation. Citizens should have access to remedies in cases of rights violations
- A State’s human rights obligations apply both within and beyond its territory ([IHREC, 2016](#))

States’ compliance with obligations are assessed in light of available resources and rights can be realised over time. However the principle of non-discrimination requires immediate implementation i.e. States must ensure the enjoyment of economic, social, and cultural rights without discrimination based on ‘race, sex, language, religion, political opinion, nationality, social origin, property, birth, and ‘other status’ which encompasses disability, health status or sexual orientation, poverty, and economic marginalisation’ (Russell et al, 2021: 6).

2.1.6 Sustainable Development Goals

The Sustainable Development Goals (SDGs) were adopted by the United Nations in 2015. Especially, SDG 7 follows the purpose to “Ensure access to affordable reliable sustainable and modern energy for all”.

Table 2 – Targets and Indicators for Sustainable Development Goal 7: Affordable and Clean Energy

Target		Indicator	
7.1	By 2030, ensure universal access to affordable, reliable and modern energy services.	7.1.1 7.1.2	Proportion of population with access to electricity Proportion of population with primary reliance on clean fuels and technology
7.2	By 2030, increase substantially the share of renewable energy in the global energy mix.	7.2.1	Renewable energy share in the total final energy consumption
7.3	By 2030, double the global rate of improvement in energy efficiency.	7.3.1	Energy intensity measured in terms of primary energy and GDP
7.4	By 2030 enhance international cooperation to facilitate access to clean energy research and technology, including renewable energy, energy efficiency and advanced and cleaner fossil-fuel technology, and promote investment in energy infrastructure and clean energy technology.		
7.5	By 2030, expand infrastructure and upgrade technology for supplying modern and sustainable energy services for all in developing countries, in particular least developed countries, small island developing States and landlocked developing countries, in accordance with their respective programmes of support.		
7. A	By 2030, enhance international cooperation to facilitate access to clean energy research and technology, including renewable energy, energy efficiency and advanced and cleaner fossil-fuel technology, and promote investment in energy infrastructure and clean energy technology	7.A.1	International financial flows to developing countries in support of clean energy research and development and renewable energy production, including in hybrid systems
7. B	By 2030, expand infrastructure and upgrade technology for supplying modern and sustainable energy services for all in developing countries, in particular least developed countries, small island developing States, and land-locked developing countries, in accordance with their respective programmes of support	7.B.1	Installed renewable energy-generating capacity in developing and developed countries (in watts per capita)

(UN, NDa)

With regard to the right to adequate housing (including access to energy services) and mobility rights - SDG 11 aims to ‘make cities and human settlements inclusive, safe, resilient and sustainable’.

Targets 11.1, 11.2, 11.3 are directly linked to the realisation of housing and mobility rights. In addition, any greenhouse gas emission reductions achieved through more sustainable housing and transport will also contribute to the achievement of targets 11.4, 11.6 and 11.7.

Table 3 - Targets and Indicators for Sustainable Development Goal 11: Sustainable Cities and Communities

Target		Indicator	
11.1	By 2030, ensure access for all to adequate, safe and affordable housing and basic services and upgrade slums	11.1.1	Proportion of urban population living in slums, informal settlements or inadequate housing
11.2	By 2030, provide access to safe, affordable, accessible and sustainable transport systems for all, improving road safety, notably by expanding public transport, with special attention to the needs of those in vulnerable situations, women, children, persons with disabilities and older persons	11.2.1	Proportion of population that has convenient access to public transport, by sex, age and persons with disabilities
11.3	By 2030, enhance inclusive and sustainable urbanization and capacity for participatory, integrated and sustainable human settlement planning and management in all countries	11.3.1	Ratio of land consumption rate to population growth rate
		11.3.2	Proportion of cities with a direct participation structure of civil society in urban planning and management that operate regularly and democratically
11.4	Strengthen efforts to protect and safeguard the world's cultural and natural heritage	11.4.1	Total per capita expenditure on the preservation, protection and conservation of all cultural and natural heritage, by source of funding (public, private), type of heritage (cultural, natural) and level of government (national, regional, and local/municipal)
11.5	By 2030, significantly reduce the number of deaths and the number of people affected and substantially decrease the direct economic losses relative to global gross domestic product caused by disasters, including water-related disasters, with a focus on protecting the poor and people in vulnerable situations	11.5.1	Number of deaths, missing persons and directly affected persons attributed to disasters per 100,000 population
		11.5.2	Direct economic loss attributed to disasters in relation to global domestic product (GDP)
		11.5.3	(a) Damage to critical infrastructure and (b) number of disruptions to basic services, attributed to disasters
11.6	By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management	11.6.1	Proportion of municipal solid waste collected and managed in controlled facilities out of total municipal waste generated, by cities
		11.6.2	Annual mean levels of fine particulate matter (e.g. PM2.5 and PM10) in cities (population weighted)
11.7	By 2030, provide universal access to safe, inclusive and accessible, green and public spaces, in particular for women and children, older persons and persons with disabilities	11.7.1	Average share of the built-up area of cities that is open space for public use for all, by sex, age and persons with disabilities
		11.7.2	Proportion of persons victim of non-sexual or sexual harassment, by sex, age, disability status and place of occurrence, in the previous 12 months

11.a	Support positive economic, social and environmental links between urban, peri-urban and rural areas by strengthening national and regional development planning	11.a.1	Number of countries that have national urban policies or regional development plans that (a) respond to population dynamics; (b) ensure balanced territorial development; and (c) increase local fiscal space
11.b	By 2020, substantially increase the number of cities and human settlements adopting and implementing integrated policies and plans towards inclusion, resource efficiency, mitigation and adaptation to climate change, resilience to disasters, and develop and implement, in line with the Sendai Framework for Disaster Risk Reduction 2015-2030, holistic disaster risk management at all levels	11.b.1	Number of countries that adopt and implement national disaster risk reduction strategies in line with the Sendai Framework for Disaster Risk Reduction 2015–2030
		11.b.2	Proportion of local governments that adopt and implement local disaster risk reduction strategies in line with national disaster risk reduction strategies
11.c	Support least developed countries, including through financial and technical assistance, in building sustainable and resilient buildings utilizing local materials		

(UN, ND_b)

2.2. Enabling Climate Justice

As outlined in D2.1, FITTER defined the *just* twin transition as a process that addresses intersectional inequalities, with fair burden-sharing, while controlling climate change and taking the dimensions of environmental, climate, and energy justice into account (adapted from [Galgóczi, 2023](#)).

In line with our definition of a just transition, we accounted for core tenets of environmental, energy and climate justice:

- **Environmental justice** is concerned with the equitable spatial and temporal distribution of the environmental burdens and the environmental benefits of systems and structures. The participation of individuals and communities most vulnerable to environmental risks in relevant decision-making processes is a core element of enabling environmental justice.
- **Climate justice** is concerned with addressing inequitable outcomes of climate change for differing peoples, communities, contexts and generations. This entails greater equity in the distribution of climate related burdens and responsibilities, as well as greater parity of decision-making and human rights protection.
- **Energy justice** is concerned with developing mechanisms to enable justice (recognitive, representative, distributive and restorative) at all stages of the energy life cycle 1) Extraction; 2) Production; 3) Operation and supply; 4) Consumption; 5) Waste Management.

We can now further streamline the FITTER definition of a just transition to more explicitly account for the digital divide and incorporate the requirement for systems redesign.

A just twin transition is a process which addresses intersectional inequalities; enables parity of participation, decision making, and human rights protection (ensuring that this is not dependent on digital access and skills); and addresses the burdens of the climate crisis by enabling the equitable spatial and temporal distribution of environmental harms and environmental benefits through systems redesign.

As highlighted in [D2.1](#), and discussed further in section 2.2.1 below, the Paris Agreement set out a goal of limiting global temperatures to 1.5C rise above pre-industrial levels. However, most climate scientists

believe that that limit is now unachievable ([Carrington, 2024a](#)). Currently global temperatures have risen by 1.2C above pre-industrial levels and this has already caused extreme weather events, with people losing their lives and/or livelihoods due to the climate crisis. As such, it is certain that the climate crisis is causing and will continue to cause harm and suffering to many. Enabling **fair burden-sharing** is thus a core concern of the just transition. This can be politically difficult to discuss and address. However, as outlined below, it is clear that a business as usual (BAU) approach that favours incumbent infrastructures, technological fixes, and established economic interests will neither deliver the required greenhouse gas emissions reduction, nor enable the achievement of social justice objectives ([Scrase and Smith, 2009](#); [Revez et al, 2021](#); [Gough 2022](#))

2.2.1 IPCC Analysis - Shared Socio-economic Pathways (SSPs), Illustrative Mitigation Pathways (IMPs) and Key Risks (KRs) in Europe

A limitation of the FITTER methodology is that the proposed risk assessment process assesses current policies as they are, and it identifies risks to disadvantaged groups in the short term. Yet, as outlined in D2.1, the *transition* to a net zero economy requires transformative, systems-wide policy design; and a *just* transition requires consideration of potential societal harms and benefits in the short, medium and long term. This is a key reason for supplementing the risk assessment process with systems inquiry.

The scientific evidence clearly demonstrates that the transition to net zero will be challenging. Neither a continuation of ‘current policies’ nor a ‘gradual strengthening’ of current policies limit temperature increases in line with the Paris Agreement.

The Paris Agreement criteria are

- a) limit warming to 1.5°C;
- b) limit warming to well below 2°C; and
- c) achieve net zero greenhouse gas emissions in the second half of the 21st century

Even with warming limited to 1.5°C the observed social and environmental harms from climate change are considerable. As outlined by Working Group II within the Intergovernmental Panel on Climate Change (IPCC)

Human-induced climate change, including more frequent and intense extreme events, has caused widespread adverse impacts and related losses and damages to nature and people, beyond natural climate variability. Some development and adaptation efforts have reduced vulnerability. Across sectors and regions, the most vulnerable people and systems are observed to be disproportionately affected. The rise in weather and climate extremes has led to some irreversible impacts as natural and human systems are pushed beyond their ability to adapt (high confidence) ([IPCC, AR6, WGII, B.1: 9](#))

The IPCC was created to ‘provide policymakers with regular scientific assessments on climate change, its implications and potential future risks, as well as to put forward adaptation and mitigation options’ ([IPCC, ND](#)). The IPCC determines the state of knowledge on climate change through regular assessments. Assessments identify areas of agreement on topics related to climate change, and areas where further research is required.

Assessment Process

To support objectivity and transparency IPCC reports are drafted and reviewed in several stages.

- A first draft of a report is prepared by chapter teams of authors based on existing evidence base
- The First Order Draft is reviewed by experts who submit comments
- The Review Editor ensures that all comments are given due consideration by author teams
- Author teams prepare a Second Order Draft and a first draft of the report’s Summary for Policymakers (SPM)
- The Second Order Draft and the SPM are reviewed by experts who submit comments
- On receipt of the final review comments author teams prepare final drafts of the full report and SPM

- The ‘final draft of the report is distributed to governments for a final round of written comments on the SPM, before governments meet in plenary session to approve the SPM line by line and accept the underlying report.
- All comments submitted through this review process are addressed by authors, and a written response is provided to each comment on the First Order and Second Order Drafts for later publication.’ (IPCC, 2015: 1)

The Treatment of Uncertainties

The IPCC have issued guidance to author teams on the consistent treatment of uncertainties (Mastrandea et al, 2010)

Experts are asked to:

Where relevant, assign confidence levels to the presented evidence ranging from ‘very low,’ ‘low,’ ‘medium,’ ‘high,’ and ‘very high.’ The confidence level increases in line with the more robust the evidence base is, and the greater the degree of scientific agreement.

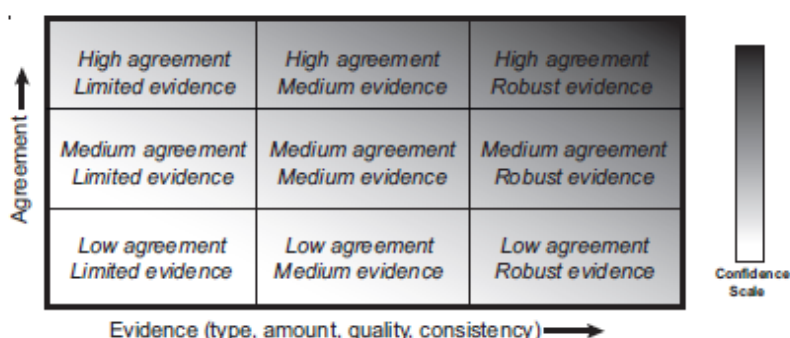


Figure 2 -A depiction of evidence and agreement statements and their relationship to confidence. Confidence increases towards the top-right corner as suggested by the increasing strength of shading. Generally, evidence is most robust when there are multiple, consistent independent lines of high-quality evidence (Mastrandea et al, 2010: 3)

Likelihood Scale	
Term	Likelihood of Outcome
Virtually certain	99-100% probability
Very likely	90-100% probability
Likely	66-100% probability
About as likely as not	33-66% probability
Unlikely	0-33% probability
Very unlikely	0-10% probability
Exceptionally unlikely	0-1% probability

Although the IPCC guidance note on the consistent treatment of uncertainties arguably requires further development, it is a ‘first step in towards helping experts from different disciplines to communicate scientific evidence and related uncertainties transparently and congruently to policymakers and the general public’ (Kause et al, 2022: 2). Of note is that the IPCC does not recommend expressing confidence levels for all types of evidence and one suggestion has been to ‘only use confidence levels when evidence is robust and agreement is high’ (Kause et al, 2022: 2). The use of ‘high confidence’ levels has increased in IPCC reports since 1990

IPCC – Sixth Assessment Report (AR6)

The IPCC produced the Sixth Assessment Report (AR6) during the sixth assessment cycle (October 2015–July 2023). It was published in three sections.

In August 2021 Working Group I contribution was released. Working Group II and III contributions were released in February and April 2022 respectively. The Synthesis Report was released in March 2023.

Working Group I modelled potential future scenarios in relation to the physical science of climate change – called **Shared Socio-economic Pathways (SSPs)**.

SSPs consider the rate at which GHG emissions are reduced and incorporate socio economic factors including population growth, urbanization, land use, wealth distribution and education.

Within one SSP ‘there can be multiple emissions scenarios that lead to different levels of radiative forcing and, therefore, global-mean temperature increase’ ([Climatedata, ND](#))

Table 4 – Shared Socio-economic Pathways (SSPs): Climate Futures and Emissions Scenarios

Shared Socio-economic Pathways (SSPs)	Scenario	Summary
1. Sustainability	Scenario 1:	Only scenario that meets the Paris Agreement goals.
	Most Optimistic	This requires a rapid transition to sustainable practices and the prioritisation of wellbeing over economic growth .
	1.5°C by 2050	
	SSP1-1.9	Net zero is reached around 2050. Temperatures peak at 1.5°C higher before declining to 1.4°C higher by 2100. Extreme weather events are more frequent, but the worst impacts of climate change are averted.
	Scenario 2	This does not meet the Paris Agreement goals.
	Next best scenario	This requires the same socioeconomic shifts towards sustainability as SSP1-1.9
	1.8°C by 2100	
	SSP1-2.6	Net zero is reached after 2050. Temperatures stabilize around 1.8C higher by 2100.
2. Middle of the road	Scenario 3	This does not meet the Paris Agreement goals.
	Middle of the road	Socioeconomic trends continue as they are with only gradual development toward sustainability and uneven development.
	2.7°C by 2100	
3. Regional Rivalry (SSP3-7.0)	SSP2-4.5	Net zero is not reached by 2100. Temperatures rise 2.7°C higher by 2100.
	Scenario 4	This does not meet the Paris Agreement goals.
	Dangerous	Socioeconomic trends continue as they are with CO2 emissions doubling by 2100. Countries prioritise own interests, fostering competition.
	3.6°C by 2100	
4. Inequality (SSP4-6.0)	SSP3-7.0	Temperatures rise 3.6°C higher by 2100.
5. Fossil-fuel development	Scenario 5	This does not meet the Paris Agreement goals.
	Avoid at all costs	Socioeconomic trends continue as they are with CO2 emissions doubling by 2050. The economy grows quickly

	4.4°C by 2100	but it is based on fossil fuel use and energy-intensive lifestyles.
	SSP5-8.5	Temperatures rise 4.4°C higher by 2100.

([Januta, 2021](#); [Hausfather, 2018](#))

With each rise in temperature the risks to nature and human health increase significantly. Even limiting temperature rise to 1.5°C results in significant environmental and social harms. As noted in the contribution by Working Group II - with regard to risks in the near term - 2021-2040:

Global warming, reaching 1.5°C in the near-term, would cause unavoidable increases in multiple climate hazards and present multiple risks to ecosystems and humans (very high confidence). The level of risk will depend on concurrent near-term trends in vulnerability, exposure, level of socioeconomic development and adaptation (high confidence). Near-term actions that limit global warming to close to 1.5°C would substantially reduce projected losses and damages related to climate change in human systems and ecosystems, compared to higher warming levels, but cannot eliminate them all (very high confidence) ([IPCC AR6 WGII SPM B.3](#)).

IPCC Working Group III focused on possible mitigation pathways known as ‘**Illustrative Mitigation Pathways**’ (IMP).

5 pathways illustrate key mitigation measures, and two further pathways illustrate high emissions that result from the continuation of current policies or from moderately increased action. The goal is to explain how our choices now will shape the extent of climate breakdown.

Table 5 – Illustrative Mitigation Pathways (IMPs): Projected Climate Outcomes Based on Policy and Socioeconomic Choices

Name	Description	Matching Category
Shifting pathways (IMP-SP)	Illustrates mitigation in the context of a broader shift towards sustainable development, including by reducing inequality, and with a phase-out of fossil fuels.	C1a
Low demand (IMP-LD)	Illustrates a strong emphasis on energy-demand reductions, and with a phase-out of fossil fuels.	C1a
Renewables (IMP- Ren)	Illustrates a future with a heavy reliance on renewable energy.	C1b
Extensive use of net negative emissions (IMP – Neg)	Illustrates a future with deployment of carbon dioxide removal that results in net negative global GHG emissions.	C2
Gradual strengthening of current policies (IMP-GS)	Illustrates a gradual strengthening of climate policies implemented in 2020, with rapid reductions mid-century and a massive reliance on net negative emissions by the end of this century.	C3b
Moderate Action (ModAct)	Illustrates a future in which, by 2030, countries meet the Nationally Determined Contributions (NDCs) they set out in 2020, but only moderate further action is taken	C6
Current Policies (CurPol)	Illustrates a future with relatively high emissions, highlighting what happens to future emissions and temperatures if we follow policies implemented in 2020.	C7

([Kloenne et al, 2023](#))

The 1.5 °C limit in the Paris Agreement ‘refers to long-term warming – the IPCC uses 20 to 30 years’ ([Kloenne et al, 2023](#)).

The ‘Shifting Pathways’ (C1a) and ‘Low Demand’ (C1a) pathways have a 50% or higher chance of limiting warming to 1.5°C (with no or limited overshoot); around a 90% chance of limiting warming to 2°C; and

reach net zero GHG emissions by 2070-2075. As such they meet the Paris Agreement goals ([Kloenne et al, 2023](#)).

However, it is increasingly unlikely any of the C1 pathways which could limit warming to 1.5°C will be pursued.

As Working Group III note, in order to remain in line with the Paris Agreement, net global GHG emissions would need to fall from 2019 levels by '43% [34–60%] by 2030 and 84% [73–98%] by 2050' ([IPCC AR6 WGIII C1.1](#)).

As of October 2025, fewer than a third of countries (62 out of 197) have sent their nationally determined contributions (NDCs) under the Paris Agreement to the UN. Neither the US nor Europe have sent their NDCs and China is alleged to have set weak targets. Of the countries that have submitted their NDCs, together their plans would cut carbon by about 10% by 2035, from 2019 levels. This falls far short of the reduction of 60% needed to limit global warming to 1.5°C ([Watts and Xipai, 2025](#); [Harvey, 2025](#))

To limit warming to 1.5°C with limited overshoot requires a rapid shift to **sustainable development, low demand lifestyles, and renewable energy**. Even with those measures

All pathways that limit global warming to 1.5°C with limited or no overshoot project the use of carbon dioxide removal (CDR) on the order of 100–1000 GtCO₂ over the 21st century. CDR would be used to compensate for residual emissions and, in most cases, achieve net negative emissions to return global warming to 1.5°C following a peak (high confidence). CDR deployment of several hundreds of GtCO₂ is subject to multiple feasibility and sustainability constraints (high confidence). Significant near-term emissions reductions and measures to lower energy and land demand can limit CDR deployment to a few hundred GtCO₂ without reliance on bioenergy with carbon capture and storage (BECCS) (*high confidence*) ([IPCC, 2018: 17](#))

Relying on heavy use of carbon removal technology (C2) risks failure if the technology cannot be scaled up in time, entails a high overshoot for up to several decades, and the consequent environmental and social harms. **Moderate Action (C6) or Current Policies (C7)** entirely fail to limit peak warming to 1.5°C throughout the 21st century ([Kloenne et al, 2023](#)).

As outlined in D2.1, current policies mean the world is on track for about a 2.7 °C rise ([Carrington, 2024a](#)). This will have negative impacts in terms of exposure to lethal heat and humidity, water shortages, the loss of coral reefs, Amazon drying, Arctic sea ice melt, sea level rise (SLR), plant and animal species extinction, and decreases in food production.

Four key risks have been identified for Europe. These are 'more severe at 2°C global warming levels (GWL) compared with 1.5°C GWL in scenarios with low to medium adaptation (*high confidence*). From 3°C GWL and even with high adaptation, severe risks remain for many sectors in Europe (*high confidence*)' ([Bednar-Friedl et al., 2022: 1819](#)). The four risks are:

1. mortality and morbidity of people and ecosystems disruptions due to heat
2. loss in agricultural production due to combined heat and droughts
3. water scarcity across sectors
4. impacts of floods on people, economies and infrastructure

Lethal Heat and Humidity

Across Europe, temperature rises will increase the frequency and severity of heatwaves and lead to a decline in the frequency of extreme cold spells. Heatwaves that used to happen about once every 50 years will become more frequent. The table below provides estimates of exposure to, and fatalities from, heat and cold waves across Europe at a range of temperature increases in 2100. (Table 6)

Table 6 - Human exposure to and fatalities from heat and cold waves in the EU and UK for different warming levels in 2100

	Base	1.5°C	2.0°C	3.0°C
People annually exposed to a 50-year heatwave	9.6 million	105 million	172 million	298 million
Annual fatalities from heatwaves	2800	30300	52400	96000
People annually exposed to a 50-year cold wave	9.6 million	5.1 million	2.8 million	1.2 million
Annual fatalities from cold waves	79	34	20	9

([Naumann et al, 2020: 1-2](#))

Loss of Agricultural Production

Rising temperatures have a significant effect on food production. Human induced climate change is estimated to have ‘reduced global agricultural total factor productivity by about 21% since 1961’ ([Ortiz-Bobea, 2021: 306](#)). In the EU ‘crop losses over the past decade were up to 30 % higher relative to predicted trends. Currently, droughts (54%), heavy rain (21%), frost (16%), and hail (9%) account for 80% of agricultural losses in the EU’ ([Jensen and Hourdin, 2025: 3](#)).

The risks of simultaneous crop failure increase disproportionately between 1.5 and 2 °C temperature increases. As such ‘surpassing the 1.5 °C threshold will represent a threat to global food security. For maize, risks of multiple breadbasket failures increase the most, from 6% to 40% at 1.5 to 54% at 2 °C warming’ ([Gaupp et al, 2019: 34](#)). After local temperatures reach 3°C without adaptation there are ‘negative yield impacts for all crops’ ([Porter et al, 2014: 505](#))

Due to heat and drought in Europe

substantive agricultural production losses are projected for most European areas over the 21st century, which will not be offset by gains in Northern Europe (*high confidence*). Yield losses for maize will reach 50% in response to 3°C GWL, especially in Southern Europe. Yields of some crops (e.g., wheat) may increase in Northern Europe if warming does not exceed 2°C (*medium confidence*). While irrigation is an effective adaptation option for agriculture, the ability to adapt using irrigation will be increasingly limited by water availability, especially in response to GWL above 3°C (*high confidence*) ([Bednar-Friedl et al., 2022: 1819-1820](#)).

Water Scarcity

[Liu et al \(2018: 275\)](#) estimate that globally, severe drought-affected populations would increase overall, and they would ‘increase in urban areas and decrease in rural areas in all global regions in 1.5 and 2 °C warmer worlds.’ (Table X)

Table 7 - Global populations exposed to water scarcity from severe drought for different warming levels in 2100

	1.5°C	2.0°C
Globally – total population exposed to water scarcity from severe droughts	+132.5 ± 216.2 million	+194.5 ± 276.5 million
Globally – total urban population exposed to water scarcity from severe droughts	+350.2 ± 158.8 million	+410.7 ± 213.5 million

Globally – total rural population exposed to water scarcity from severe droughts	-217.7 ± 79.2 million	-216.2 ± 82.4 million
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([Liu et al, 2018](#): 275)

In Europe

Risk of water scarcity will become high at 1.5°C and very high at 3°C GWL in Southern Europe (*high confidence*) and increase from moderate to high in Western Central Europe (*medium confidence*). In Southern Europe, more than a third of the population will be exposed to water scarcity at 2°C GWL; under 3°C GWL, this risk will double, and significant economic losses in water- and energy-dependent sectors may arise (*medium confidence*). For Western Central and Southern Europe, and for many cities, the risk of water scarcity will be strongly increasing under 3°C GWL. Adaptation becomes increasingly difficult at 3°C GWL and above, due to geophysical and technological limits; hard limits are likely first reached in parts of Southern Europe ([Bednar-Friedl et al., 2022](#): 1820).

Impacts of floods

Sea-level rise (SLR) will continue for centuries under all emission scenarios. Minimum estimates of SLR can be made, and mitigation plans can be put in place. However, there is significant uncertainty regarding the maximum levels of SLR that need to be planned for. As noted by the IPCC

The average rate of sea level rise was 1.3 [0.6 to 2.1] mm yr⁻¹ between 1901 and 1971, increasing to 1.9 [0.8 to 2.9] mm yr⁻¹ between 1971 and 2006, and further increasing to 3.7 [3.2 to 4.2] mm yr⁻¹ between 2006 and 2018 (*high confidence*). Human influence was *very likely* the main driver of these increases since at least 1971. Evidence of observed changes in extremes such as heatwaves, heavy precipitation, droughts, and tropical cyclones, and, in particular, their attribution to human influence, has further strengthened since AR5. Human influence has *likely* increased the chance of compound extreme events since the 1950s, including increases in the frequency of concurrent heatwaves and droughts (*high confidence*) ([IPCC, 2023: 5](#))

Under emissions pathways consistent with the Paris Agreement the

likely level rise is 0.15–0.23 m by 2050 and 0.28–0.55 m by 2100; while for the SSP5-8.5 GHG emissions scenario it is 0.20–0.29 m by 2050 and 0.63–1.01 m by 2100 (*medium confidence*). Over the next 2000 years, global mean sea level will rise by about 2–3 m if warming is limited to 1.5°C and 2–6 m if limited to 2°C (*low confidence*) ([IPCC, 2023](#): 18).

Sea level rise has significant negative geopolitical, environmental, and human security impacts.

Globally ‘between 4 million to 72 million people are expected to migrate due to SLR over the twenty-first century across a range of scenarios. The minimum is associated with the 5th percentile of the lowest SLR scenario (SSP1–1.9), and the maximum with the 95th percentile of the SSP3–7.0 scenario’ ([Ballesteros et al., 2025](#): 04).

Current estimates of annual damage from coastal flooding in Europe is ‘around EUR 1.25 billion but could increase by 2–3 orders of magnitude if coastal adaptation is only maintained to its current level.’ ([van de Wal et al, 2024](#): 2). As the IPCC note

risks to people and infrastructures from coastal, riverine and pluvial flooding will increase in Europe (*high confidence*). Risks of inundation and extreme flooding will increase with the accelerating pace of SLR along Europe’s coasts (*high confidence*). Above 3°C GWL, damage costs and people affected by precipitation and river flooding may double. Coastal flood damage is projected to increase at least tenfold by the end of the 21st century, and even more or earlier with current adaptation and mitigation (*high confidence*). Sea level rise represents an existential threat for coastal communities and their cultural heritage, particularly beyond 2100 ([Bednar-Friedl et al., 2022](#): 1820).

As outlined in [D2.2](#), to avoid the worst impacts of rising temperatures globally, transformative change is required. The IPCC have also highlighted that layering environmental policies on existing social policies and leaving the underlying reproduction, production and consumption systems in place keeps us on track for a

temperature increase of 2.7 °C. This is more than double the current increase to 1.2°C - at which negative environmental, social and economic impacts are already occurring.

2.2.2 Employing a Systems Lens to Enable Climate Justice

As outlined in this chapter States are duty bearers when it comes to defending human rights. There is an explicit right to adequate housing outlined in international law and in the European Social Charter. Rights to employment, health, education and social amenities are acknowledged to be dependent on rights-based urban and transport planning. Without sufficient proximate services and access to transport, disadvantaged communities remain disconnected from services and opportunities. The [European Pillar of Social Rights](#) (2017) and the [UN Sustainable Development Goals](#) (2015) include energy as an essential service which everyone is entitled to access. Further, temperatures are rising globally, and transformative action is required to limit global warming. Under current plans and commitments, the goal of limiting global warming to 1.5 °C is not possible. This will have serious and irreversible negative impacts in Europe and globally. From an environmental and social justice perspective a business-as-usual approach is unsustainable.

In [D2.2](#) we focused on proposed changes to the dominant economic paradigm. We contrasted understandings of care work, sources of wealth for individuals, sources of revenue for government, the environmental costs of economic activities, and proposed basic services provision and consumption patterns, under three paradigms – green growth, doughnut economics and a foundational economy approach. We argued that recognition of the political nature of technology, and achievement of the social control of technology, are necessary in order to enable sustainable, democratic and inclusive futures ([Sætra, 2023b](#): 262; [Bennahum et al 1998](#)). Our goal was to provide research partners with a good understanding of available paradigms and their strengths and limitations with regard to the achievement of a *just* twin transition.

We have since concentrated on further assessing the evidence base for transformative change. The scale and urgency of the challenges faced are significant. In a recent update to the Doughnut Economics framework Andrew Fanning and Kate Raworth ([2025](#)) detail how far off course the objective of meeting the needs of all, within the means of a living planet is. They highlight how:

- As income levels increase in a country *social shortfall improves* but *ecological overshoot worsens* (going beyond the safe space for humanity).
- The richest 20% country cluster, which includes 15% of the world’s population, contributes a disproportionately large 26–73% share to global ecological overshoot on an annual basis, depending on the indicator (median: 44%). This wealthy group holds a relatively small 0–21% share of global social shortfall (median: 2%). Of the case study countries Spain, Italy, Germany and Ireland are in this group.
- The middle 40% country cluster, which includes 43% of the world population, contributes a 27–58% share to global ecological overshoot (median: 52%) and holds a 29–53% share of global social shortfall (median: 33%). Of the case study countries Portugal and Hungary are in this group.
- The poorest 40% country cluster, which includes 42% of world population, contributes a negligible 0–18% share to global ecological overshoot (median: 4%). This poorer group holds a disproportionately large 27–97% share of global social shortfall (median: 63%).
- Global gross domestic product has doubled between 2000 and 2022. However modest progress has been made in reducing social shortfalls worldwide and ecological overshoot has increased rapidly. They note that our ‘collective pressure on the planet has substantially worsened since the early 2000s and now transgresses at least six of the nine planetary boundaries’ ([Fanning and Raworth, 2025](#): 48).

Fanning and Raworth call for

a deep renewal of both economic theory and practice, including by overcoming nations’ structural dependency on GDP growth, so that they can instead reorient towards ecologically regenerative and socially

distributive economic policies and outcomes. Such reorientation must acknowledge country-specific differences in priorities and responsibilities, given the disproportionate contribution to global ecological overshoot by the richest, and the disproportionate share of population in deprivation among the poorest, as shown by the Doughnut's country-cluster analysis ([2025: 53-54](#))

This call is in line with the two IPCC illustrative mitigation pathways that could potentially meet the Paris Agreement criteria. They involve a shift towards sustainable development, the prioritisation of social justice objectives and low demand lifestyles. Working Group III in the IPCC also highlight the utility of a systems lens and transformative actions, positing that

Employing a systems lens to address the 'root causes' of unsustainable development pathways (such as dysfunctional social or economic arrangements) rather than the 'symptoms' (dwelling quality, vehicle efficiency, etc.) can trigger the non-linear change needed for a transformation to take place ([Denton et al, 2022: 1737-1738](#))

They posit that transformative outcomes could be achieved by exploring the synergies between key objectives such as climate change adaptation, mitigation actions, increasing biodiversity, and achieving social equity. However, they caution that data on the nature and potential of these synergies is still under development.

In [D2.1](#) we outlined the need for a systems lens to enable policy makers to identify and address the root causes of the production of high emissions and reproduction of inequalities. However, given the initial scope of the FITTER project – an assessment of the green and digital transition across four sectors with a focus on both producers and consumers – it was not possible for us to translate that recommendation into a workable methodology. Following on from the submission of D2.2 the scope of the project has narrowed to three sectors with a focus on consumers. This enabled us to identify and adapt a systems inquiry methodology developed by the OECD ([2021](#), [2022](#)) for use in FITTER which is outlined in the next chapter by way of a supplement to the risk assessment process.

3. Overview of Relevant Methodologies

The purpose of D2.1-D2.3 is to develop a shared understanding at consortium level of problem context, problem complexity, key concepts, and relevant methodological approaches.

Enabling a twin transition to a net-zero economy, and ensuring that social justice objectives are achieved within that transition, are interconnected, highly complex processes.

It has proven challenging to adapt the risk assessment approach outlined in the Description of Action with the requirement of the identification and implementation of transformative actions that address root causes, as identified in the literature review and in consultation with sectoral experts (see [D3.2](#)). To assist policy makers identify the root causes of the production of high emissions and inequalities, the research team suggested supplementing the risk assessment framework with systems inquiry. A specific approach – as outlined by the OECD ([2021](#), [2022](#)) – was proposed relatively late in the project (M16). This meant that a full systems inquiry approach could not be incorporated into existing and future tasks. Understandably, FITTER task leaders are primarily committed to delivering on the risk assessment approach as outlined in the Description of Action. Although they understand the need for systems lens, the extent to which they can reasonably incorporate it into their tasks remains limited.

To further advance a consortium-wide shared understanding of the strengths and limitations of relevant methodological approaches this chapter provides a high-level overview of the purpose, strengths and limitations of systems inquiry, risk assessment, and impact assessment.

3.1. Systems Inquiry

A systems inquiry approach proposes that individual choices largely result from the systems in which people are embedded and the mental models that shape those systems.

Systems inquiry theorists argue that policy design primarily focused on awareness-raising i.e. encouraging individuals to make better choices, while not seeking to restructure a system and the rules which make such better choices difficult or unattractive, cannot achieve its objectives. A systems inquiry approach to policymaking proposes that public authorities need to deliberately redesign systems to ‘promote and facilitate other choices and trigger large-scale behavioural change which is otherwise unlikely’ ([OECD, 2022: 13](#); [Systems Innovation, ND](#)). Systems inquiry is concerned with identifying the ‘root causes’ of unsustainable practices such as dysfunctional social or economic arrangements. There are three stages to adopting a systems inquiry approach to policy design.

Table 8 – Stages of Systems Inquiry for Transformative Policy Design

1	Envision & Challenge	Envision goals and the patterns of behaviour a properly functioning system fosters; Challenge mental models underlying poorly functioning systems;
2	Understand & Determine	Understand why current system is not achieving envisioned goals and patterns of behaviour; Determine whether implemented and planned policies have the potential to redesign the system;
3	Prioritise & Scale Up	Prioritise and Scale Up the policies which can redesign systems to foster desirable patterns of behaviour

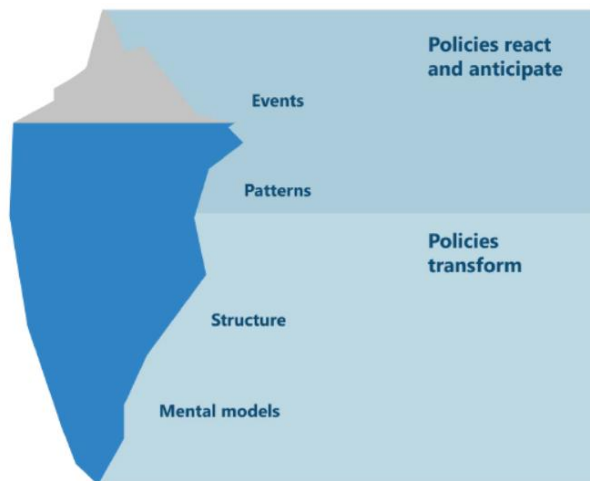
Four tools can assist with taking a systems inquiry approach to policy design.

1. **The iceberg model** serves as a visual guide to a systemic perspective
2. **Causal loop diagrams** illustrate the system’s interconnections and the dynamics at the source of the patterns of behaviour that policies seek to reshape
3. **Stock and flow analysis** illustrate the transformative potential of policies affecting flows given current stock levels in the system

4. **The leverage points framework** ([Meadows, 1999](#)) provides a guide to assessing the transformational potential of policies ([OECD, 2022](#): 16; [Systems Innovation, ND](#))

3.1.1 The Iceberg Model

Figure 3 – The Iceberg Model



Thinking about how systems work in terms of an iceberg can assist with differentiating ‘symptoms’ from ‘root causes’.

At the top of the iceberg, above the water level, are the ‘events’ we see and experience day to day such as traffic jams, an increasing number of people who are unhoused, and high greenhouse gas emissions. Reactive policies are designed to decrease the harm of observed events. They do not address root causes.

If these ‘events’ repeat over time they become ‘patterns’, located just below the surface of the water. A standard approach to policy making is often focused on looking at past patterns, in order

to predict future patterns, and then aiming to curb the negative effects of those patterns. However, there can be a lack of analysis with regard to what produces the ‘events’ and ‘patterns’ we are experiencing. Anticipatory policies seek to decrease the harm of existing trends. They do not address root causes either.

Deeper under water is the structure of the system. In simple terms this refers to the ‘rules of the game’. These rules structure who has power and how they can use their power. It is also the level at which information flows, with some people well informed, others less informed, and yet others misinformed or disinformed. Transformative policies seek to redesign the way the parts of a system are organised and interrelated such that repeated events (e.g. traffic congestion, homelessness, high GHG emissions) are ‘designed out’ of the way the system functions. They can potentially address root causes.

At the bottom of the iceberg, near the seabed are the dominant mental models. These refer to the mindset or paradigms out of which the system – its goals, structure, rules – arise. Neverending economic growth is arguably the current dominant whole-system goal although it is subject to increasing challenges from economists, environmentalists and politicians. Enabling a change in mindsets or paradigms can be highly transformative of a system as it involves developing a critical understanding of current and alternative values.

Identifying policies with high transformative potential requires a deep understanding of the system the policy is trying to transform, as well as how the policy affects the system ([OECD, 2022](#): 15)

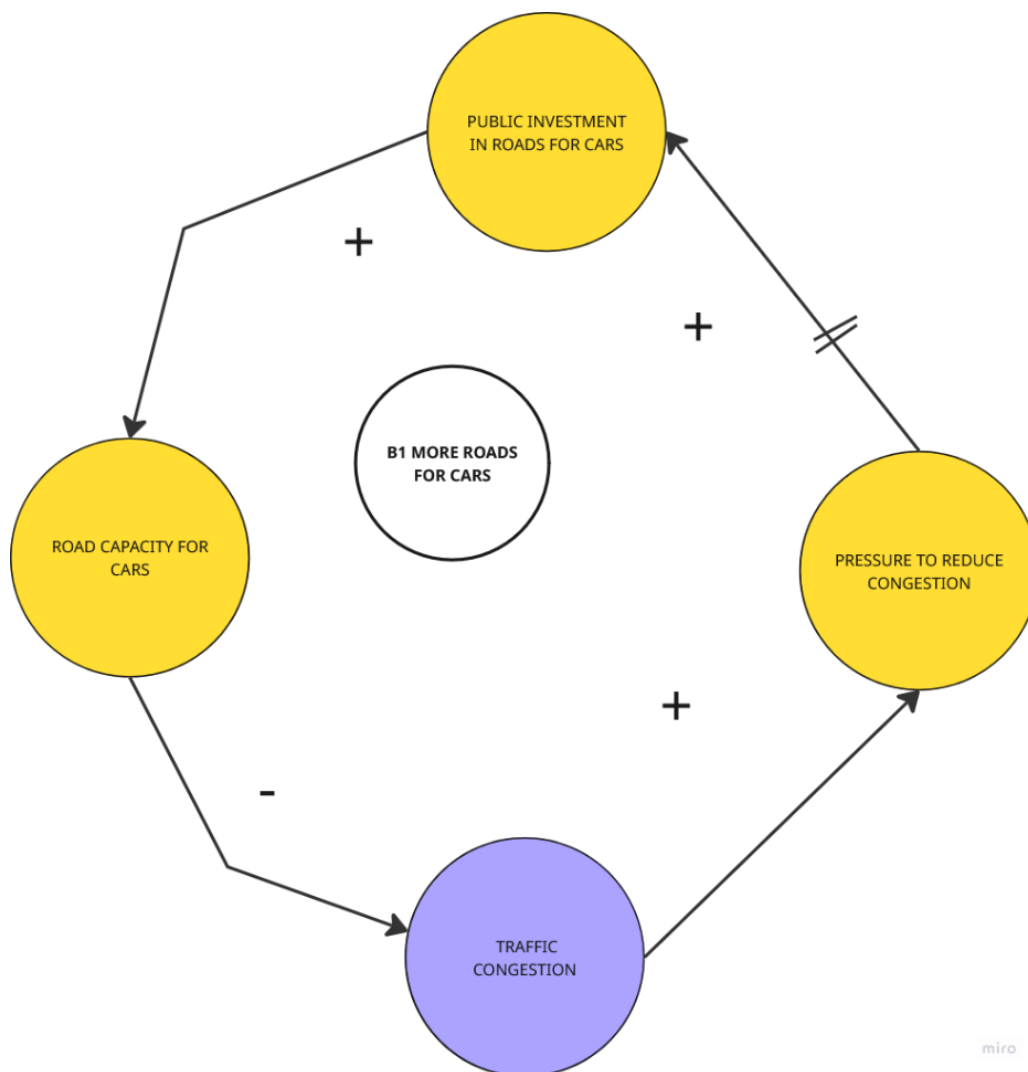
3.1.2 Causal Loop Diagrams

A causal loop diagram is a map that represents a system’s structure by its causal relationships.

- Text variables represent the individual elements of the system
- Arrows link the variables.
 - Variables can vary in the same direction i.e. an increase in one variable leads to an increase in the next; a decrease in one variable leads to a decrease in the next
 - Alternatively, variables can vary in the opposite direction i.e. an increase in one variable leads to a decrease in the next (or vice versa)

- Two links crossing an arrow indicates a delay i.e. it will take time for the change in one variable to cause a change in the variable it points to.
- A feedback loop links several variables together in a circle. The variables feed into each other leading to circular causal chains i.e. a variable affects a second variable which affects a third variable which affects the first variable again.
- Feedback loops can be reinforcing (R) or balancing (B).
 - In a reinforcing feedback loop each of the variables feed into one another in the same direction
 - In a balancing feedback loop variables affect each other in opposite directions
- The interactions between several feedback loops and time delays can produce more complex behaviours e.g. oscillations, s-shaped growth and overshoots ([OECD, 2022: 42](#))

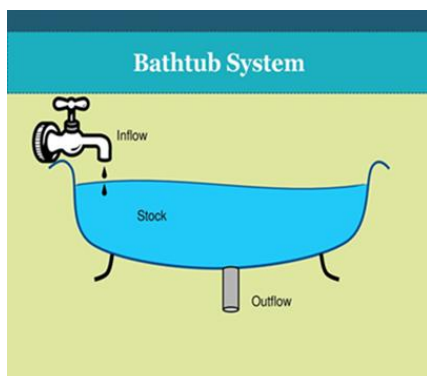
Figure 4 – Example of a balancing feedback loop



An example of a balancing feedback loop is shown in the figure above. When there is traffic congestion, public pressure to reduce congestion increases. After a delay this leads to increased public investment in roads for cars. As the road capacity for cars increases this leads to a (temporary) decrease in traffic congestion.

3.1.3 Stock and Flow Analysis

Figure 5 – Bathtub system



Stocks and flows can be analysed using a bathtub analogy. The standing stock is the water in the bathtub. This can be physical e.g. housing stock, and non-physical e.g. the belief that housing is an investment asset.

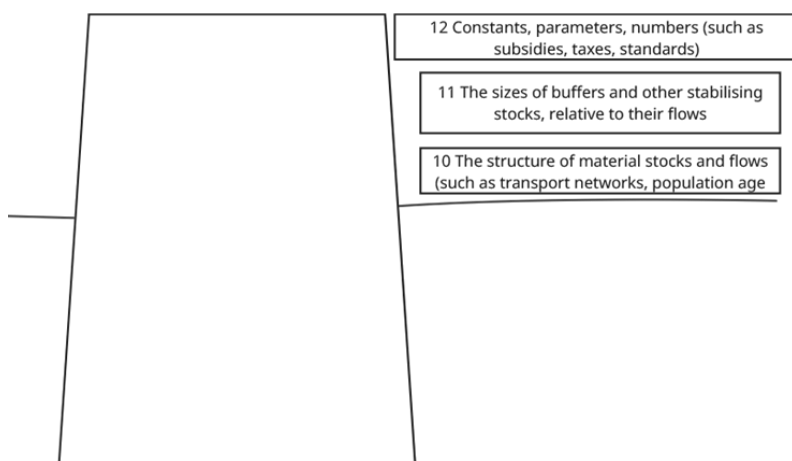
Inflows are represented by the tap(s) on the bathtub. Outflows are represented by the drain(s). Policy interventions – whether they are inflows (increasing standing stock) or outflows (decreasing standing stock) take time to accumulate their effects. Policy makers can use control of the inflow and outflow to build the standing stock that serves the goal of the system. However simple stocks and flows - e.g.

your bank account balance (standing stock), your income (inflow) and your bills (outflows) - when plumbed together with all the accounts in a bank and the wider financial system that that bank is part of, make up incredibly complex systems that are very difficult to map (Meadows, 1999: 4-5). As any system is complex it is also difficult to identify leverage points, defined as 'places within a complex system [...] where a small shift in one thing can produce big things in everything' (Meadows, 1999: 1).

3.1.4 Leverage Points Framework

To assist policy makers identify leverage points Meadows (1999) developed a typology of places to intervene in a system, rating them from 1-most effective to 12-least effective. These have been further divided in line with the elements of the iceberg model (events, patterns, structures, models) (Meadows, 1999; OECD, 2022: 56)

Figure 6 – Leverage points at the level of events



The least effective leverage points are at the top of the iceberg at the **level of events** (no. 12-10).

If the policy maker does not wish to change how the system functions they can concentrate on these actions and measures.

12. Constants, Parameters, Numbers

- These policy measures slow down or speed up the existing inflows and outflows to the standing stock. For example, changes to income tax brackets can make a difference to

low-income earners in the short-term but they do not change how the system works.

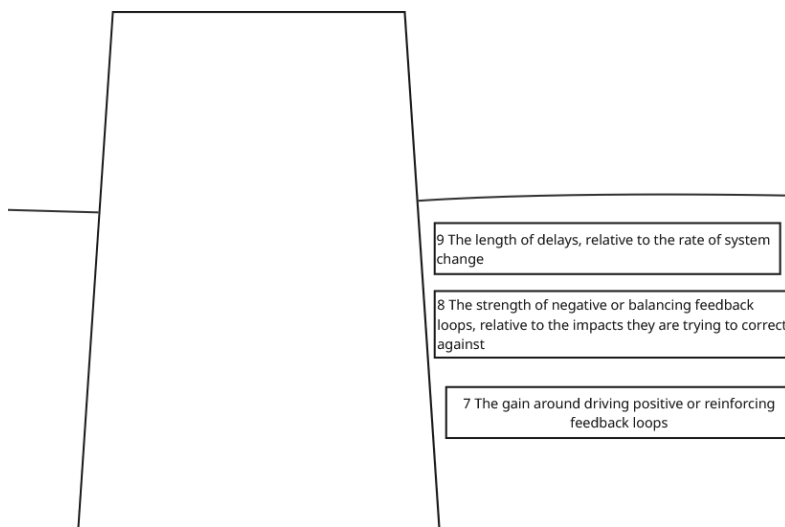
As Meadows notes, they are the same taps, plumbed into the existing standing stock, and turned according to the same old information, goals and rules. On their own, these measures are not going to change the system. However, 'parameters can become leverage points when they go into ranges that kick off one of the items later on this list' (Meadows, 1999: 6).

11. The sizes of buffers and other stabilising stocks, relative to their flows - Stocks that are big relative to the inflow and outflow are more stable than stocks that are small relative to the inflow and outflow. This is why many people have savings and try not to live paycheque to paycheque. Building up a big stabilising stock is termed a buffer. An example of a buffer is the addition of extra lanes to a motorway in order to ease traffic congestion.

To stabilise a system policy makers can increase the capacity of standing stock. However, if the buffer is too big it is inflexible, and it can cost a lot to build or maintain. Buffers (existing standing stock) are ‘usually physical entities, not easy to change’ ([Meadows, 1999: 6](#))

10. The structure of material stocks and flows (such as transport networks, population age structures) - Some stock-and-flow structures are either very difficult (e.g. the roads network) or impossible (e.g. the US baby boom post WWII) to change. As Meadows notes ‘physical structure is crucial in a system, but rarely a leverage point, because changing it is rarely simple. The leverage point is in proper design in the first place. After the structure is built the leverage is in understanding its limitations and bottlenecks and refraining from fluctuations or expansions that strain its capacity’ (1999: 8).

Figure 7 – Leverage points at the level of patterns



The next set of leverage points are at the **level of patterns**, just beneath the water level. These are not generally effective at transforming system functioning.

9. The length of delays, relative to the rate of system change – As systems are complex it takes time to map the state of the system. There can be information delays, meaning your map is out of date if elements within the system change in significant ways that nevertheless take time to understand and to map.

Delay length could be a leverage point but it is often not easily changeable. It is generally easier to slow down the growth of an existing system so that new ideas or technologies have sufficient time to be sustainably embedded, than it is to expect the existing system to pivot quickly in line with new ideas or technologies ([Meadows, 1999: 9](#)). A system cannot respond effectively to short-term changes when it has long-term delays e.g. significant increases in electricity demand from data centres can cause disruption and may destabilise the whole system because it takes years to build additional energy installations.

8. The strength of negative or balancing feedback loops, relative to the impacts they are trying to correct against - A negative or balancing feedback loop is the process through which the system self-corrects to get back in line with its goal when deviations from that goal are identified. As Meadows states ‘the strength of a negative loop – its ability to keep its appointed stock at or near its goal – depends on the combination of all its parameters and links – the accuracy and rapidity of monitoring, the quickness and power of response, the directness and size of corrective flows’ (1999: 10).

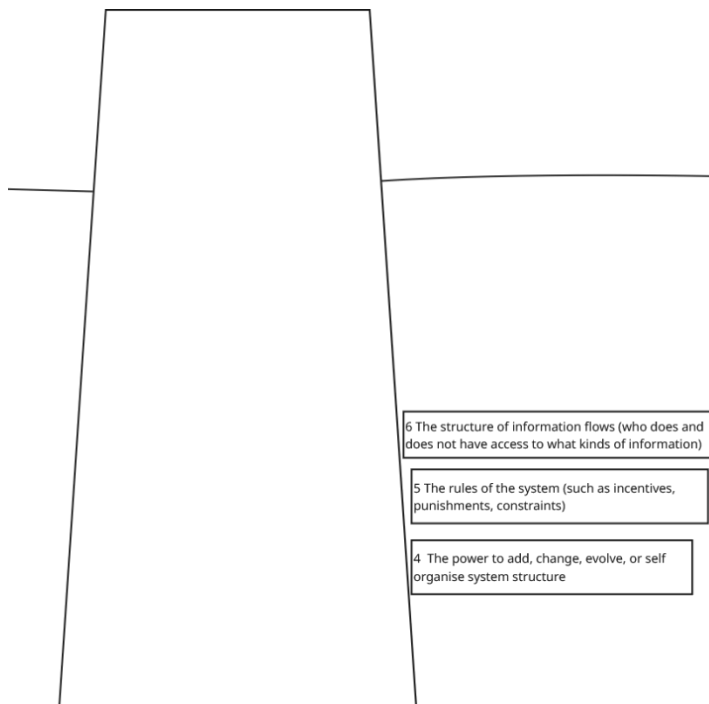
Negative feedback controls need to be adequate to the impact they are designed to correct. For example, the government can pass a Freedom of Information Act to increase transparency, but if the office assigned to process requests is under-resourced the issues of secrecy and corruption may not be effectively addressed.

7. The gain around driving positive or reinforcing feedback loops - A positive or reinforcing feedback loop is self-reinforcing. When a reinforcing feedback loop replicates much faster than balancing loops can respond, chaos can ensue. Although a reinforcing feedback loop is a source of growth of a particular phenomenon, an unchecked reinforcing feedback loop will ultimately destroy itself.

An example of a positive or reinforcing feedback loop is the ability of wealthy people to hire experts who can structure their finances so that they earn interest, pay less tax, and can afford private healthcare and private education. This is a ‘success for the successful’ loop. We can address this in two ways. One way is

to set up a parallel negative feedback loop, for example social welfare programmes for disadvantaged people. However, these tend to be weak and have little to no impact on the positive feedback loop. To address widening inequalities, it is better to directly weaken or reduce the positive loop e.g. through the introduction of inheritance tax, wealth taxes, the development of high-quality universal services (leverage point 5) ([Meadows, 1999: 12](#)).

Figure 8 – Leverage points at the level of structures



At the **structural level** of the rules of the game and information flows, the leverage points (6-4) can enable effective transformation of system functioning.

6. The structure of information flows (who does and does not have access to what kinds of information) - Whereas a parameter adjustment is the strengthening or weakening of an existing loop (e.g. subsidies, taxes, emissions reduction standards) – delivering information on how the existing system actually operates is a new loop which can cause significant behaviour change.

An example Meadows provides is the requirement that US factories report their emissions every year per the Toxic Release Inventory (1986). No new standards were introduced as part of this requirement, but some factories reduced

their emissions significantly in response - because of corporate shame ([Meadows, 1999: 13](#)).

Meadows' argument is that adding or restoring information can be a powerful intervention in a system, and it is much easier than rebuilding physical infrastructure.

Sufficient feedback on how a system is actually functioning is necessary to (ideally) prevent, or at least identify, malfunction(s). Such feedback needs to be adequate to the task though i.e. targeted at the right audience(s) and structured to generate sufficient behaviour change. Adding or restoring key information flows can ensure those who make decisions are accountable for those decisions.

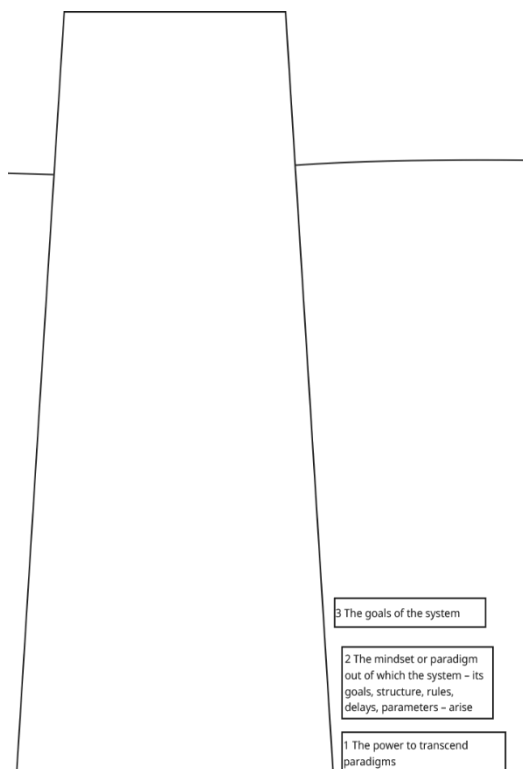
5. The rules of the system (such as incentives, punishments, constraints) - The rules of the system – constitutions, treaties, trade agreements – structure how systems operate. In a system grounded in an accessible, stable, and enforceable legal process the design of rules is a relatively high leverage point.

Changing the rules of the game has the power to significantly restructure how a system operates and the behaviours of individuals within it. Conversely structuring the rules such that they do not prioritise social and environmental interests can have significant negative impacts. As noted in D2.2, [Ankersmit and Partiti \(2020\)](#) demonstrate there are significant shortcomings in the EU's approach to promoting the sustainable extraction of raw materials as part of Free Trade Agreements (FTAs). Shortcomings include the exclusion of interests other than that of EU industry in dedicated raw materials chapters; the almost exclusive focus on trade liberalization commitments over trade regulation in these chapters; and the separation of economic interests from social and environmental interests in raw materials extraction by moving the latter to more general, secondary and weaker sustainable development chapters.

4. The power to add, change, evolve, or self-organise system structure - Self-organisation means changing the goals or mindset of a system by adding new physical infrastructures or making new rules.

An example might be the increasingly influential role social media has played in politics since the mid-2010s.

Figure 9 – Leverage points at the level of mental models



The leverage points (3-1) at the **level of mental models** are very effective at enabling the transformation of system functioning.

3.The goals of the system - The goal of a system is a high leverage point and identifying whole-system goals is an important exercise.

As outlined above limitless economic growth is arguably the current, dominant, whole-system goal although it is subject to increasing challenges from economists, environmentalists and politicians.

The economic program initiated by Ronald Reagan and Margaret Thatcher is an example of a process which was designed to significantly shift in the goals of the system. Meadows argues that there is high leverage in ‘articulating, meaning, repeating, standing up for, insisting, for better or for worse, upon new system goals’ (1999: 17)

2. The mindset or paradigm out of which the system – its goals, structure, rules, delays, parameters – arise - Every society holds a set of beliefs about how the world works. Our beliefs generate shared social agreements, system goals, information flows, feedbacks, stocks and flows. Although the

ubiquity, power, and permanence of these beliefs seem inescapable, they can be changed.

Changing a societal paradigm is very difficult but it is also a highly effective means of changing whole-system goals and operations. Achieving a change in paradigm entails highlighting the anomalies and failures of the old paradigm, demonstrating the value of a shift in paradigm, working with active change agents and speaking to ‘the vast middle ground of people who are open-minded’ ([Meadows, 1999: 18](#)).

1. The power to transcend paradigms - For Meadows the most effective leverage point in changing how systems operate is to fully understand that no paradigm is ‘true’.

Each paradigm or set of beliefs provides a very limited understanding of how things work. That said, to achieve system goals some set of beliefs is necessary. Policy makers can use a paradigm which best serves the purpose of enabling a just transition whilst understanding the limits of the chosen paradigm and remaining open to alternative paradigms.

3.1.5 Stages of Systems Redesign

The OECD ([2021](#); [2022](#)) propose three stages of systems redesign.

The first stage – Envision – entails shifting our mindset from means (e.g. earn more money; achieve higher GDP) to ends (e.g. live comfortably in safe, sustainable, and inclusive communities; enable societal wellbeing including climate stability). This first shift enables us to envision health and wellbeing through low-demand systems rather than automatically equating a high energy-use and high materials-use society as a prerequisite for human life quality.

As outlined in chapter 2 this assumption fundamentally misses the extraordinarily high social and environmental costs of high demand societies.

Envisioning can entail asking two questions. First what does human health and wellbeing require? Second, how can the conditions for health and wellbeing be created sustainably? A sample outline is provided below:

Table 9 – Envisioning Health Wellbeing through Systems Redesign in Transport, Energy, and Housing

	Transport	Energy	Housing
Wellbeing requirements	Access to services and opportunities (e.g. consumption, work, health services, leisure) with ease – ‘including by not have to travel long distances, or not having to travel at all’ (OECD, 2021 : 20)	Access to energy to meet daily needs with ease	Affordable, accessible, adequate (including climate resilient) and secure housing
Sustainable conditions for wellbeing	Clean air, clean water, global fairness and equity. Safe and convenient active and/or shared modes including public transport. Balance between proximity and mobility.	Clean air, clean water, global fairness and equity. Access to renewable energy sources	Clean air, clean water, global fairness and equity. Rights based access to affordable and accessible housing for all Security of tenure

The second stage – Understand & Determine - entails a mindset shift from analysing and aiming to fix ‘parts’ of a system - to analysis of systems functioning.

This shift allows us to understand the system dynamics driving unsustainable outcomes. Policy makers can then focus on reversing unsustainable dynamics through redesigning the system. Essentially, we need to identify:

- i) key dynamics in the system’s structure that are leading to unsustainable results (i.e. the vicious cycles);
- ii) relevant actors in the system; and
- iii) barriers to systemic change (e.g. mindsets, policies in place, governance, budget allocation, monitoring frameworks, power dynamics) ([OECD, 2021: 19](#)).

A policy focus on immediate and urgent issues has led to decades of policies that only react to events or anticipate patterns. Although reactive policies can minimise some harms they cannot address root causes. Looking at historical data, policy makers can often predict future trends. Anticipatory policies aim to reduce the harms of those trends, but again, they neither identify nor address root causes. In contrast transformative policies seek to enable a shift in the ‘rules of the game’ and our mental models, such that patterns of behaviour that produce the envisioned results (wellbeing and climate stability) are encouraged. Problems are ‘dissolved’ by dealing with the root causes.

To analyse the potential impact of a policy it is necessary to consider its intent and its place within the system. For example, although the intent of a policy may be transformative - if due regard is not given to the existing infrastructure - it may have very limited impact. As the OECD note, to ‘identify a policy’s transformative potential—that is to say, the actual effect a policy can have on current systems—understanding the system’s feedback loop structure is fundamental’ ([2022: 54](#)).

To gain an understanding of system functioning we must map

- the ‘interconnections or causal relationships that produce the results at the tip of the iceberg’ (OECD, 2022: 54) via causal loop diagrams (tool 2); and
- the physical and mental lock-in within the system i.e. existing standing stock, in-flows, and out-flows (tool 3)

Once we understand the relative magnitudes of existing stocks - and we understand the system’s unsustainable dynamics – we can assess a policies’ transformative potential in two ways:

- (i) Potential mismatches between the intent and ambition of the policy and the magnitude of unsuitable stock can be identified. For example, in a country where the ‘lock-in’ of car-purposed infrastructure is established - additional investment in public transport will have minimal transformative impact.
- (ii) Stock and flow analysis reminds us that we do not only need to ‘add’ additional elements – we can also take away or reallocate unsuitable infrastructure. Too often policy decisions are about the changing levels of investment in various parts of the system, rather than on actively reducing undesired stock

Having identified the intent of existing policies it is necessary to understand how ‘deep’ the basin of attraction is. This illustrates the level of embeddedness of unsustainable system dynamics due to existing physical (infrastructure) and non-physical (mental models) stock.

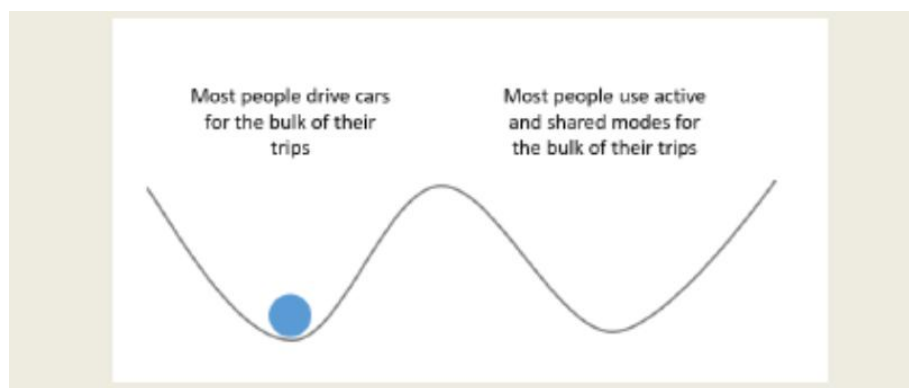


Figure 10 - Sample Basin of Attraction (OECD, 2022: 52)

For example, the existing situation where most people drive cars for the bulk of their trips is a result of accumulated past policies:

- Investment in car-purposed road infrastructure
- Lack of investment in public and active transport modes infrastructure
- Car-centric communication efforts

Reactive and anticipatory policies do not move the blue ball in the figure above from the current situation toward the situation we require if we are to decrease emissions and address social inequalities. With reactive and anticipatory policies, cars remain highly attractive, GHG emissions are not significantly reduced, and enforced car ownership is not addressed.

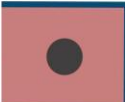
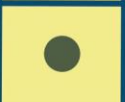
Transformative policies with low to medium transformative potential may slow down the growth of a car-dependent system but they do not achieve the required outcomes to move the blue ball from the left to the right. For example, carbon prices push the ball towards the right by increasing the cost of driving, but other factors such as safety, access, and time savings mean that cars remain highly attractive. As such, GHG

emissions do not decrease significantly, and, as low-income earners pay disproportionately high costs for policies such as carbon pricing, inequalities increase.

Basins of attraction illustrate several characteristics of complex systems:

1. Complex systems can have more than one basin of attraction and different patterns of behaviour can emerge in line with policy design
2. In non-linear systems change can be rapid once the **conditions for action** are transformed
3. The deeper the basin of attraction for unsustainable system dynamics, the greater the need to prioritise transformative policies with high transformative potential i.e. policies with low to medium transformative potential will not adequately restructure the rules of the game and trigger the behaviour change required to address climate breakdown and social inequalities ([OECD, 2022: 53](#)).

Having understood current lock-in and the basins of attraction we can then classify each policy according to intent (reactive, anticipatory or transformative) and then to transformative potential (low, medium, high) given the physical lock-in of current systems.

		Transformative potential		
		Low	Medium	High
Policy intent	Reactive			
	Anticipatory			
	Transformative			

The third stage – Redesign- entails policy design that prioritises system goals (i.e. societal wellbeing including climate stability) and creates the conditions for the achievement of those goals, taking due regard of system lock-in. Policies with high transformative potential, when implemented at scale, have the potential to flip the ball from the left to the right such that GHG emissions reduce and inequalities are addressed.

Figure 11 - The transformative potential of reactive, anticipatory and transformative policies ([OECD, 2022: 50](#))

3.1.6 Strengths and Limitations of Systems Inquiry

Systems inquiry maps out a given system – it is primarily concerned with getting a view of the overall workings of the system and the major relevant factors to be considered before conducting a more detailed analysis

Systems inquiry is a response to the now widespread recognition that a just transition requires systems change.

Strengths of Systems Inquiry

- An understanding of how the system is, how it works, the underlying structure rather than just the observable outcomes.
- An understanding of why it is important not to attempt quick fixes that are unsustainable and quickly out of date once the context changes
- A holistic vision which prevents a narrow focus on any one part of the system or any one solution.
- The ability to track the impacts of a policy intervention across the system – providing a much better understanding of what is and is not working
- A systems map which can be continuously updated as the context changes

- By creating system maps collectively, we can start to come to some shared understanding; some shared reference point. This enables better communication – people can see how their actions impact on the whole system

Limitations of Systems Inquiry

- Systems inquiry does not provide a detailed analysis of the associated risks or impact of specific policy interventions

3.2 Risk Assessment

Risk assessment is widely used in public policy making. It encompasses the “assessment and management of potential risks that could adversely impact society, particularly in the context of governance and decision-making (Flynn, 2021). Risks are generally understood as the potential negative effects of an option (Hollenstein, 2021), including policy options. In relation to the green transition specifically, although multiple benefits have been identified, some risks remain, particularly in terms of rising energy prices or the reconfiguration of labour markets (Vrontisi et al., 2024). As policies rarely deal with certainties, risk analysis can be a valuable tool for reducing the likelihood of harmful or hazardous outcomes. Risk analysis is often used in areas such as public health hazards, workplace accidents, failure of financial institutions or terrorist attacks (OIA, 2023). Environmental policy making has adopted quantitative risk assessment for more than 30 years, with an increasing attention to the human health damage from toxic chemicals (Russell and Gruber, 1987).

Risk assessment has been defined as “a reductive analytical tool used to evaluate the risk (probability, consequence, perceived importance etc.) of a specific event” (Evans et al., 2006). More specifically, risk has also been conceptualized as “the product of the probability of an adverse event (extensive margin) and the severity of persistence of its impact (intensive margin) (Kaplan and Garrick, 1981 in Hekkelman et al., 2025). The definition of risk in this context consists of two elements: the consequence of an event and its seriousness, and the probability that these consequences will occur (Boholm et al., 2015).

3.2.1 Risk Assessment Tools and Techniques

Techniques for assessing societal risks typically involve evaluating the likelihood and severity of risks, as well as identifying effective mitigation strategies. Key methodologies for risk analysis include brainstorming, sensitivity analysis, Delphi surveys, and decision tree analysis - all of which contribute to a comprehensive understating of risk, particularly in domains such as health, environmental issues and safety (Flynn, 2021).

In relation to climate action policies, Capuder et al. (2020) propose a hybrid approach, with PESTLE (Political, Economic, Social, Technological, Legislative and Environmental Aspects), SWOT (Strengths, Weaknesses, Opportunities and Threats), and Integrated Risk Management (IRM) used for a detailed analysis of the drivers for electrification of transport. Their approach includes the following steps, as illustrated by the diagram below:

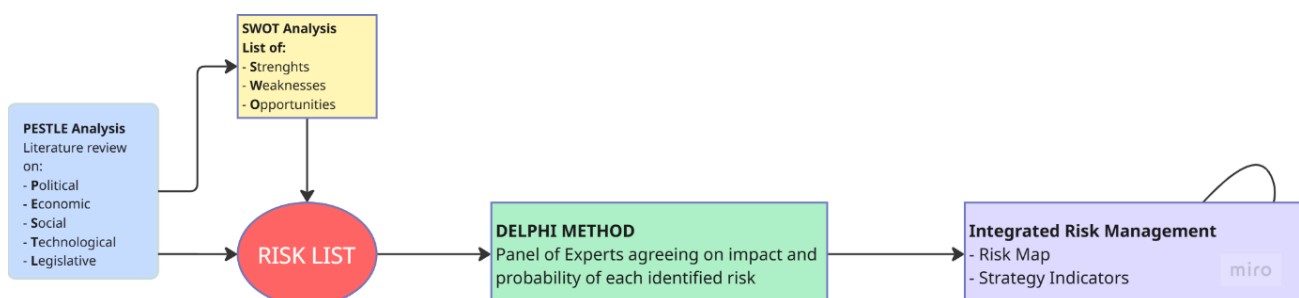


Figure 12 – Integrated Risk Management (IRM)

This model explores the factors behind the uptake of electric vehicles with various social aspects accounted for.

However, the model has serious limitations when applied to the evaluation of transformative policies. This is particularly an issue when accounting for structural inequalities and analysing the differential impacts of the proposed policy on specific groups, especially through the lens of double advantage/double disadvantage - a concept that highlights how certain groups may disproportionately benefit from or be burdened by policy changes. In other words, the method frames risks as obstacles to the uptake of the EVs (such as cost, infrastructure, consumer hesitancy etc) rather than as potential catalysts for deepening existing social disparities or creating new ones. In other words, it overlooks how these risks might interact with systemic inequities to reinforce exclusion or privilege.

To address these shortcomings, [Mercure et al. \(2021\)](#) proposed a risk-opportunity approach (ROA) as theoretically suited with the demands of structurally transformative change. Using the example of EV adoption, the authors argue that policy effectiveness is highly dependent on the market and regional context. Their ROA model draws inspiration from the UK offshore wind success story: iterative and adaptive policy making. This means that policy decisions are revisited and refined over time. At each step of the way, uncertainty is reduced by new market information, which allows to re-evaluate options on the basis of ever-better informed projections using non-linear vehicle adoption models - models that account for complex feedback loops and tipping points in consumer behaviour and market evolution ([Mercure et al., 2021](#)).

In the EU, risk assessment is carried out across the Commission and various EU agencies, spanning a wide range of policy areas, particularly those related to natural disasters, climate change, security, human/animal/plant health, environment, IT systems functioning, financial markets, energy supply and air traffic. Public health-related risk assessments are among the most prominent as they relate to exposure to chemical substances on the one hand, and biological hazards (including infectious diseases) on the other hand.

The EU Impact Assessment toolbox identifies risk assessment as consisting of three elements: hazard, exposure and vulnerability, with risk conceptualised as the product of these factors ($\text{Risk} = \text{Hazard} \times \text{exposure} \times \text{vulnerability}$). This formulation highlights that risk represents both the severity of potential hazards and the likelihood of its occurrence. Tool #14 within this toolbox offers a step-by-step guide to risk analysis. It outlines key concepts of risk assessment and provides guidelines on how risk assessment can contribute to the Commission's impact assessment process.

The tool #14 recommends three steps to be taken for the Risk Analysis:

1. Identify and characterize the hazards and the inherent properties of the agent or phenomenon in terms of potential negative effects (on population or environment), establish the casual relationship between the hazard and its effects, describe the negative effect and determine its severity (...) and dose-effect relationship
2. Assess the likelihood of occurrence of the hazard
3. Characterize the risk: based on the results from the previous steps, determine quantitatively, and if not possible, qualitatively, the level of risk under given assumptions and uncertainties.

Risk is thus defined as the chance, whether high or low, that a hazard will actually cause somebody or something harm and the probability that a person or something will be harmed or experience an adverse effect if exposed to a hazard. Therefore, the risk is a function of both the hazard and the potential likelihood/exposure to hazard. It can be identified through a wide range of evidence, including previous studies, expert opinions, data analysis, with specific models and experimental data applied to chemical risks, while climate change risk identification involves scientific models and climate scenarios (customized to sectoral and geographical context). Risk assessment, in this context is a process or method to identify a hazard that has the potential to cause harm and to analyse risk associated with that hazard (assessing what

is the likelihood of exposure to hazard and what are the likely impacts of exposure if hazard happens). Risk assessment feeds into risk management.

An advantage of risk assessment is its applicability to a wide variety of settings without the need to change methodology as well as a high degree of transparency ([Hollenstein, 2021](#)). Risk analysis can also support comparative assessments, enabling policy makers to prioritize risks and allocate resources accordingly ([Flynn, 2021](#)).

3.2.2 Strengths and Limitations of Risk Assessment

While Risk Assessment (RA) and Risk Analysis offer valuable frameworks for evaluating potential hazards and informing policy decisions, they also present several limitations, particularly in the context of complex, transformative policy challenges. One key weakness lies in the ambiguous nature of policy goals. These goals are often broadly defined, requiring further interpretation before they can be meaningfully assessed through RA methodologies ([Evans et al., 2006](#)). This ambiguity complicates the process of translating policy questions into scientifically testable hypotheses, thereby limiting the analytical precision of RA tools.

Additionally, RA struggles to accommodate the transformative nature of certain policy interventions. Policy interventions aimed at systemic change (such as decarbonisation, social equity, or technological transitions) often involve deep structural shifts that RA is not well-equipped to model. The diversity of stakeholder interests further complicates this landscape, introducing conflicting priorities and values that are difficult to reconcile within a risk-based framework ([Geels et al., 2016](#)).

Moreover, transformative policy design is inherently fraught with uncertainty. Studies highlight the unpredictable nature of costs, benefits, and path-dependent outcomes, which challenge the assumptions of linearity and predictability that underpin many RA models ([Hughes et al., 2013](#)). These uncertainties make it difficult to generate robust forecasts or to evaluate long-term trade-offs.

Importantly, RA tools are typically designed to assess discrete events (e.g., financial crises) or specific behaviours (e.g., adoption of electric vehicles). As such, they are ill-suited for examining more complex, systemic issues, particularly those related to the reproduction of intersectional inequalities. RA tends to focus on **proximate risks** rather than the deeper structural causes of advantage and disadvantage, such as institutional bias, socio-economic stratification, or historical marginalisation.

In summary, while RA can be a useful tool for short-term, event-driven analysis, its limitations become apparent when applied to long-term, transformative policy contexts.

Strengths of Risk Assessment

- **Identifies immediate social impacts:** Enables policy makers to recognise the short-term negative consequences for individuals or groups who are unable to access the benefits of a given policy intervention
- **Evaluates the risk of a *specific event*:** Assesses the risk of a defined event by considering two key dimensions - its consequences and seriousness (intensity and extent), and the likelihood of occurrence ([Evans et al., 2006](#). [Boholm et al., 2015](#))
- **Supports transparent short-term assessments:** Provides a clear and structured approach to evaluating short-term negative consequences of events with a high degree of transparency in decision-making ([Hollenstein, 2021](#)). However, this strength may not be applicable to more complex social problems
- **Enables prioritisation and resource allocation:** Assist policy makers in ranking risks and allocating resources toward the most pressing or probable threats ([Flynn, 2021](#))
- **Integrates with Impact Analysis:** Can be embedded within broader impact analysis frameworks to illuminate sources of uncertainty regarding potential policy outcomes. It is particularly useful during early stages such as problem definition and impact consideration ([OIA, 2023](#)).

Limitations of Risk Assessment

- **Inability to assess systemic policy effectiveness:** Risk assessment does not evaluate whether a policy intervention is effective within the broader context of systemic transformation, such as the twin transition. It fails to identify the root causes of inequalities or determine whether a policy disrupts entrenched systems to enable progress toward net-zero goals.
- **Unsuitability for long-term, complex transitions:** Typically designed for discrete events, risk assessment struggles to address extended, multi-dimensional processes like the twin transition. These involve evolving dynamics and systemic change that require more adaptive and holistic evaluation frameworks.
- **Limited scope for addressing non-disaster social challenges:** Risk assessment is not equipped to tackle complex social issues that are not tied to specific crises, such as public health disparities, environmental justice, or financial system reform. These challenges require broader, intersectional approaches.
- **Difficulty translating abstract policy questions into testable hypotheses:** Many policy questions, especially those involving ethics, values, or systemic change, cannot be easily reduced to scientifically testable formats. This limits the utility of risk assessment in evaluating transformative policy options (Evans et al., 2006).
- **Challenges in quantifying risk magnitude:** Although risk assessment can provide estimates, the inherent uncertainty in risk quantification makes precise measurement difficult. Even when risks appear numerically defined, significant ambiguity remains (OIA, 2023).

3.3. Impact Assessment

Impact Assessment (IA) has become a cornerstone of policymaking across the European Union. All major EU policy initiatives are now subject to IA - a structured, evidence-based process that enables policymakers to evaluate the potential consequences of proposed measures. IA considers the likely effects on individuals, society, the economy, and the environment, and allows for adjustments to mitigate adverse outcomes. It is applied across all levels of EU policymaking and is central to the European Commission's Better Regulation Agenda ([European Commission, 2021](#)).

Formally introduced in 2002, the Commission's IA system adopts an integrated approach, assessing environmental, social, and economic impacts. This reflects broader governance trends, particularly the influence of New Public Management (NPM), which gained prominence in Europe and North America during the 1990s. NPM emphasised transparency, accountability, and efficiency in public administration ([Sauer, 2018](#)). IA embodies these principles by promoting evidence-based policymaking and enhancing regulatory quality. The rise of IA is also closely linked to the increasing complexity of governance systems ([Verschuuren, 2009](#)).

Typically, the IA process follows a sequence of steps similar to those used in Environmental Impact Assessment (EIA): identifying the problem, setting objectives, developing policy options, and analysing their potential impacts ([Adelle & Weiland, 2012](#)). The process is guided by scientific methods and aims to establish "evidence-based criteria for policy advice in the larger endeavour of ex-ante policy analysis," with the terms "analysis" and "assessment" often used interchangeably (Sauer, 2018). In the EU context, ex-ante IA is defined as "an attempt to provide, in advance of legislating, a coherent analysis of the reasonings that lie behind, and foreseeable effects of any proposed measures or policy initiative" ([European Parliament, 2015: 1](#)).

Since its formal introduction, between 25 and 135 IAs are conducted annually by the European Commission, each reviewed by the Regulatory Scrutiny Board, an independent body that ensures the quality and robustness of the analysis (European Parliament, 2015; European Commission, 2021). Within

this framework, various forms of impact assessment are deployed to address specific dimensions of policy. Social Impact Assessment (SIA), for example, focuses on the consequences of policies for individuals, organisations, and social systems. It is defined as “the process of identifying the future consequences of a current or proposed action which are related to individuals, organisations or social macro-systems” ([Becker, 2001](#)). Environmental Impact Assessment (EIA) and economic analyses are also routinely used, often combining quantitative and qualitative methods to ensure comprehensive evaluation.

A particularly important tool within the EU’s IA landscape is Gender Impact Assessment (GIA), which supports gender mainstreaming across policy domains. The European Commission introduced its first GIA framework in 1997/98, recognising that policies perceived as gender-neutral may still produce unequal outcomes. As the original guidance stated: “Policy decisions that appear gender neutral may have a differential impact on women and men, even when such an effect was neither intended nor envisaged”. ([European Commission, DG Employment, Equal Opportunities Unit, 1997/1998, p.2](#))

Although the tool was initially limited to employment policies and not mandatory, it marked a significant step toward institutionalising gender-sensitive analysis within the Commission. The European Institute for Gender Equality (EIGE) later expanded this approach, offering detailed guidelines and methodologies for conducting GIA. EIGE defines GIA as an ex-ante evaluation that anticipates whether a law, policy, or programme may reduce, maintain, or increase gender inequalities. It recommends a dual-pronged method: assessing both the current gender dynamics in a policy area and the projected impacts post-implementation. The process must be structured, analytical, and documented, with the aim of improving policy design to prevent unintended consequences and promote transformative change ([EIGE, 2017: 8](#)).

GIA is especially relevant in the context of the EU’s twin transition, where structural inequalities risk being reproduced if not proactively addressed. By embedding gender analysis early in the policy cycle, GIA strengthens governance, enhances institutional learning, and contributes to more inclusive and effective policymaking.

At the national level, Member States apply their own approaches to impact assessment, often incorporating gender and social dimensions. In Ireland, for instance, the Department of Social Protection conducts evaluations through its Social Inclusion Division, which promotes tools such as Poverty Impact Assessment (PIA) and SIA. A notable example is the use of the SWITCH model, developed by the Economic and Social Research Institute (ESRI), to assess the distributional effects of budgetary measures. This microsimulation model, based on EU-SILC data, evaluates welfare and direct tax impacts, though it excludes indirect taxes and social expenditure ([Department of Social Protection, 2024](#)).

Gender Impact Assessment is also used by national governments to support gender mainstreaming. The first formal GIA model was developed in the Netherlands in 1994 by Mieke Verloo and Conny Roggeband, inspired by EIA methodologies and informed by Giddens’ structuration theory ([Verloo, 2001](#); [Woodward, 2004](#)). These early efforts laid the groundwork for broader adoption across Europe, with national administrations increasingly integrating GIA into their policy evaluation frameworks.

Together, these EU and national-level practices reflect a growing institutional commitment to embedding impact assessment into the policymaking process. While IA is widely promoted as a tool for improving regulatory quality and fostering more inclusive, evidence-based decisions, its effectiveness - particularly in addressing structural inequalities and supporting the twin transition - remains subject to important limitations. These include challenges in assessing transformative potential, navigating complex multi-level issues, evaluating long-term impacts, ensuring objectivity, and overcoming data gaps.

3.3.1 Impact Assessment Tools and Techniques

At the EU level, IA is guided by the Better Regulation Toolbox ([EC, 2023](#)), which defines IA as “a process comprising a structured analysis of policy problems and corresponding policy responses”. It develops objectives and alternative policy options and assesses their impact. It also considers subsidiarity, proportionality of options, and how to monitor and evaluate the policy in the future. It helps to develop the Commission’s policy response to a certain policy problem by providing the evidence base for, and the

impacts of, various options. According to the Toolbox, an ex-ante evaluation should include the following elements:

- Problem analysis and needs assessment, including alignment with Sustainable Development Goals (Tool #8)
- EU added value
- Policy management objectives
- Policy options and associated risks
- Expected results and impacts
- Monitoring and evaluation mechanisms

To address structural inequalities, particularly those related to gender, Tool #6 of the Toolbox, ‘Evaluating gender equality as a horizontal principle’, provides a two-stage approach: (1) assess the impacts of the intervention on gender equality; (2) evaluate the extent to which gender has been mainstreamed in the design, implementation and monitoring of the intervention.

The tool emphasises that policy impacts are never neutral and must be assessed for differential effects on women and men in all their diversity. It provides guiding questions across six thematic areas where gender and intersecting inequalities are most evident:

While the Gender Impact Assessment (GIA) tool is primarily focused on gender equality, its methodological steps can be adapted to evaluate other forms of inequality. In the context of climate action policies, the European Institute for Gender Equality (EIGE) has developed the [GREENA Toolkit](#), which outlines key considerations for assessing such policies through a gender lens. The toolkit adopts an ecofeminist and transformative approach, aiming to address both the systemic drivers of gender inequality and the structural causes of climate and environmental degradation.

It emphasises the interconnectedness of the climate crisis and structural gender inequalities, highlighting how both are rooted in systems of oppression that disproportionately affect structurally disadvantaged groups of women and men through intersectional inequalities ([EIGE, 2024: 2](#)). This transformative framework seeks to identify the underlying drivers of inequality and guide policy evaluations that advance gender equality, environmental sustainability, and a care-based economy—ultimately centering a gender-just future and a healthy planet.

To achieve these goals, the toolkit outlines essential elements for conducting a gender-responsive evaluation of policies aimed at delivering a “sustainable future for all”:

- **Responsiveness to gender equality and environmental sustainability:** Evaluations should aim for transformation by understanding gender not merely as unequal access to resources, but as shaped by deeply embedded social norms and power structures that perpetuate both gender and environmental injustices.
- **Gender analysis and intersectional approach:** Critical for identifying differential impacts between and among women and men, while accounting for their relative positions in society and the distribution of resources, opportunities, constraints, and power. Intersectionality helps uncover how gender interacts with other identities to produce unique experiences of discrimination.
- **Sex-disaggregated data and gender statistics:** Essential for generating evidence-based insights into gendered impacts—including environmental outcomes—and for increasing the visibility of gender inequalities.
- **Participatory approach:** Evaluations must incorporate the perspectives of those affected, using culturally responsive methods that reflect diverse values and beliefs. Marginalised groups, such as migrant women and men, should be proactively included to ensure their voices are heard.
- **Comprehensive evaluation design:** This includes understanding the complexity of social change, applying mixed-methods tailored to context, ensuring gender expertise and balanced evaluation teams, and using gender-sensitive language throughout.

In parallel, the Initiative for Climate Action Transparency (ICAT) provides a structured methodology for assessing the impact of renewable energy policies. ICAT's guidelines offer step-by-step instructions for evaluating different impact areas, with a focus on greenhouse gas (GHG) emissions and transformational change.

The core assessment principles are included in the table below.

Table 10 - ICAT Gender Impact Assessment Framework: Core Questions and Relevance to Twin Transition Policies

Thematic area	Specific Questions	Relevance to twin transition policies
Participation in the labour market and economy	How is control over assets distributed among women and men in the intervention's policy area? Did the intervention consider differential access to, and control of, resources for women and men and adopt gender-specific objectives or strategies? Was gender segregation in the labour market or education addressed? Did the intervention perpetuate existing gender gaps?	Persistent gender income gaps increase women's risk of poverty, including energy poverty. Green transition policies—such as those promoting energy-efficient housing—must consider disparities in homeownership, access to green finance, and relevant skills.
Participation in public life and decision making	What is the representation of women and men in decision making roles and positions of power within the sector? How did the intervention affect representation? What barriers were addressed, and how effectively? Did the intervention improve access to leadership training for women?	For example, the representation of women at UN climate change conferences has been historically low, which can significantly affect the development of equitable climate policies
Gender stereotypes, gender roles and responsibilities	What gender stereotypes and social norms exist in the intervention context? Did the intervention reinforce or challenge these norms? Were unintended consequences identified? Were stakeholders consulted to detect gender bias? Was staff trained to recognise gender norms?	Men and women often share similar concerns about climate change, but their engagement with climate action can be shaped by social norms, roles, and access to resources. Differences in behaviour—such as energy use or prioritization of economic versus environmental goals—are influenced by societal expectations rather than inherent traits. Recognizing these gender-based norms and stereotypes is essential for designing inclusive and effective climate policies.
Care, including paid and unpaid care work	Did the intervention make any assumptions about traditional care roles? Did it promote equitable sharing of responsibilities and time use?	Unequal care responsibilities (both paid and unpaid) disproportionately affect women and limit their ability to participate in green and digital transition opportunities. Policies that

	Were barriers due to care commitments considered in the design and implementation of the intervention?	promote flexible energy use or digital services must consider how time poverty and household labour divisions impact access and engagement. Without addressing these dynamics, twin transition policies risk reinforcing gender inequalities instead of promoting inclusive and sustainable change.
Gender-based violence	Did the intervention address gender-based violence concerns? Were specific objectives adopted to tackle it? Did the intervention exacerbate or perpetuate risks? Were safety and harassment issues considered? Was support provided to victims of gender-based violence? Did the intervention challenged misconceptions regarding gender-based violence among its participants/beneficiaries?	Safety concerns -such as those related to public transport, for example- often influence women's travel choices. These are frequently overlooked in transport planning, and EU-level data on women's safety remains limited.
Institutions, law, policies and budgets	Did the intervention align with laws, policies, regulations, strategies and institutional practices that promote gender equality? Were existing challenges identified and addressed? How did the intervention respond to institutional barriers?	Institutional frameworks are key to ensuring that green and digital policies benefit everyone equally. Despite the transport sector's central role in the European Green Deal, gender segregation in employment persists. The EU Gender Equality Strategy 2020–2025 calls for gender mainstreaming across all policy areas, including those driving the twin transition. To avoid reinforcing existing inequalities, interventions must align with gender equality laws, apply inclusive governance, and allocate resources through gender-responsive budgeting.

ICAT also recommends the following steps to be followed while conducting the renewable energy impact analysis:

Phase 1: Describe the policy to be assessed -> Decide whether to assess an individual policy or a package of policies -> choose ex-ante or ex-post assessment

Phase 2: Identifying impacts: Identify GHG impacts (changes in emissions) -> define the GHG assessment boundary -> define assessment period -> identify sustainable development and transformational change impacts.

In this context, transformational change is defined as “fundamental, sustained systemic change that disrupts established high-GHG emissions development pathways and contributes to zero-carbon development, in line with the goals of the Paris Agreement and the SDGs” ([ICAT, 2020](#)).

3.3.2 Strengths and Limitations of Impact Assessment

In the context of the FITTER-EU analysis, it is essential to understand both the strengths and limitations of IA. The following points outline what IA can and cannot do, particularly in relation to complex policy challenges such as the twin transition.

Strengths of Impact Assessment

- **Structured and systematic process:** IA provides a formal framework for evaluating the implications of policy options on individuals, communities, and the environment. It allows for adjustments to mitigate potential negative economic, social, or environmental consequences.
- **Methodological rigour:** IA enables policymakers to assess the potential impact of a chosen policy option using both quantitative and qualitative methods in a structured, analytical way.
- **Evidence-based policymaking:** IA incorporates scientific and technical tools to establish “evidence-based criteria for policy advice in the larger endeavor of ex-ante policy analysis” (Sauer, 2018).
- **Support for transformative policy design:** In specific cases, such as Gender Impact Assessment (GIA), IA can contribute to more inclusive and transformative legislation, helping to strengthen gender equality through improved policy design (EIGE).
- **Governance principles:** IA considers subsidiarity and proportionality, helping to determine whether policy decisions are appropriate at a given level of governance and how they should be monitored and evaluated over time.
- **Anticipation of unintended consequences:** IA can help identify and mitigate potential negative side effects of policy interventions before they are implemented.

Limitations of Impact Assessment

- **Limited capacity to assess systemic transformation:** IA can indicate whether a policy is effective in relation to the twin transition, but it cannot determine whether the intervention is substantively transformative—i.e., whether it restructures systems to address deep-rooted inequalities and achieve net-zero goals.
- **Challenges with complex, multi-level problems:** Existing guides and toolkits are not fully equipped to address “wicked” problems – i.e., those that are deeply interconnected, span multiple governance levels, and involve structural inequalities affecting disadvantaged groups.
- **Difficulty in evaluating long-term impacts:** IA often struggles to account for the long-term effects of policy options within a dynamic and evolving policy landscape. For example, assessing how a shift to renewable energy interacts with housing market trends requires broader systemic analysis.
- **Subjectivity and contextual bias:** IA is not a neutral or purely objective tool. Practitioners conduct assessments within specific political and institutional contexts, shaped by their own values and assumptions. Unlike cost-benefit analysis, IA does not offer a standardised metric for comparison.
- **Data limitations:** IA may be constrained by the availability and quality of data—particularly disaggregated data needed to understand intersectional inequalities. This can limit the depth and accuracy of the analysis.

For example, the Dublin City Council Climate Action “[Climate Readiness Toolkit](#)” (DCC, 2024) offers a practical IA tool tailored to climate policy. While highly useful for reactive and anticipatory planning, it does not enable practitioners to enact transformative change or address the root causes of inequality. It should therefore be seen as a complementary instrument rather than a comprehensive solution.

3.4 Methodological rationale

Risk assessment, when integrated with systems inquiry, becomes a powerful tool for uncovering the deeper structural dynamics (such as feedback loops, institutional inertia, and root causes) that shape both environmental and social outcomes of twin transition policies. This methodological integration enables policymakers to move beyond narrow evaluations of impact and toward identifying strategic intervention points that can disrupt unsustainable patterns while addressing intersectional inequalities.

The rationale for combining these approaches stems from a recognition that indicators alone are insufficient for guiding transformative policy. While indicators offer measurable insights into policy performance and progress, they often fail to illuminate the systemic constraints, power asymmetries, and leverage points that determine whether a policy will lead to meaningful change or else reinforce the status quo.

By mapping system dynamics (i.e., causal relationships, reinforcing cycles, and structural bottlenecks) and assessing risks to marginalised or vulnerable groups, the FITTER approach seeks to support policy design toward interventions with high transformative potential. This means prioritising actions that not only mitigate immediate harms but also reconfigure the underlying systems that produce inequality and ecological degradation in the first place.

In sum, rather than relying solely on conventional, indicator-based policy appraisal, this methodology integrates systems inquiry to uncover structural dynamics and guide more transformative, equity-focused policy design. This calls for a shift from reactive, incremental adjustments to ambitious, justice-oriented transformation, even when such a shift demands greater analytical complexity. In doing so, FITTER promotes a transition strategy that is not only ecologically sustainable but also socially equitable, ensuring that the benefits of change are shared and the burdens are not disproportionately borne by those already disadvantaged.

4. Unsustainable Dynamics – Case Study Sectors

Due to time and resource constraints, it has not been possible for the research team to conduct a full literature review of unsustainable system dynamics and relevant sectoral policies across the three sectors. Each sector is complex and the list of policies relevant to each sector within the relevant case study countries is extensive.

In this chapter we provide a sample outline of a high-level systems inquiry for the transport sector as developed by the OECD (2021; 2022). We also provide a high-level review of relevant research in the housing and energy sectors. Our purpose here is to provide initial guidance only on some of the root causes of the generation of high emissions and the reproduction of inequalities in the three sectors, as identified in the literature. Further research would be required to ascertain the applicability of these root causes and unsustainable system dynamics within the case study contexts.

4.1 Transport

The Avoid-Shift-Improve (ASI) framework is well established in the literature as the pathway toward reducing transport emissions. As outlined in D2.2, studies demonstrated that ASI strategies can account for 40-60% of transport emission reductions, and at lower costs than ‘improve’ strategies only ([SLOCAT, 2021](#)). Initially developed in Germany in the 1990s, the ASI approach includes three hierarchical pillars:

- AVOID: Reduce the frequency and distance of car trips. This could be achieved through integrated urban planning (e.g. 15- minute neighbourhoods), encouraging remote work and the digitalisation of relevant services
- SHIFT: Prioritise and develop sustainable transport infrastructure e.g. public and active transport (cycling or walking) infrastructure, road space reallocation away from private vehicles
- IMPROVE: Electrification of transport prioritising high occupancy vehicles - this step should supplement the first two, rather prioritise the electrification of private vehicles ([TUMI, 2023](#))

The need for moving away from a ‘private vehicle ownership logic’ was also highlighted by some of the transport experts interviewed within the FITTER project

Interviewees argue that the current governance approach to mobility is constrained by outdated paradigms. The transition is largely being pursued within a private vehicle ownership logic, which is increasingly seen as incompatible with long-term sustainability goals. Experts stress the need for a fundamental shift towards integrated mobility systems and public transport-oriented planning, going beyond technological fixes ([Viñé and Armayones, 2025: 167](#))

The IPCC has concluded that rapid growth in demand for energy and materials significantly reduces the chances of achieving stringent mitigation targets. Such an approach (high demand, high material use) exacerbates related environmental and social challenges.

4.1.1 The Goal of Sustainable Accessibility

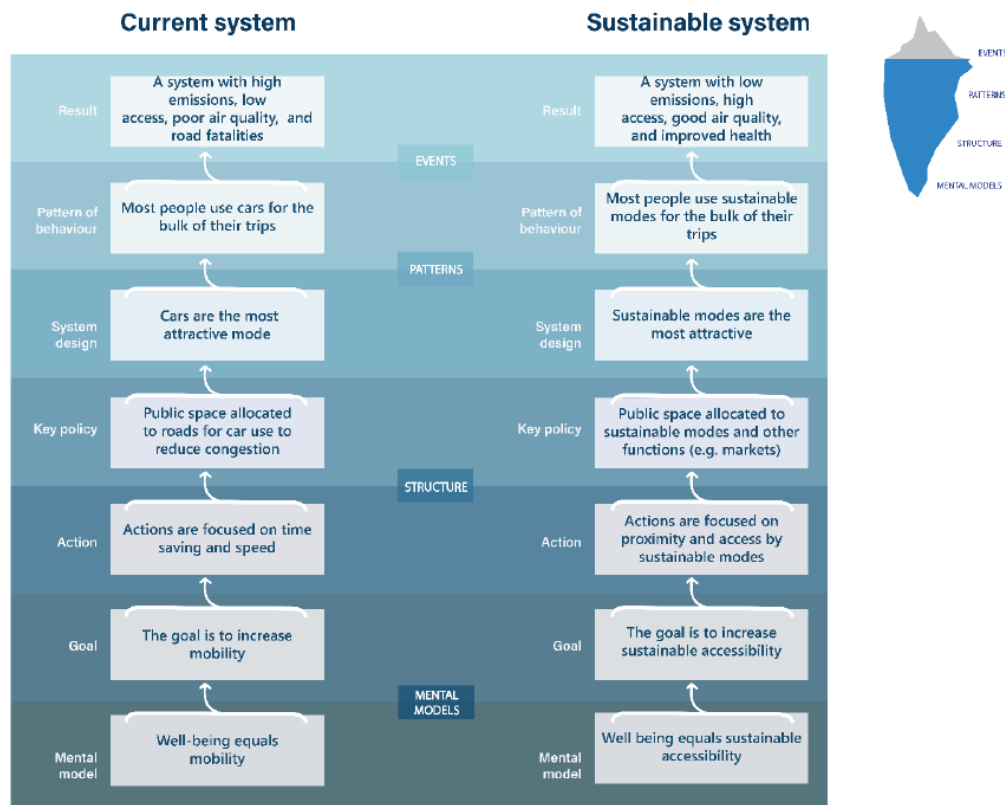


Figure 13 – Differences between current and sustainable mobility systems (OECD, 2022: 25)

The image above – developed by the OECD (2022: 25) - illustrates the differences between a car dependent system and a system which has ‘sustainable accessibility’ as its goal.

In a car dependent system wellbeing is equated with being able to travel from A to B quickly. The goal is to increase mobility (the ability to travel from A to B quickly). Policy actions are focused on enabling time saving and speed. As traffic volume grows so too does to traffic congestion. To ease congestion more public space is allocated to car-purposed infrastructure. The system thus ‘designs in’ cars as the most attractive mode of transport. Most people use cars for the majority of their trips and the result is a ‘system with high emissions, low accessibility, poor air quality and road fatalities’ (OECD, 2022: 25).

In a system which holds sustainable accessibility as its goal, policy actions are focused on creating proximity (e.g. 15-minute neighbourhoods) and access by sustainable modes (e.g. active, public and shared transport modes). Public space is allocated to sustainable transport modes and compact development is prioritised. The system thus ‘designs in’ the attractiveness of sustainable transport modes. Most people use sustainable transport modes for the bulk of their trips. The result is a ‘system with low emissions, high accessibility, good air quality and improved health’ (OECD, 2022: 25).

Before conducting a systems inquiry it is important to identify what the goal of the system currently is (and what beliefs, norms and assumptions underly that goal). Policy makers need to then envision what the goal of a sustainable, accessible, affordable and healthy system ought to be (and what beliefs, norms and assumptions would underly that goal).

The needs of populations which are more vulnerable to structural harm (e.g. people with disabilities, carers, the elderly, members of minoritised communities, rural dwellers) should be given adequate consideration and ‘accessibility for all’ should be enabled.

4.1.2 Unsustainability of high-demand, high material use transport systems

The pollution from the extraction, refining and burning of petroleum is multi-dimensional and includes water contamination, soil degradation, air pollution and habitat destruction.

Although the batteries for electric vehicles require more minerals, they can be recycled at high recovery rates, potentially reducing the need for mining over time. However, the manufacturing process for vehicle batteries is material and resource intensive and a full assessment of environmental risks and the implementation of environmental protection measures is required.

To create a sustainable and ethical electrified transportation system we need to reduce mining impacts as per the guidance from the [United Nations](#), the [Initiative for Responsible Mining Assurance](#), and the [OECD](#). Through the European Critical Raw Materials Act the European Union has taken some steps toward ‘encouraging the recovery of critical raw materials from extractive waste facilities and increasing efforts to mitigate adverse impacts with respect to labour rights, human rights and environmental protection’ ([EC, 2023](#)). (To note, an assessment of the scope and effectiveness of this Act was beyond the scope of this deliverable.) As outlined by some experts in [D3.2](#)

Environmental considerations remain under-addressed in the current energy regime. While technological efficiency has improved, its foundational principles have not changed. Interviewees express concern that the dominant focus on renewables often overlooks broader ecological impacts, particularly biodiversity. Batteries are cited as a key example, with lithium extraction carrying heavy environmental costs. Without integrating biodiversity and ecosystem concerns into mainstream planning, the transition risks reproducing extractive patterns under a green label ([Viñé and Armayones, 2025: 144](#))

Designing a low demand system; shifting as many people as possible to shared, public and active travel modes; where a private vehicle is required, using smaller vehicles that require less energy to run; and reusing and recycling batteries, can all contribute to reducing demand for material extraction and resource use, and the negative environmental impacts of that. However, in Europe, recycling capacity remains underdeveloped. A recent study found that ‘current recycling practices are not profitable: transport can account for up to 70% of total recycling costs, chemical processing infrastructure requires investments of around 23 euros per kg of input material, and many recycling plants operate at less than 10% of their capacity due to insufficient battery return volumes’ ([Cattani et al, 2025: 1](#)). As outlined by some experts in [D3.2](#)

Interviewees also note that the sector’s transformation is deeply dependent on access to critical materials and minerals, many of which are not produced in Europe. Without breakthroughs in material science or large-scale battery recycling systems, this dependency may delay the transition. The connection between energy and mobility is also underlined as central to planning the transition effectively ([Viñé and Armayones, 2025: 168](#))

Buses and other high-occupancy modes are a more efficient use of battery materials than private electric vehicles. A recent US study found that there is potential for reducing up to 66% of projected lithium demand in 2050 if US cities move rapidly toward multimodal transportation systems:

A transportation strategy that foregrounds mass transit, cycling, and walkability—combined with a circular economy approach to raw material recovery and reuse— implies an overall reduction in new lithium mining and localized vehicle pollution, improvements in transportation equity, and a hastening of urgent planetary decarbonization. By reducing lithium demand relative to the most alarming forecasts, such a pathway provides a clear alternative to prevailing transportation scenarios, which rely on fast-tracking new mines, intensifying environmental pressures on landscapes subject to mining, destroying Indigenous lands and livelihoods, or increasing pressure on globally fraught supply chains in order to power an expansive, electrified car fleet ([Riofrancos et al, 2023: 43](#))

The ‘optimistic’ scenario for transport, developed in [D3.2](#), similarly noted that a best case is where

Mobility-as-a-Service platforms allow access to shared, on-demand fleets in every major city. These platforms integrate booking and payment, as well as routing, across trains, buses, shared EVs, micromobility, and demand-responsive shuttles, replacing much of the friction once found between modes. While private cars still exist, they are fewer and cleaner, and they are integrated in a system optimised for connectivity and efficiency ([Viñé, and Armayones, 2025: 194-195](#))

Thus far, transportation strategies have focused on **decoupling emissions** while maintaining a high mobility car-dependent system as this is assumed to be linked to wellbeing and freedom. Policy has thus focused on ‘fixing’ just one part of the system (high emission vehicles). This analytical approach considers parts in isolation and looks for the solution to the problem linked to specific part(s). However studies show that focusing on improving vehicle technology only reduces emissions by about 44% in 2050 relative to 2015; but a reduction of 76% is possible in a scenario where ‘technological improvements are embedded in a wider policy package promoting the use of active modes and shared/high-occupancy vehicles, including important changes in urban planning (i.e. systems redesign)’ (Fulton et al, 2017 and Fulton 2018 cited in [OECD, 2021: 34-35](#)). Other studies have reached similar conclusions ([ITF, 2021](#)).

4.1.3 Transport – Unsustainable System Dynamics

The OECD have identified three core unsustainable dynamics in the passenger transport sector.

Induced car demand

Induced demand for road travel can be broadly defined as ‘the increment in new vehicle traffic that would not have occurred without the improvement of the network capacity’ ([WSP and RAND, 2018: 2](#)).

This is illustrated in the causal loop diagram below. Starting with the variable in the centre ‘Traffic Congestion’ – as more people drive this increases, leading to pressure to reduce congestion. For decades, this has been responded to with increased public investment in roads for cars, which in turn increases the road capacity for cars. With all else being equal, the ‘traffic congestion’ is decreased (B1) although this decrease is temporary. With the decrease in congestion the attractiveness of driving increases. The number of users of public, shared and active transport decreases and there is an increase in the number of cars in the region (B2A). As the attractiveness of driving increases, the average distance driven by car each day increases (B2B). Both lead to increased traffic volume which eventually leads to an increase in ‘traffic congestion’ – at which point the cycle starts all over again.

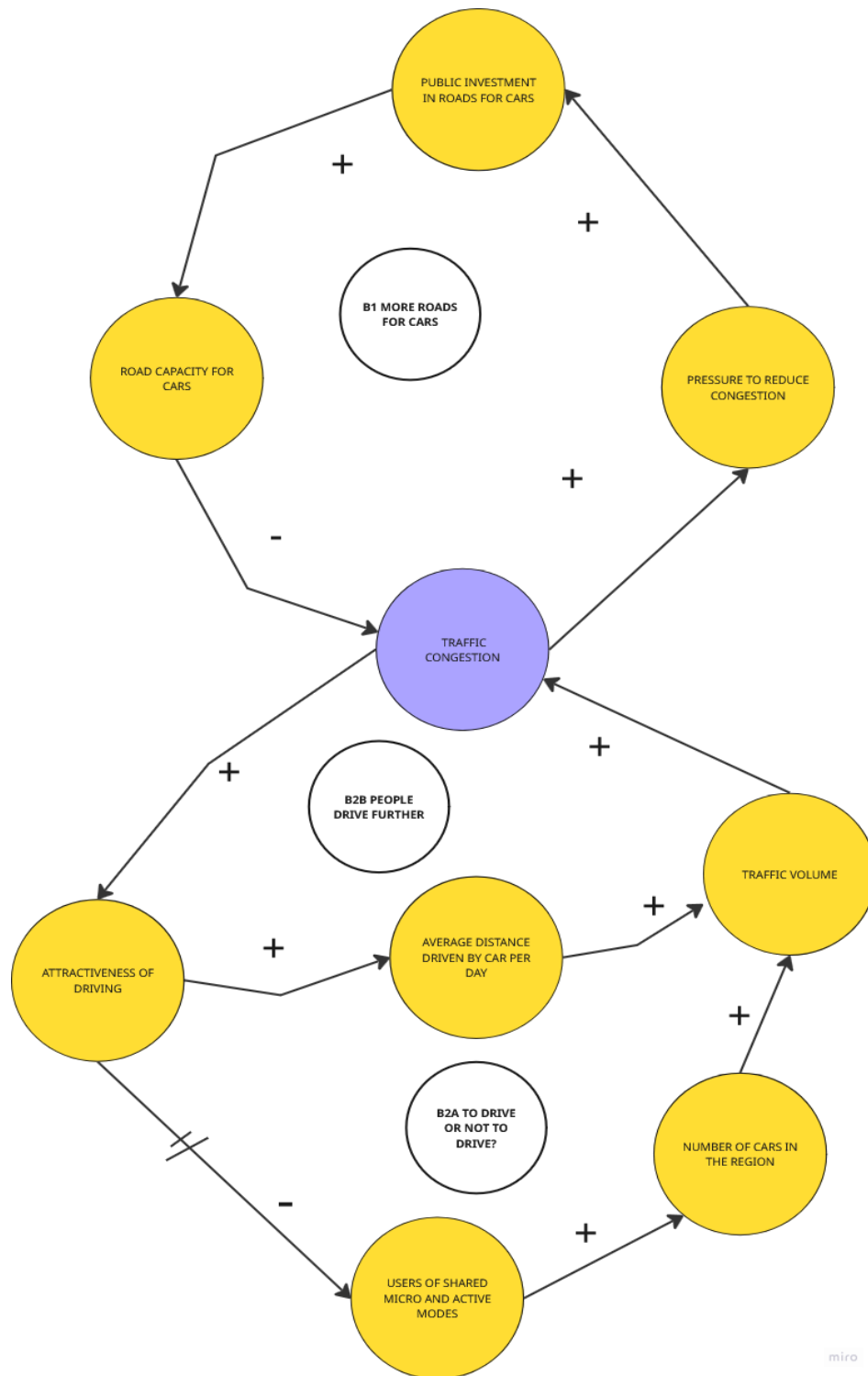


Figure 14 - System Dynamics underlying induced car demand (adapted from [OECD, 2022: 43](#))

Urban sprawl

Urban sprawl is characterised by low density housing and facilities that are ‘discontinuous, strongly scattered and decentralised’ (OECD, 2022: 46). It can be the result of a lack of integrated spatial planning, the prioritisation of roads for cars, and poor public and active transport development. People who live far away from healthcare, education, shopping employment and leisure are car dependent.

This is illustrated in the causal loop diagram below (Figure 15). As we noted above in the induced car demand diagram – as traffic congestion increases, so too does political pressure to reduce it. This has led to public investment in roads for cars and increase capacity for cars. As the road capacity increase so too does the catchment area i.e. the area from which the city or town can be accessed within a reasonable time budget. Previously inaccessible sites can become suburbs with an associated scattered development pattern. These suburbs and one-off developments often have limited public transport and other services (health education leisure etc). The declining density of population and the limited number of local services and places of interest, increases the attractiveness of driving, and this links back into the other dynamics associated with increased traffic volume in the induced car demand diagram.

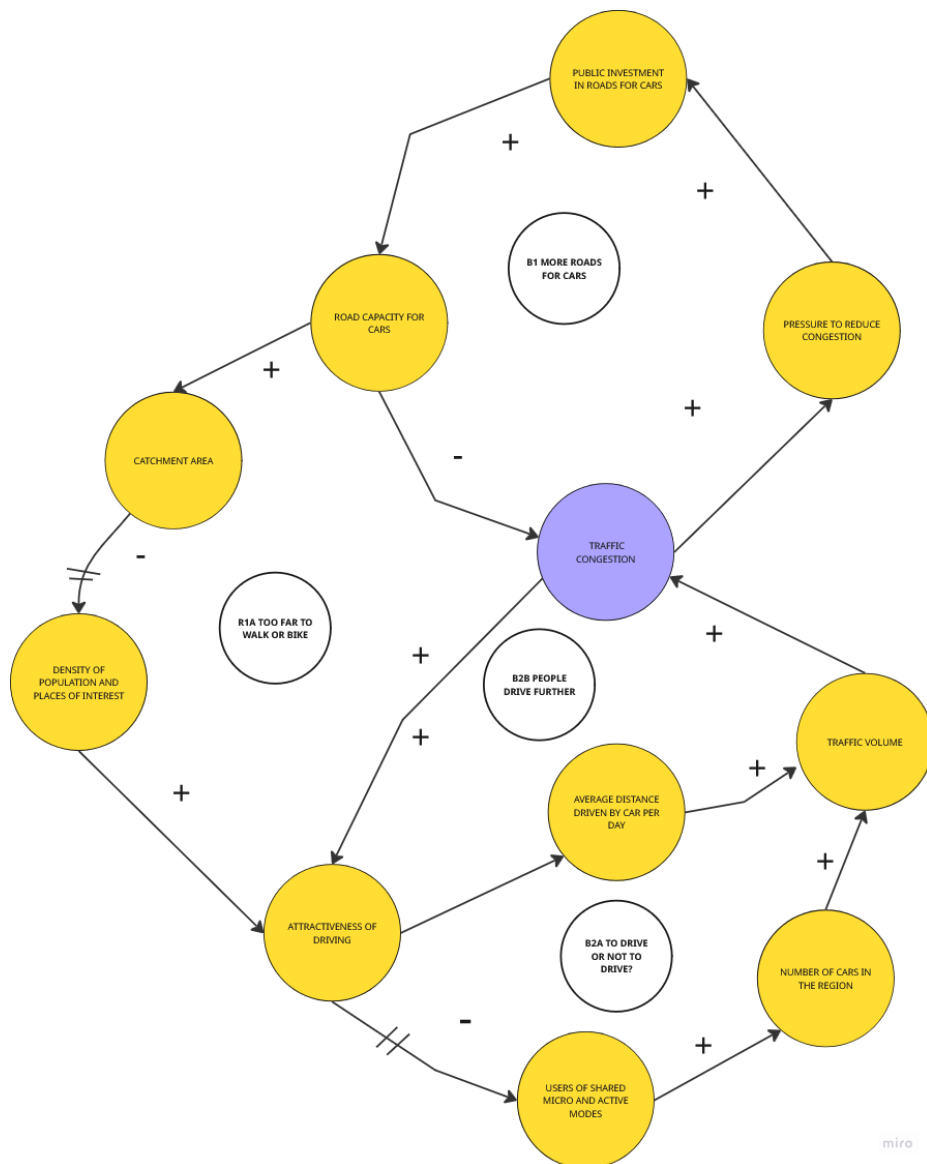


Figure 15 - System dynamics underlying urban sprawl (adapted from [OECD, 2022: 46](#))

Sustainable modes low-attractiveness trap

Public investment in sustainable transport modes competes with public investment in road for cars. In most countries across the EU there have been decades of priority investment in 'car-purposed road infrastructure, car-centric communication efforts and a lack of investment in public and active mode infrastructure' ([OECD, 2022: 52](#)).

Taking the figure above and adding key dynamics to it we can see how the increased road capacity for cars has led to less space for active, shared and public transport (R1C; R1D). This reduces the adequacy of active mobility and of public and shared transport. This reduced adequacy of alternatives increases the attractiveness of driving. It also leads to a decrease in the number of users of public, shared and active transport which reinforces the shortage of revenue and investment for public transport. This further decreasing the adequacy of public and shared transport.

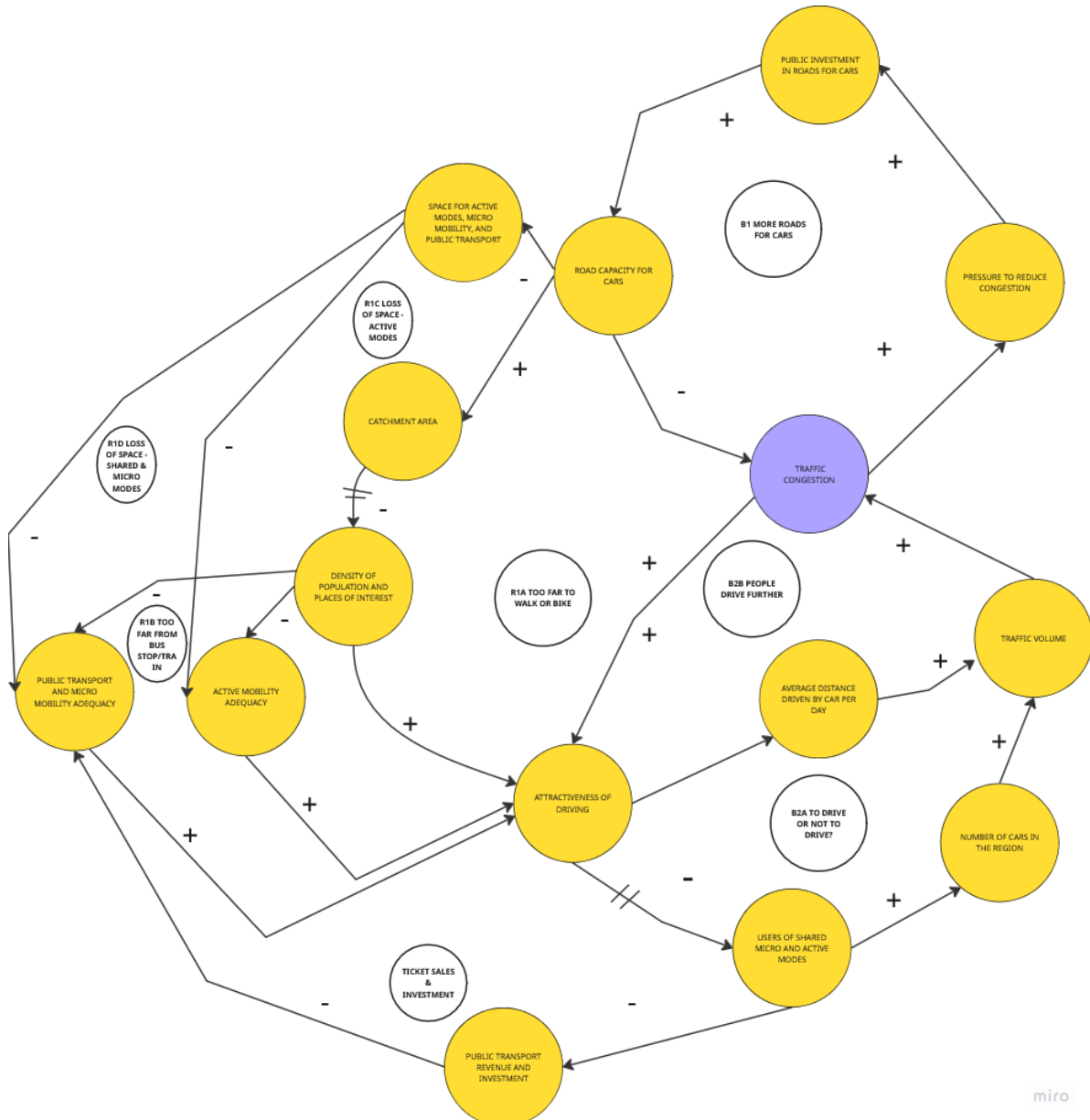


Figure 16 - System dynamics underlying the sustainable modes low-attractiveness trap (adapted from OECD 2022: 48)

4.1.4 Transport Poverty

These environmentally unsustainable dynamics - induced car demand, urban sprawl without adequate public transport, and underfunded public, shared and active transport modes - also contribute to widening inequalities and to social injustice.

Transport poverty refers to a 'lack of adequate transport services necessary to access general services and work, or to the inability to pay for these transport services' (Kiss, 2022: 1). Elements that lead to transport poverty include:

- Lack of nearby services and opportunities (i.e. proximity not considered in spatial planning) – this is also linked to too much time spent travelling (time poverty)
- No or limited transport availability (underdeveloped active, shared and public transport infrastructure, lack of multi modal options, low frequency of available services)
- Inaccessible transport (for disabled people for example) - even when services exist, they may not be designed inclusively. Disabled people, older adults, or parents with young children often encounter physical barriers (e.g., lack of ramps, elevators, or low-floor buses) and informational barriers (e.g., poor signage or digital-only ticketing).
- Inability to meet the cost of transport - Rising fares, fuel prices, and insurance costs mean that transport can consume a significant share of household budgets. For low-income groups, this can lead to difficult trade-offs between mobility and other essentials such as food or heating. Transport poverty here overlaps with energy poverty, creating compounded vulnerabilities.
- Forced car ownership – where no practical alternative to a private car exists but people struggle to pay the high costs of car ownership. This “forced car ownership” traps people in a cycle of debt and financial stress, while also contributing to environmental pressures.
- Dangerous or unsafe transport options - Where infrastructure is poorly maintained or where traffic conditions are hazardous, people may avoid walking, cycling, or using public transport. Safety concerns (ranging from accidents to personal security on late-night services) discourage use and deepen exclusion. ([Kiss, 2022](#); [Cludius et al, 2024](#))

The prioritisation of decoupling strategies in policy packages does not achieve wider social justice and wellbeing objectives. The prioritisation of private car users has resulted in:

- Lock-in of car dependent systems
- The mental and physical health benefits that can come from active modes are largely untapped
- Pedestrians and cyclists are not allocated sufficient and appropriate road space thus decreasing the attractiveness of those modes (e.g. cycle lanes that are too narrow and unprotected from traffic, pedestrian overbridges and subways across and under motorways). Many measures for pedestrians require people to go up and down steps - limiting accessibility for non-car users and particularly for people with additional mobility needs. They also take increased time and effort which may be circumvented by some pedestrians crossing at highly unsafe places.

As the ITF (2021) demonstrate

policies focused on ensuring access to car use (...) can cause forced car ownership, in particular for lower income households living in areas with fewer alternative transport options. In these cases, private vehicles, and their associated costs, become the only way to satisfy people’s needs, locking them into car dependency and transport-related costs ([ITF, 2021](#) cited in [OECD, 2021](#): 37).

As noted in [D3.2](#)

Social dimensions of the current mobility regime show significant tensions between urban and rural populations. Existing transition strategies tend to prioritise urban contexts, where the benefits of electrification, shared mobility and infrastructure upgrades are more readily implemented, while rural communities risk being left behind. Perceptions of exclusion and unfairness are growing, especially among those who feel disconnected from the goals of the twin green and digital transition. These dynamics are contributing to political polarisation and raise important challenges for the design of inclusive transition pathways ([Viñé and Armayones, 2025: 169](#))

As noted in D3.3 the Gilets Jaunes movement in French was not about rejecting climate action but rather objecting to a regressive flat fuel tax which burdened ‘lower-income, car-dependent people in rural areas while urban elites were less affected’ ([Martin, 2019](#) cited in [Armayones and Viñé, 2025b: 12](#)).

To move away from policy measures that only address individual parts of the system and toward policy measures that enable transformative change – by intervening effectively in the system - it is necessary to understand the unsustainable dynamics that are currently operative.

4.1.5 Stock and Flow Analysis

In 2022 5617 billion passenger-km were travelled in the EU-27. This is an increase of 24.9% of passenger-km since 1995 which represents a compound annual growth rate (CAGR) of 0.8%. Although the 2022 figure is a slight decrease on passenger-km travelled prior to the COVID 19 pandemic it represents a progressive return to pre-COVID levels of passenger-km travelled (EEA, 2024). Indeed, the recent MIX-FF55 scenario from the European Commission (EC, 2021) forecasts that overall transport activity will grow. The scenario indicates that there will be a further increase to 6487 billion passenger-km travelled in 2030 (EEA, 2024).

Fuel combustion per EU inhabitant in transport was 16% higher in 2019 than in 1990. This ‘suggests that the average use of powered transport per inhabitant in the EU increased at a faster pace than any improvements achieved in terms of fuel efficiency’ (Eurostat, 2025: 43). This high level of activity, combined with the high modal share of private vehicles (70% for EU 27 in 2022 (Eurostat, 2025)), make reducing the environmental impacts of the transport sector extremely challenging. Although there has been an increase of travel by train and shared transport across the EU since 2020, passenger cars remain the dominant modal choice (Table 4.1)

Table 11 - Modal split of passenger transport (% based on passenger-kilometres)

	Year	Train %	Passenger cars %	Motor coaches, buses and trolley buses %	Seagoing vessels %	Aircraft %	Total
EU 27	2020	5.1	82.1	6.9	0.3	5.7	100.1
	2023	7.1	70.6	7.2	0.4	14.7	100.
Spain	2020	3.4	81.7	5.4	0.3	9.2	100.
	2023	5.8	66.4	6.9	0.5	20.5	100.1
Italy	2020	3.9	81.3	10.4	0.4	4.0	100.
	2023	6.1	74.7	9.0	0.6	9.7	100.1
Hungary	2020	5.5	73.0	14.2	:	7.3	100.
	2023	8.1	61.3	11.8	:	18.7	99.9

In current systems people use their car for the majority of trips for two reasons - the long distances between people and places, and the allocation of road space primarily to motorised vehicles which increases their convenience (OECD, 2021: 8). The choice to drive a car is not only the result of individual preferences, but rather the result of transport and urban systems primarily organised to facilitate travel by motorised vehicle (cars, motorbikes).

The number of passenger cars relative to the size of the population is the motorisation rate for road passenger vehicles. Of the case study countries Italy has the second highest motorisation rate in the EU (701 per 1000 inhabitants) while Hungary has the third lowest motorisation rate (447 per 1000 inhabitants). The motorisation rate has increased across the EU, and in Italy and Hungary, since 2021.

Table 12 - Passenger cars - per thousand inhabitants (Eurostat, 2025a)

	2021	2022	2023	2024
EU 27	561	564	570	576
Spain	554	553	551	544

Italy	675	682	694	701
Hungary	418	426	435	447

Since 2009 there has been an increase in the density of motorways in all the case study countries and most significantly in Hungary.

Table 13 - Density of motorways (km per 1000 km square of land area, 2009 and 2023)

	2009	2022	% change
Spain	28	31.54	12.64
Italy	22	25.38	15.36
Hungary	14	20.46	46.14

(Eurostat)

Since 2009 there has been relatively no change in the density of railway lines in Italy and Hungary, but Spain has seen an increase of 7.77%.

Table 14 - Density of railway lines (km per 1000 km square of land area, 2009 and 2023)

	2009	2023	% change
Spain	30	32.33	7.77
Italy	56	56.28	0.5
Hungary	81	81.04	0.05

(Eurostat)

The number of new vehicles sold has increased year on year across the EU since 2022. The majority of new vehicles sold in the EU run on petroleum products. The % of new electric vehicles bought in the case study countries remains low.

Table 15 - New passenger cars by type of motor energy 2022-2024 (Eurostat, 2025b)

		Petroleum products %	Diesel %	Electricity %	Alternative energy %	Total (n)
EU 27	2022	57	19	11	13	10,490,773
	2023	56	16	13	15	12,203,159
	2024	59	15	12	15	12,182,058
Spain	2022	68	23	4	5	883,262
	2023	69	18	5	8	1,039,109
	2024	70	16	5	8	1,126,555
Italy	2022	60	23	4	13	1,384,650
	2023	62	22	4	13	1,646,956
	2024	64	19	4	13	1,624,825

Hungary	2022	73	18	4	5	116,338
	2023	72	17	5	6	113,667
	2024	69	18	7	7	130,176

The share of household consumption expenditure in the EU for the purchase of vehicles was the same in 2022 as it had been in 2012. The shares for the operation of personal transport equipment and for transport services were slightly lower ([Eurostat, 2025: 56](#)).

Table 16 - Share of household consumption expenditure on transport

EU 27	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Transport	12.9	12.6	12.7	12.6	12.6	12.9	13.2	13.2	11.6	12.0	12.5
Purchase of vehicles	3.2	3.1	3.2	3.4	3.6	3.7	3.8	3.8	3.8	3.6	3.2
Operation of personal transport equipment	7.6	7.4	7.3	6.9	6.7	6.9	7.1	7.1	6.4	6.9	7.4
Transport services	2.2	2.2	2.2	2.2	2.2	2.3	2.3	2.3	1.4	1.5	1.9

[Eurostat](#)

In 2023 environmental transport taxes in the EU were equivalent to 0.37% of GDP. The relative importance of environmental transport taxes has remained relatively steady since 2020 in the case study countries.

Table 17 - Environmental taxes on transport (% of GDP)

	2020 (% of GDP)	2021 (% of GDP)	2022 (% of GDP)	2023 (% of GDP)
EU 27	0.43	0.40	0.38	0.37
Spain	0.23	0.23	0.22	0.21
Italy	0.59	0.57	0.52	0.53
Hungary	0.49	0.47	0.52	0.48

(Eurostat [Transport](#) and [GDP](#))

Given the high modal share of passenger cars in the case study countries, the increase in investment in motorways, and comparatively lesser investment in rail infrastructure, it is plausible to hypothesize that policy interventions in the case study countries are oriented more toward enabling mobility rather than toward enabling sustainable accessibility.

This policy focus has possibly contributed to the increased investment by individuals in private vehicles. To note there may be significant regional or urban/rural differences in terms of investment in car purposed infrastructure and public, active and shared transport infrastructure.

To conduct a systems inquiry we need to understand the systems physical lock-in, in-flows and out-flows. In car-dependent systems the stock, in-flows and out-flows tend in the direction of:

- Stock – large fleet of ICEV (hard to change), large road infrastructure for cars, minimal road infrastructure for active travel, underdeveloped rural public and shared transport services
- In-flows – investment in additional car-purposed road infrastructure, limited investment in public transport, new car sales

- Out-flows – road space reallocation away from private cars (i.e. a reduction in the car-purposed road infrastructure)

Over the years the transport infrastructure in many EU countries has become dominated by private cars because car-purposed road infrastructure is a priority investment for States.

4.1.6 Sample Assessment - Policy of Subsidy for Private Electric Vehicles

As outlined in Chapter 3, once we understand the relative magnitudes of existing stocks - and we understand the system's unsustainable dynamics – we can assess a policies' transformative potential in two ways:

- i. Potential mismatches between the intent and ambition of the policy and the magnitude of unsuitable stock can be identified.
Currently in a country where the 'lock-in' of car-purposed infrastructure is established - additional investment in public transport will have minimal transformative impact. As the OECD note, the same effort, made before such car-purposed infrastructure was 'locked-in' could have a much higher transformative power (2022: 55)
- ii. Stock and flow analysis reminds us that we do not only need to 'add' additional elements – we can also take away or reallocate unsuitable infrastructure. Too often policy decisions are about the changing levels of investment in various parts of the system, rather than on actively **reducing undesired stock**

Sample Assessment

Current infrastructure –

- high level of car purposed infrastructure
- good public transport services in city centre
- limited public transport availability in suburban areas
- almost no public or shared transport services in rural areas and small towns or villages
- poor active travel infrastructure across the region/nation

The structure of the EV policies in the case study countries assume continued private vehicle purchases and use. As such they can be classified as **anticipatory** policies – they look at past patterns (growing private car use), assume their continuation, and seek to address the environmental harms (high emissions from ICEVs).

In a car-dependent context i.e. where there is a high level of car purposed infrastructure, and limited or poor active and public transport infrastructure, a subsidy for electric vehicles has a **low potential** to be transformative.

As the OECD note electric vehicle incentives for private cars

leave the car-dependent structure of the system intact and may lock it in further by reinforcing induced car demand, urban sprawl, car-centric mental models and the belief that changes in technology are sufficient to achieve climate targets (potentially undermining political support for transformative climate policies) (OECD, 2022: 60)

Electric vehicles may alter the type of vehicle sold (i.e. fewer ICEVs and more EVs) and this can lead to a reduction in local emissions. However, it does not transform the overall dynamic of induced car demand which is what leads to ever growing private car use. Private vehicles remain a highly attractive transport mode – in some cases they are the only mode available – and so most people continue to use their cars for the bulk of their trips (OECD, 2022: 60). The car-centric mental 'lock-in' remains intact.

Car fleets are 'hard to change' stock and thus a low leverage point (Meadows, 1999). New car sales are an 'in-flow' to the system, but they are small in number compared to the existing stock of vehicles. As the OECD note even

if the sale of new combustion vehicles were banned tomorrow, and those sales were entirely replaced by EV sales, it would still take time for the fleet [to be] predominantly electric. This is because the stock of combustion vehicles is high and cars are expensive goods that last for decades ([OECD, 2022](#): 63)

The International Energy Agency note that, according to stated policies, globally just over one in four cars will be electric by 2035 ([IEA, 2024: 104](#)).

EV incentives keep the attractiveness of private vehicle use intact - and may even increase it. They also arguably work against enabling the goal of 'sustainable accessibility' in other respects:

EV incentives for private car use may reinforce the belief that the existing transport infrastructure does not need to be transformed

When climate actions are targeted at decreasing fuel emissions from existing (high) levels of private vehicle use, the belief that the transport system – as it is – is fit for purpose is reinforced. Provisions for private charging points can reinforce the idea that 'everyone is entitled to have a private car and a dedicated space for it, and that the duty of the government is to facilitate ease of use, further reinforcing car-centric mindsets' ([OECD, 2022](#): 62). However, the evidence base strongly suggests otherwise. Reinforcing the belief that high levels of private car use is environmentally and socially feasible, undermines public understanding of, and political support for, transformative transport policies.

EV incentives for private car use may reinforce the idea that the green transition is unjust

Existing EV policies are expensive to run and predominantly benefit high-income households and businesses who can afford to switch to an EV and have the space to install charging stations. There is some evidence to suggest that EVs may represent a household's second or third car ([Bauer, 2018](#); [Caulfield et al., 2022](#) cited in [OECD, 2022: 62](#)). Lower income households, people in apartments and people in rented accommodation often lack the money and/or space and decision-making authority (re install of charging station) to switch to an EV ([Aryanpur et al., 2022](#); [Caulfield et al., 2022](#); [Mukherjee and Ryan, 2020](#) cited in [OECD 2022](#): 62).

When EV incentives for private use are prioritised - combined with policies such as carbon pricing, road pricing and low emission zones - and no or minimal provisions are made for an increase in accessible and affordable public and shared transport modes, high-income households are able to adjust and maintain high mobility levels, while middle and low-income households pay higher costs and have greater difficulty accessing necessary services and opportunities.

EV incentive may reinforce urban sprawl

EV incentives reduce the cost of driving and increase the attractiveness of detached and large houses in suburban areas. Some actions (such as requiring car parking spaces and EV charging stations in new developments) perpetuate car-dependency and urban sprawl. As more space is needed to deliver housing with car parking spaces, housing development on the outskirts of urban areas where land is cheaper is incentivised ([OECD, 2021](#)). There are several negative impacts of urban sprawl including:

- Longer commuting distances between residences, place of work and amenities which leads to increase car dependency, lower levels of active travel, higher GHG emissions, and more noise and air pollution.
- Greater land use which often comes at the expense of green spaces. Green spaces around urban areas are required to provide ecological benefits such as fresh air, water filtration and leisure activities
- Encroachment on agricultural land threatening food supply and livelihoods.
- The provision of additional infrastructure and new services is required, duplicating what is already available within existing built-up areas. This incurs direct cost to the State.
- The State does not benefit from economies of scale in relation to key services such as schools and hospitals ([Indecon, 2024: 6](#)).

EV incentives for private car use increases the risk of a rebound effect

The rebound effect is where vehicle usage increase after purchase of an EV. [Ojeda-Diaz et al \(2025\)](#) caution that quantifying increased vehicle use and the underlying mechanisms that lead to it remains difficult. That said, in their review of 38 studies that explored changes in vehicle use - '30 reported an increase in usage after EV adoption, six found no differences, and one reported both outcomes' (2025: 15). Some studies suggest that the operation cost advantage of EVs, and the desire for payback on investment, could be factors that lead to increased vehicle usage after EV adoption ([Orsi, 2021](#); [Caulfield, Carroll and Ahern, 2020](#) cited in [OECD, 2022: 61](#)). The rebound effect may increase traffic congestion, energy consumption and GHG emissions.

4.1.7 Sample Policy – Road Space Reallocation

Road space reallocation is a process through which road and street space is taken away from cars and allocated to 'different transport modes and functions beyond transport (e.g. recreation, bus lanes, markets)' ([OECD, 2022: 78](#)). Some estimates say that nearly 80% of public space in cities in Europe is allocated to cars and associated infrastructure (roads, parking spaces) ([EC, 2022](#)). This disproportionate allocation of public space increases the attractiveness of cars leading people to choose to drive. The over allocation of public space to cars has had additional negative effects including:

- A 10% decline overall between 1990 and 2015 in areas accessible within walking distance from an arterial road
- A decline in intersection density – an indicator of the ease with which a person can shorten travel distances by active travel
- An increase in the size of city blocks – reducing walkability, making cities less bicycle friendly, and fostering traffic congestion ([UN-Habitat, 2016](#) cited in [OECD, 2021: 62](#))

To reduce GHG emissions and improve wellbeing and social connectedness we need to increase the amount of public space allocated to sustainable transport modes and functions other than transport including 'public seating, green spaces, markets, restaurant terraces or playgrounds' ([OECD, 2022: 80](#)). This has multiple environmental and social benefits such as increasing the attractiveness of public and active transport, increasing the attractiveness of urban areas, potentially reducing urban sprawl over time, and supporting the development of 15-minute cities ([OECD, 2022: 80](#)).

The OECD categorises road space reallocation as **transformative** with **high-potential** to transform the car-dependent system. This categorisation is based on the fact that it is an action which understands that **traffic volume** is a primary problem (not traffic congestion). This action 'aims to transform the system by creating the conditions for shared transport and active modes to become the most attractive. By doing so, the policy intends to trigger behavioural change and lead to disappearing traffic' ([OECD, 2022: 79](#)).

Road space reallocation leads to the *development of a new system by breaking a crucial link in the car-dependent system*

When traffic congestion increases there is often political pressure to further increase the allocation of space to roads and car users. By not doing this – and instead allocating road space to alternative uses – road space reallocation breaks an important link in the car-dependent system. When policy focuses on liberating space for sustainable modes, these become more attractive and people use them for the bulk of their trips, the catchment area reduces and the pressure to further enhance sustainable modes increases.

Road space reallocation addresses the *over dimensioned stock of car-purposed capacity*

By reducing the stock of road capacity for cars the cycle of car dependency is broken. Further the stock of liberated space is increased and this can lead to a cycle that is both environmentally and socially advantageous. It is an inexpensive and technically straightforward action and can thus be categorised as a 'high leverage point' – a small change which leads to large changes in the system. The benefits can be seen quickly and include 'buses are no longer stuck in traffic (being provided with an exclusive lane) and thus move faster than cars; children ride their bikes to school as cycling becomes safer; seniors ride e-bikes or walk more often as benches are available and walking is pleasant' ([OECD, 2022: 80](#)) Road space reallocation

also increases the spread and uptake of e-bikes and e-scooters. These modes fill public transport gaps and ‘significantly reduce the cost of electrification of transport’ ([OECD, 2022](#): 81).

Road space reallocation may facilitate *a change in mental models*

Once people have positive experiences within car free zones, they may become more supportive of further enhancing sustainable transport modes through more road space reallocation which leads to reduced road capacity for cars and an increase in liberated space. However, this dynamic is not guaranteed, and multiple factors influence public acceptance of road space reallocation. Amenities and sustainable transport modes need to be accessible and affordable to increase the likelihood of positive experiences with road space reallocation ([OECD, 2022](#): 81).

4.1.8 Transport – Conclusion

A big and growing fleet of electric vehicles for private use generates significant environmental and social impacts from mining and the production of batteries. It also maintains the unattractiveness of shared and active travel, negatively impacting air quality and people’s wellbeing. As demonstrated in [D2.1](#) and [D2.2](#), a ‘strong focus of climate action therefore needs to be on redesigning systems so that – in their functioning – they require less energy and materials and produce less emissions while improving wider well-being goals’ ([OECD, 2021](#): 11). Systems inquiry enables policy makers to identify the unsustainable system dynamics which underlie key issues such as high traffic volume and the low attractiveness of sustainable transport modes.

4.2 Energy

Energy systems are central to everyday life. The industrial revolution created a reliance on fossil fuels and led to the intensive extraction of natural resources, triggering unprecedented economic, social and environmental changes. The global energy mix is still dominated by fossil fuels which comprise about 80% of total energy supply (coal and coal products (27.8%); oil and oil products (30.2%); natural gas (22.7%)); and account for about 60% of electricity generation (coal (35.5%); oil (2.6%); natural gas (22.1%)) ([IEA, 2023](#)).

However, factors such as the environmental harm of fossil fuel extraction, production and use; rising cost of fossil fuels; the need to reduce reliance on energy imports; and increased awareness of the impacts of the climate crisis are changing energy systems.

Clean technologies such as solar power, wind turbines, hydropower and geothermal systems can mitigate some of the harms associated with energy production and consumption. That said, they too require the extraction of natural resources, highlighting the need for whole-system sustainable practices from mining to production to use, recycling and waste disposal.

4.2.1 The Goal of Sustainable Energy Systems

The goal of a just energy system could be to shift to a participatory renewable energy system wherein both the costs and benefits of energy services are fairly distributed. An energy system which reflects the principles of environmental justice and energy democracy could entail:

- Robust environmental protection laws that prioritize the wellbeing of communities
- Achieving universal energy access by investing in
 - Utility reform – to include community participation in governance; investments in community-scale clean energy
 - Rooftop solar – to include a commitment to distributed energy resources including the prioritisation of rooftop solar and battery storage in vulnerable communities
 - Community energy – to include community ownership and control, project scale, economic benefits and location aligned with community values

4.2.2 Unsustainability of fossil fuel dominated, high-demand energy systems

Rapid decarbonisation of the grid and electrification of the global economy has been identified by the IPCC as a key measure to reduce greenhouse gas emissions. This is required because energy sector activities ‘from extraction, conversion, intermediate and final use, accounts for roughly three quarters of greenhouse gas emissions, mainly due to the combustion of coal, natural gas, and oil products; most of this combustion is used today to produce electricity’ (UNECE, 2021: 9).

High emissions

A life cycle assessment of technologies used for electricity generation found that fossil fuels generate much higher greenhouse gas emissions than renewable energy sources (UNECE, 2021).

Table 18 - Lifecycle environmental profile of various technologies

Technology	GHG Emissions
Coal power	751 gCO ₂ eq./kWh (minimum); 1095 g CO ₂ eq./kWh (maximum)
Natural gas combined cycle plant	403–513 g CO ₂ eq./kWh; between 92 and 220 g CO ₂ eq./kWh with CCS
Nuclear power	5.1–6.4 g CO ₂ eq./kWh
Hydropower	6 to 147 g CO ₂ eq./kWh (site specific)
Solar technologies - CSP	27 to 122 g CO ₂ eq./kWh
Solar technologies - photovoltaics	8.0–83 g CO ₂ eq./ kWh
Wind power - onshore	7.8 and 16 g CO ₂ eq./kWh
Wind power - offshore	12 and 23 g CO ₂ eq./kWh

(UNECE, 2021)

Cost of fossil fuels

By 2023 an estimated 96% of new, utility-scale solar photovoltaics (PV) and onshore wind capacity had lower power generation costs than new coal and gas plants. Approximately 75% of new wind and solar PV plants offered cheaper power than existing fossil fuel facilities globally. In 2024 the ‘global average electricity generation cost from solar PV was 41% cheaper, and onshore wind 53% cheaper, than the least-cost new fossil fuel-fired power plant (UN, 2025: 1). On average project lead times for solar PV and onshore wind are 1-3 years; for coal- and gas-fired power plants are up to 5 years or more; and for nuclear power plants are 10-15 years (UN, 2005: 1).

Energy Security

The ratio between net imports and gross available energy (quantity of energy need to meet a country’s energy needs) is called ‘energy dependency’. This ratio indicates the extent of dependency on energy imports. There has been a decrease in primary energy production in the EU and an increase in primary and secondary energy imports over the past decades. In 2023 demand was highest for oil and petroleum products of which 94.9% were imported. 90% of natural gas and 40.8% of solid fossil fuels used were imported.

Overall, the trend from 1990 to 2000-2005 saw an increase in import dependency, from 50.0% to 56-58%. This level of import dependency has been kept around 56-60% throughout the years, with two exceptions in 2019 (60.5%) and in 2022 (62.5%). The figure stood at 58.4% in 2023 (Eurostat, 2025)

With the exception of Hungary each of the case study countries have a higher level import dependency than the EU level. A heavy reliance on imported energy makes countries vulnerable to supply disruption. As noted by some experts in D3.2:

The macroeconomic landscape around energy is shaped by dependency on fossil fuels, geopolitical tensions, and the volatility of global supply chains. The European transition to renewables is motivated in part by economic vulnerability, specifically, dependence on cheap Russian oil, which created price instability during international crises. Another structural issue is the economic risk of investing in regions with conflict potential, like rare earth deposits in Ukraine. Political and commercial interests also influence energy markets globally, particularly in relation to fossil fuel trade ([Viñé and Armayones, 2025: 107](#))

Table 19 - Net energy imports of total energy supply (2024)

	Imported Energy Supply %
Germany	70
Ireland	87.5
Portugal	74.8
Spain	78.4
Italy	78.9
Hungary	49.4

([IEA, 2024](#))

Growth in Electricity Demand

Alongside the growth in the use of electric vehicles, heat pumps and AI, demand for electricity will also increase. An 'estimated total of 9 million new battery electric vehicles and 11 million new heat pumps are expected to become operational by 2026 in the European Union ([IEA, 2024: 20](#)). The IEA (2024) estimate that electricity consumption for data centres in 2026 will be 30% higher than 2023 levels in the EU with Ireland and Denmark making up 20% of the expected increase ([2024: 20](#)).

The European data centre market is valued at USD 42.98 billion (2023) and this is expected to grow to USD 64.5 billion by 2029 ([McArdle and Terras, 2025: 3](#)). Current AI systems are highly resource intensive, with a query from ChatGPT requiring 2.9 watt-hours of electricity compared to 0.3 watt-hours for a Google search. Training AI is also resource and energy-intensive. A training run of Pathways Language Model by Google, at a data centre where 89% of energy comes from carbon-free sources, resulted in emissions equivalent to those 'produced by a fully occupied commercial jet during 1.5 flights across the United States' ([McArdle and Terras, 2025: 4](#)).

Data centres are adding significant demand to the power grids in Europe. There are concerns that this demand may prolong the use of coal and gas infrastructure in Europe. As of April 2024, there were '82 data centres in Ireland, with a further 14 under construction and planning approved for 40 more' ([Ryan Christensen, 2024](#)). With due regard for the significant uncertainties related to the growth of energy demand by data centres, a recent study posited that, by 2030

Europe (the EU and the UK) could see an additional 287 TWh of electricity demand from data centres, assuming high demand growth. If even 61% of this demand is met with fossil fuels, nearly 85 TWh of additional electricity could be generated by burning fossil gas in the EU in 2030. This could lead to the EU's GHG emissions rising by up to 121 million tons of CO₂e. If, on the other hand, the demand is met with renewables, emissions would fall to almost zero. However, assuming this renewable energy is taken from the already-planned growth in renewable energy generation in the EU, this would mean nearly 20% of new renewables in the EU would need to go to data centres, which could limit access to renewables in other sectors. If not enough renewable energy projects are built across the EU to meet all demand, renewables would fail to displace overall fossil fuel use in the economy ([McArdle and Terras, 2025: 7](#)).

As such the energy demand from data centres risk making climate commitments very difficult or unachievable for some countries and risks diverting renewable energy away from other services.

Climate Targets

[The Revised Renewable Energy Directive \(2023\)](#) raised the EU's binding target for the share of renewable energy in the overall energy mix to a minimum of 42.5% by 2030 but aiming for 45% ([EC, ND](#)). The current share of renewable energy in the energy mix is outlined below.

Table 20 - Energy Mix 2024

	% Coal/ Coal Products	% Oil/ Oil Products	% Natur al Gas	%Nu- clear	% Hydro- power	% Solar Wind and other renewabl es	% Biofuels and waste	% Total - Renewables
Europe	12	33	24	11	3	6	11	20
Portugal	0	45	17	0	7	12	20	38
Italy	2	36	40	0	4	8	11	23
Germany	16	34	27	0	1	8	13	22
Spain	2	44	21	13	3	10	7	20
Ireland	3	47	33	0	1	8	8	16
Hungary	3	32	31	18	0	5	11	16

([IEA, 2024](#))

Although Portugal is closest to achieving the target, the share of renewable energy in the overall energy mix across the case study countries falls far short of the target of 42.5%.

4.2.3 Energy Poverty

As outlined in D2.1, energy poverty is when a household reduces 'its energy consumption to a degree that negatively impacts the inhabitants' health and wellbeing' ([EC, ND](#)). It is driven by low income, a high proportion of household expenditure on energy, and the low energy performance of buildings and appliances.

Energy justice is concerned with developing mechanisms for more just production, transportation, distribution and use of energy.

The concept of energy poverty was introduced to EU law in 2009 in the common rules for the internal electricity market ([2009/72/EC](#)). It is key to the EU's efforts to enable a just and fair transition.

Research indicates that those already experiencing energy poverty during winter may also struggle in summer. Intense heat waves are making summer energy poverty a growing concern across Europe. Heat and cold bring significant public health, economic and social challenges. Keeping cool in summer requires cross-departmental innovation in areas such as spatial and urban planning and the greening of spaces.

As outlined in section 3 the European Pillar of Social Rights includes energy as an essential service which everyone is entitled to, and the right to help and protection against forced evictions.

As outlined in [D2.1](#), several groups have been identified as vulnerable to energy poverty in the context of green transition in the Global North. These include: low-income households, people from ethnic minorities and indigenous people, immigrants and those with low proficiency in the local language, older people, disabled persons, those with ill-health, women, young people and full-time students, single-parent families, socially isolated people and those living alone, as well as large households and multi occupancy households ([Jessel et al., 2019](#); [Middlemiss, 2022](#)).

In order to assess primary energy poverty, a set of indicators was developed at a European level, according to which a household is facing energy poverty if:

Its share of income spent on energy services become larger than twice the national median

Its income after energy costs drops below poverty line and the share of its income spent with energy is above the national median

Its energy spending is lower than the half the national median energy expenditure

It cannot warm the house during cold season ([Redemaekers et al., 2016](#)).

The Energy Efficiency Directive (EED) ([European Commission, 2023](#)) includes updated and more comprehensive interpretations of energy poverty with a cross-cutting approach that integrates the impacts of economic and health crises into the understanding of energy poverty. It recognizes the diverging impacts of it on different societal groups, such as women, people with disabilities, older persons, children and racialised people ([Clancy, et al., 2022](#)).

[The Clean Energy for All Europeans package \(2019\)](#) includes obligations to identify, monitor and address energy poverty through the National Energy and Climate Plans. The package details the rights of consumers as regards contracts, opportunities for energy communities, the role of active consumers, and the entitlement to smart meters.

Immediate concerns related to the affordability of energy can be addressed through measures such as:

- Energy vouchers
- Heating allowances
- Social tariffs
- Tax reductions

Although such measures are necessary in times of crisis they only address the outcome of underlying unjust structures. To address the root cause of energy poverty other measures are required, including:

- Investment in renewable energy sources
- Investment in energy efficiency
- Diversifying energy options for consumers
- Targeted and accessible retrofitting supports
- Incentivising energy demand reduction

As outlined in the [Commission Recommendation on energy poverty \(2023\)](#) structural measures have

long-lasting impacts and support the Union's objective of a fair energy transition. They should therefore be prioritized along with complementary relevant social measures. The measures taken by Member States should also aim at diversifying the energy options for customers while preventing vulnerable customers from getting locked in fossil fuels and particularly fossil gas, coal, and oil and risking having to face increasing network and administrative costs in the course of the transition. Therefore, addressing energy poverty and ensuring a fair transition depends crucially on the policy mix chosen ([EC, 2023](#))

The FITTER focus is on the install of solar PV by residential dwellers. While this is a positive development several barriers exist that may negatively impact already disadvantaged groups.

- Upfront investment required – may not be possible for those on low or fixed incomes
- Information/Support schemes/Maintenance information generally accessed online – may be inaccessible to those with low levels of digital literacy
- Limited grid capacity in some areas – rural dwellers could be negatively impacted
- Limited space to install solar PV – apartment dwellers, nomadic people who live in caravans may be negatively impacted
- No agency to install solar PV – tenants may not have permission to install ([Sovacool et al, 2022](#))

4.2.4 Energy - Unsustainable System Dynamics

The transition to Renewable Energy Systems (RES) in Europe is now progressing, with several elements identified as important points in the shift.

Table 21 - Key Shifts in the Transition from Fossil Fuel to Renewable Energy Systems in Europe

Traditional (fossil fuel dominated)	Transition to
Few large power plants	Many small power producers
Centralised, mostly national	Decentralised, cross-boundary
Based on large power lines and pipelines	Including small-scale transmission and regional supply compensation
Top to bottom	Both directions
Households as passive consumers (only paying)	Households as active and participating in the system

(adapted from [Sharma, 2025](#))

The ability to install solar PV or other small scale renewable technologies is dependent on a range of factors which, depending on context, could include access to relevant information and supports, digital literacy, ability to invest upfront costs, space, and/or home ownership status.

Some people do not have or have limited means of installing on-site renewable energy technologies. For those, access to community energy systems may be more viable and lead to greater returns and benefits. As outlined in [D3.2](#), interviewees noted that although there are many benefits to community energy, systems bureaucratic and financing barriers remain in many EU countries.

Table 22 - Community Energy Projects – Potential Benefits and Barriers

Benefits		Barriers	
Environmental	- Reduction in GHG emissions - Increased supply of renewable electricity	Infrastructure	- Grid access costs (especially in rural areas) - Limited availability of grid (low grid capacity)
Economic	- Increased security of supply of energy (larger scale than individuals) - Creation of local jobs	Policy and support framework	- Administrative burden (grant applications and project requirements) 100% community ownership
Financial	- Cheaper electricity for households - Potential reduction in energy poverty	Financial	- Access to additional finances required at initial and subsequent stages - Grid connection costs
Social	- Capacity building for local areas - Increased acceptance for renewables/energy transition	Social	- Skills and knowledge (technical) – lack of experts - Reliance on volunteers - Planning and social acceptance

The shift to a decentralised energy system requires proactive inclusion of all stakeholders and across dimensions (environmental, social, technological, economic). As noted by expert interviewees ([D3.2](#))

Concerns also emerge about governance blind spots: biodiversity is often excluded from energy plans, technical professions remain siloed from environmental concerns, and gender divides persist in cooperative leadership. Meanwhile, energy literacy is expected to grow at the local level, with municipalities developing plans and battery projects expanding. Still, energy community engagement remains uneven, with time constraints and marginalisation limiting participation for many. A more inclusive and stable governance approach is needed: one that bridges technical and environmental domains, simplifies processes, and ensures energy transition policies support all scales of initiative, not just the largest actors ([Viñé and Armayones, 2025: 141-142](#))

Expert interviewees also noted that

Decentralisation and prosumer models are often mentioned as desirable, but current regulations do not provide sufficient incentives or clarity. Some experts stress that justice must be a central goal of public subsidies; yet in practice, wealthier groups are often the main beneficiaries. Stronger, more targeted governance frameworks are needed to ensure that public support mechanisms are fair, transparent and effective ([Viñé and Armayones, 2025: 125](#))

A potential equity issue that is emerging relates to the payment of grid maintenance, operation and expansion costs. The modernisation of the electricity grid is a key part of the green transition across Europe. Grids are financed from multiple sources including public funding, private capital, and electricity tariffs for industry and households (Avallone et al, 2025).

Network tariffs support the cost of operating, maintaining and expanding the grid. According to the most recent CAN report on grids ([Wettingfeld and Zerzawy, 2025](#)), consumers in Ireland face the highest grid costs, paying almost €450 per year, with high grid tariffs per kWh. For other case study countries, Hungary has the lowest cost (€150 per year), with the rest in the middle: Portugal approx. €180, Germany approx. €260 and Spain at nearly €300 per year. However, this should be considered as a share of income rather than a stand-alone figure. As such the highest burden is experienced by households in Spain and Hungary.

The shift to the use of renewable energy sources entails a shift to a decentralised system wherein many people are both electricity consumers and producers - often called ‘prosumers’. People who produce energy via solar PV, wind turbines, home charging stations for recharging electric cars, or other means, can consume the electricity produced and/or feed it back into the national grid. Progress on this shift is uneven and the regulatory framework is out of step with the transition. As some expert interviewees noted:

While the vision of citizen-led production and exchange has been circulating for years, its implementation remains limited, with prosumer participation still seen as marginal. However, financial innovation is gaining ground, with expectations that decentralised energy schemes and differentiated tariff structures will expand, especially in countries like Austria. Despite this, fossil fuels remain highly profitable and continue to benefit from unaccounted externalities. For new models to thrive, interviewees suggest that regulation must adapt; for example, by setting fair contributions to grid infrastructure or enabling community-level revenue sharing ([Viñé and Armayones, 2025: 141](#))

Increased take up of small-scale production of electricity will have knock on effects on the collection of grid operation, maintenance and expansion costs, particularly for those consumers who cannot install solar PV or other means of small-scale energy production. This is because ‘network charges’ are often passed on to consumers through the energy suppliers standing charge and unit rate.

The example of the breakdown of costs within a bill for 100 euro in Ireland is outlined in the table below.

Table 23 - Breakdown of Electricity Bill Components in Ireland: Implications for Grid Cost Equity

Cost Type	Sample Charge (euro)	Explainer
Fuel Generation	40	40-45% of the cost of electricity is for the fuel used to generate electricity

Eirgrid Transmission Use-of-system (TUoS) charges	35	30-35% of the cost of electricity is made up of network charges. Energy suppliers pay to use the transmission network (power plant to substations) and the distribution network (substations to homes).
ESB Networks Distribution Use-of-System (DUoS) charges		Consumers are not charged these explicitly. They are passed on through the energy suppliers standing charge and unit rate.
Government charges - Value Added Tax (VAT) (9%) and Public Service Levy	10	9% VAT rate applies to gas and electricity. The Public Service Obligation (PSO) levy is charged to all electricity customers in Ireland. The money raised is used to support the renewable energy sector in Ireland. The levy is currently €1.46 excluding VAT a month for the 2025/26 period. Every residential electricity customer is subject to the same charge, no matter how big or small their energy usage is.
Supplier charges	10	8-10% of the cost of electricity pays for supplier overhead and profits.
Capacity market charge	5	Renewable energy is highly variable so if demand peaks or renewable energy production decreases, back-up electricity is used from gas plants, interconnectors and battery storage. This back- up is paid for through the capacity market. Consumers are not charged this explicitly. It is passed on through the energy suppliers standing charge and unit rate.

([Cassidy, 2025](#))

The potential impact of some consumers becoming prosumers, leading to a smaller consumer base paying higher network charges, was outlined in a study by Agnew et al. ([2018](#)). Using a case study of Queensland Australia, they explored how the installation of solar PV, combined with residential battery energy storage systems, enables consumers to reduce the amount of electricity used from the network; shift the times that they use electricity from the grid; or disconnect from the grid entirely. The authors sought to understand how the increasing use of battery energy storage, by a high number of consumers with solar PV installations, could lead to supply chain inefficiencies and influence the end-use consumer in terms of access to reliable and affordable electricity.

Fixed network costs in Queensland made up 50% of the cost of electricity (compared to Ireland 30-35%). Arguably this charge alone was a real incentive to install solar PV. However, as Agnew et al note, as more people installed solar PV and then produced electricity, demand for electricity from the grid decreased. This led to suppliers charging a higher unit rate to a *smaller customer base*. As the unit cost of electricity increased, more consumers were incentivized to install solar PV or engage with other energy efficiency measures (R1).

At the same time, peak demand is only minimally met by solar PV, and peak demand is the primary driver of network investment. This means that ‘while total use of the system declines due to PV uptake, the same level of network capacity is still required to ensure reliability at peak times’ ([Agnew et al, 2018](#): 2367).

Another point of interest is that a scheme providing 44 c/kWh for any electricity sent back to the grid (up to 2028) was set up – called the premium FiT. It subsequently closed to new entrants, but the authors note that more than 60% of PV households have access to premium feed-in-tariff (FiT) (as of 2017). Up to 2028 these households are incentivized to export the electricity they generate to the network rather than using it in their home or charging a battery.

The combined effect of increasing electricity prices, falling costs for solar PV, and government subsidies to support the installation resulted in high PV penetrations. This was difficult to manage for network service providers. To slow down the growth of PV, they introduced several measures (B1) which led to distrust and frustration as outlined below.

In response to high PV penetrations, reduced demand and a range of technical integration issues, network service providers have introduced PV connection restrictions in certain locations and increased fixed costs to achieve more equitable cost recovery (loop B1). These interventions serve to reduce both the financial case for new residential PV installations and the financial returns for pre-existing PV systems. Whilst utility intervention in this respect is understandable, it serves to increase consumer frustration and mistrust of the electricity sector ([Agnew, 2018](#): 2367)

Agnew et al define the introduction of PV connection charges, increasing fixed charges on residential solar and introducing new fees (to slow down the growth of PV installation) as 'fixes that fail'. The increased costs levied on solar PV delayed new installations and reduced the value of existing installations. Agnew et al posit that although this may have been reflective of costs and may have improved the equity of network tariffs - it increases dissatisfaction with network utilities. The additional costs also

reduce the consumer's ability to proactively and autonomously manage their electricity costs. This could increase electricity consumption and peak demand, which could further increase electricity costs. Over time, these dynamics reinforce consumer desire to reduce reliance on network utilities and when financial motivations align, increase uptake of non-network solutions, ultimately increasing the severity of the original problem ([Agnew et al, 2018](#): 2370)

Agnew et al (2018) note how increases in the use of residential PV with battery energy storage presents challenges to existing actors.

- Electricity generators will see decreased demand and flatter load profiles from the residential sector
- Distribution network service providers will need to meet minimum service and reliability standards even as overall residential grid consumption declines but peak demand remains stable or increases

Even for those customers who have no financial incentive to invest in battery storage and thus disconnect further from the grid (as they are selling back to the grid at a favourable rate) a range of non financial incentives may apply. These include environmental concerns, concerns about price rises, concerns about disruption to electricity supply, dissatisfaction with incumbent utilities (due to price rises), and desire for great self-sufficiency and/or grid independence.

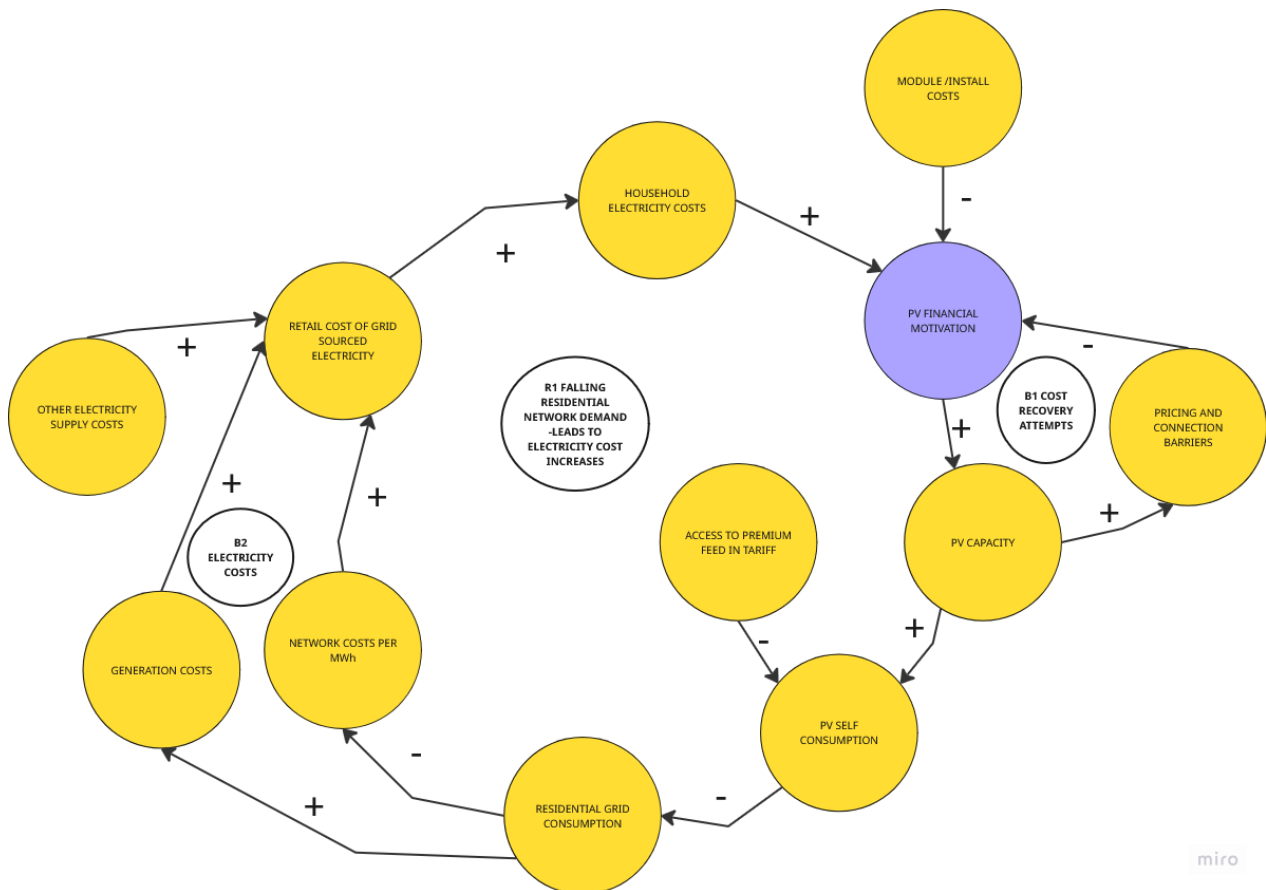


Figure 17 - Impact of Solar PV on Grid Costs and Charges (adapted from Agnew et al, 2018: 2368)

From a just transition perspective there are three cohorts of consumers of interest.

- First there is the cohort that installed solar PV early on. These ‘early adopters’ have several advantages – they can make more money by selling back to the grid (via premium feed in tariff (FiT)); they did not pay high fixed charges initially; and they are less reliant on the grid.
- Second there are those who installed solar PV at a later date (‘later adopters’). If they did so after the premium FiT scheme closed to new entrants, they make less money selling back to the grid. They could also have been charged connection costs and the high fixed charges for solar that were later introduced. However, as they have solar PV they are less reliant on the grid.
- Finally there are the ‘non adopters’. These people are unwilling or unable to install solar PV – they may be tenants, on a low or fixed income, have insufficient space, or lack information and digital literacy. They are entirely reliant on grid electricity and subject to price increases from network service providers.

Although both early adopters and later adopters were charged increased fixed costs on residential solar – thus increasing their dissatisfaction with utility service providers – they are still advantaged compared to the ‘non adopters’.

- The early adopters remain doubly advantaged (up to 2028) as they retain access to premium FiT.
- The later adopters are advantaged as they are not entirely reliant on the grid.
- The non-adopters without capacity to install solar PV are doubly disadvantaged – they are reliant on electricity from the grid and so subject to price increases and impacted by any disruption to the grid, and they have no opportunity to benefit from feed in tariffs.

This example demonstrates the importance of understanding the energy transition from a systems perspective. Fixes that fail solve a problem in the short-term but the unintended consequences ultimately increase the severity of the original problem (consumer desire to reduce their reliance on grid electricity;

network service providers need to finance grid costs). Network service providers need to recoup the costs of grid maintenance, operation and expansion but they need to do so in an equitable manner while incentivizing the uptake of renewable energy production and use. Agnew et al argue that their study demonstrates the increasing power of (some) pro-/consumers. They argue that ‘incumbent efforts to maximize revenues under the current regulatory framework by impeding competition or frustrating customer preferences are likely to reinforce drivers for greater battery uptake and disruption to the existing electricity supply system in the longer-term’ (2018: 2372).

Further research is required to ascertain the degree of applicability of these dynamics to the case study countries. It would also be important to investigate the impact of community energy products on grid costs and benefits to consumers. More generally, the findings of Agnew et al. demonstrate the need for fair burden sharing. There is a need to balance the funding requirements of publicly owned assets such as the electricity grid, with enabling equitable access to renewable and affordable energy as we transition from a centralized to a decentralized energy system.

4.2.5 Stock Overview

Household energy consumption expenditure has risen in all case study countries and across the EU since 2018.

Table 24 - Final energy consumption expenditure of households - Total k€/capita

	2018	2019	2020	2021	2022
Hungary	7	7.57	6.94	7.62	8.69
Portugal	13.59	14.16	12.49	13.43	15.94
Spain	15.57	15.71	13.15	14.57	16.61
EU27	16.08	16.48	15.26	16.4	18.3
Italy	18	18.21	16.25	17.52	19.99
Germany	20.3	20.89	19.96	20.85	22.74
Ireland	20.04	20.53	18.37	20.46	23.64

(Eurostat)

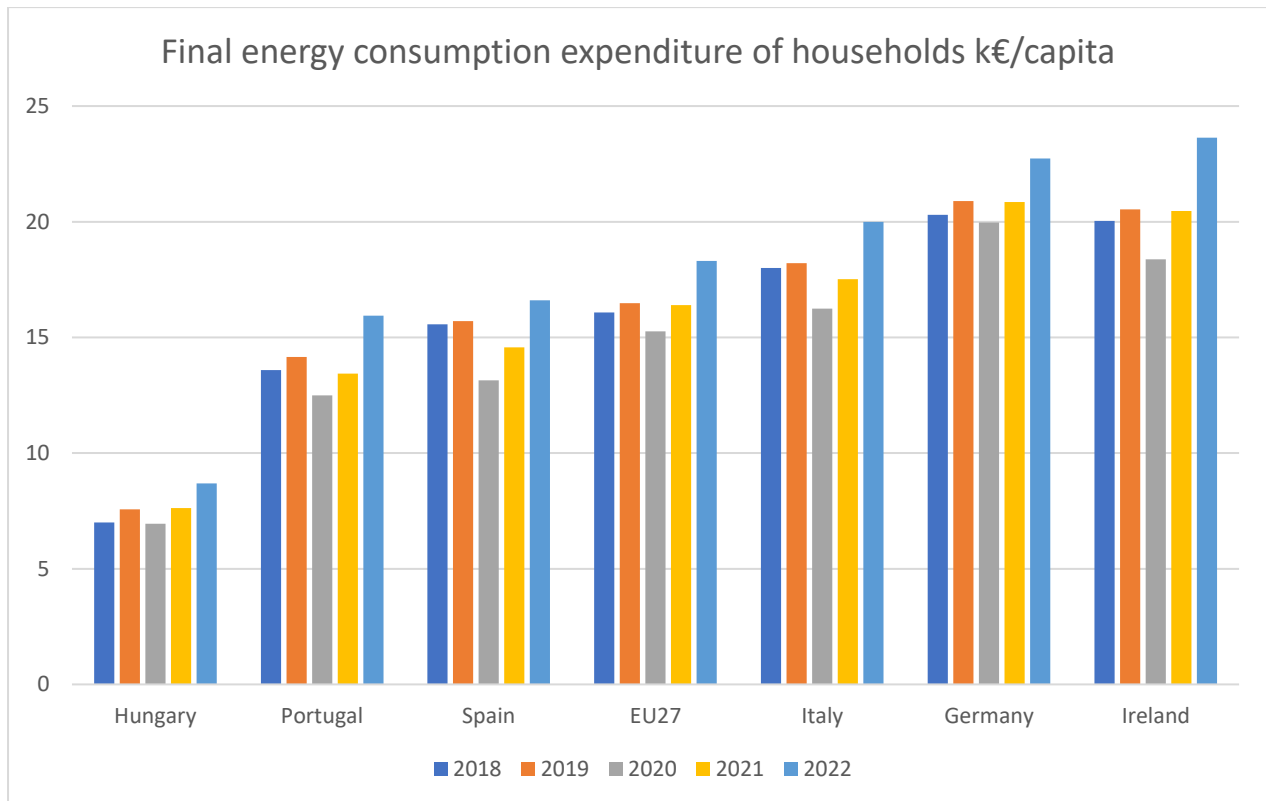


Figure 18 - Final energy consumption expenditure of households - Total k€/capita (Eurostat)

Development of the share of energy from renewable sources has been particularly slow in Ireland, Hungary and Italy (Table 4.2.5).

Table 25 - Share of Energy from Renewable Sources 2014-2023 (% share)

	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Ireland	9	9	9	11	11	12	16	13	13	15
Hungary	15	14	14	14	13	13	14	14	15	17
Italy	17	18	17	18	18	18	20	19	19	20
Germany	14	15	15	15	17	17	19	19	21	22
EU 27	17	18	18	18	19	20	22	22	23	25
Spain	16	16	17	17	17	18	21	21	22	25
Portugal	30	31	31	31	30	31	34	34	35	35

(Eurostat)

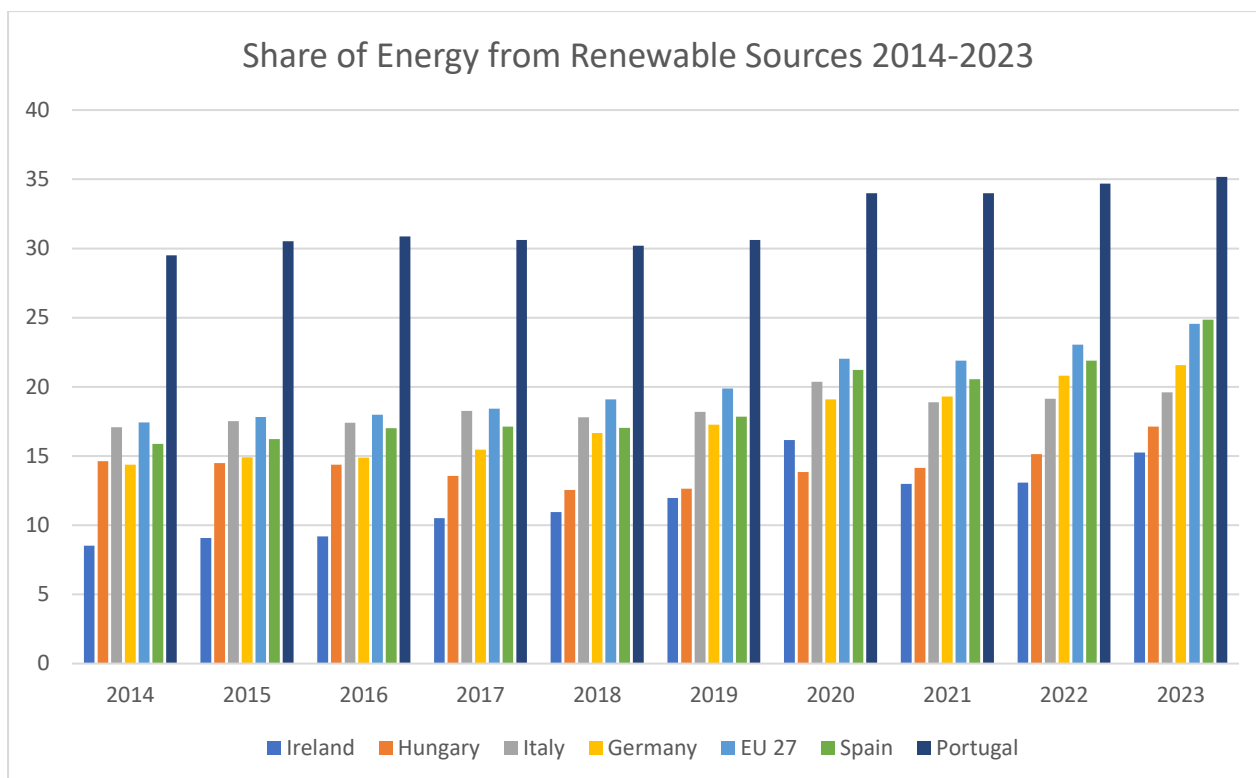


Figure 19 - Share of Energy from Renewable Sources 2014-2023 (%) (Eurostat)

Table 26 shows electricity prices for households without taxes and levies (basic) and with all taxes and levies (full) in the EU and in the case study countries. The reduction in costs for households in Ireland since 2022 is reflective of the energy credit for all domestic electricity accounts, introduced by the government in 2021, and first paid in 2022.

Table 26 - Electricity Prices for Household Consumers 2020-2024

	2020-S2	2021-S2	2022-S2	2023-S2	2024-S2
EU27 - Basic	0.1297	0.1554	0.2360	0.2244	0.2175
EU27 - Full	0.2153	0.2415	0.2768	0.2847	0.2887
Germany - Basic	0.1451	0.1596	0.2333	0.2882	0.2797
German-Full	0.3006	0.3234	0.3357	0.4020	0.3943
Ireland - Basic	0.2179	0.2420	0.3823	0.4590	0.3729
Ireland - Full	0.2616	0.2974	0.3225	0.3794	0.3046
Spain - Basic	0.1260	0.1878	0.2966	0.2068	0.1882
Spain Full	0.2298	0.2816	0.3350	0.2347	0.2408
Italy- Basic	0.1331	0.1760	0.3192	0.2565	0.2232
Italy - Full	0.2153	0.2360	0.3641	0.3347	0.3111
Hungary - Basic	0.0794	0.0790	0.0853	0.0891	0.0812

Hungary - Full	0.1009	0.1003	0.1084	0.1132	0.1032
Portugal - Basic	0.1138	0.1164	0.1972	0.2012	0.1693
Portugal - Full	0.2133	0.2170	0.2222	0.2299	0.2626

(Eurostat)

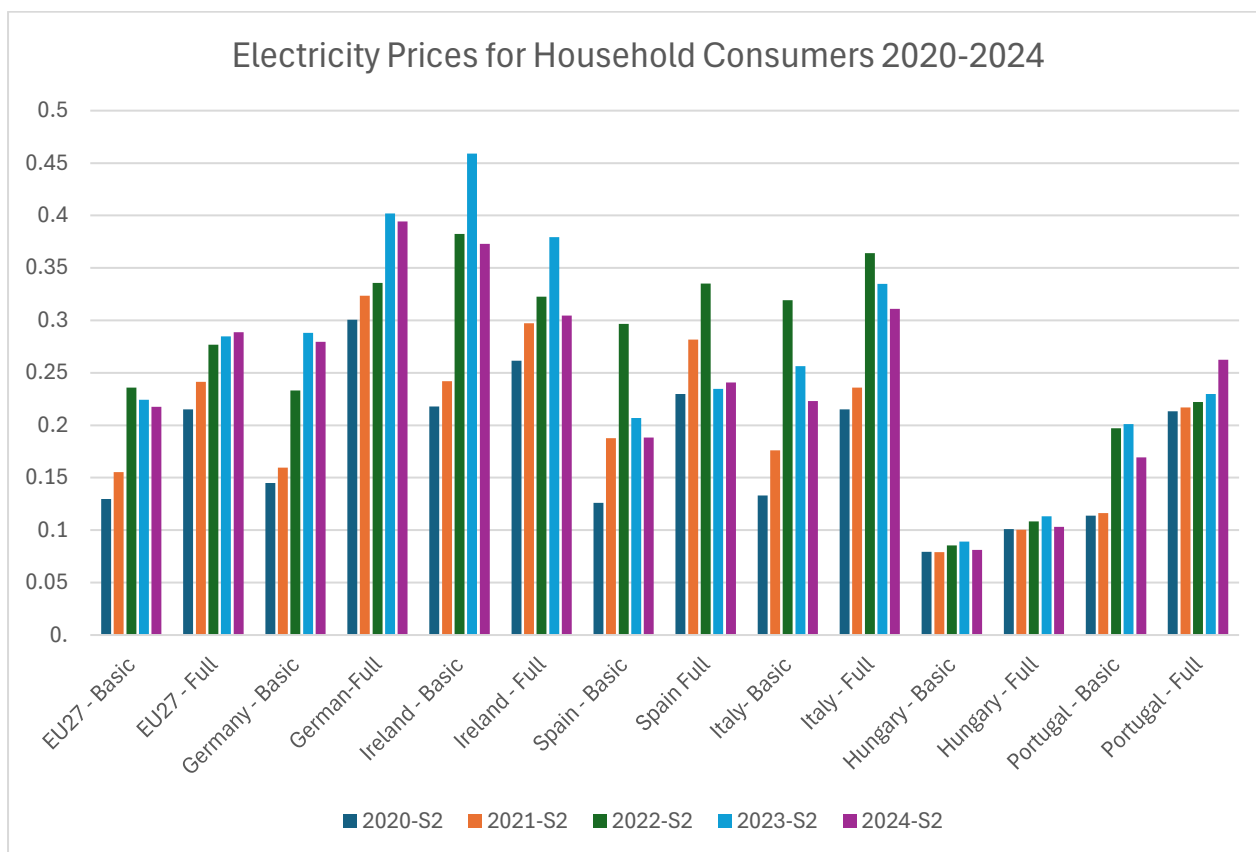


Figure 20 - Electricity Prices for Household Consumers 2020-2024

The share of population living in private households who are unable to afford to keep the home adequately warm, regardless of whether the household actually needs to keep it adequately warm is called the 'Inability to keep home adequately warm' indicator.

Across the EU 10.6% of the population were unable to keep their home adequately warm. The highest shares of people unable to keep their homes adequately warm in Europe are in two FITTER countries - Spain and Portugal (Eurostat, 2025) (Table 4.2.7)

Table 27 - Inability to keep home adequately warm (2021-2024)

	2021 %	2022 %	2023 %	2024 %
Germany	3.3	6.7	8.2	6.3
Spain	14.3	17.1	20.7	17.6
Hungary	5.4	4.6	7.1	6.1
Ireland	3.4	6.8	7.2	4.9

Italy	8.1	8.8	9.5	8.6
Portugal	16.4	17.5	20.8	15.7

(Eurostat)

4.3 Housing

The built environment provides essential services and exists within and is dependent on the natural environment. Using systems inquiry, we can identify and understand interdependent systems and deliver more effective interventions which lead to improved outcomes for people and planet. We can rethink our relationship to the built environment and design infrastructure which is

- Powered by renewable energy and designed to enhance the natural world
- Regenerative – part of a circular economy that eliminates waste
- Supportive of people and planet – meeting people’s needs and reflective of the values of sustainability, equity and wellbeing

As part of this, a FITTER focus is on envelope efficiency upgrades for residential buildings so that they can be heated and cooled in sustainable ways.

4.3.1 The Goal of Adequate Housing for All

A rights-based approach to housing acknowledges that housing is far more than bricks and mortar and actively addresses inequality in housing to enable the provision of adequate housing for all.

As outlined in chapter 2 the right to adequate housing is recognised in international human rights instruments, with the dimensions of adequate housing detailed in the ICESCR - access, affordability, security, cultural adequacy, habitability, and location/environment.

Adequate housing is a key precondition for human dignity - providing shelter, security, and the basis for participation in society and the achievement of people’s full potential. The provision of adequate housing for all leads to a more just, inclusive and sustainable society.

Situating the right to adequate housing within planetary boundaries entails envisioning housing systems that are decarbonised and resource-efficient, powered by renewable energy, and designed to be regenerative — reducing ecological harm while actively supporting human dignity, social inclusion, and planetary health (Ermgassen et al., 2022; Horn et al., 2025).

4.3.2 Unsustainability of high-demand, high material use housing

Energy consumption in European buildings is very high, accounting for 44.2% of total final energy use. This total ‘includes all different energy carriers used for the operation of buildings, like fossil fuels (e.g., onsite combustion of natural gas, oil or solid fuels for space heating) and/or electricity for all building operations’ (Balaras et al, 2024: 2). Direct emissions from EU-27 buildings generally account for about a tenth of total emissions. The operation of buildings has additional impacts due to the indirect emissions associated with electricity generation (Balaras et al, 2024: 2).

Buildings in the EU account for 36% of energy related GHG emissions. Currently ‘about 35 per cent of the EU’s buildings are over 50 years old and almost 75 per cent of the building stock is energy inefficient. At the same time, the average annual energy renovation rate is only about 1 per cent’ (Environment Ireland, 2024).

Between 2005 and 2022 GHG emissions from EU buildings fell by 34%. This fall is driven by ‘higher energy efficiency standards for new buildings, efficiency improvements in existing buildings, measures to decarbonise the electricity sector and heating systems, and warmer temperatures’ (EEA, 2024).

4.3.3 Housing Cost Overburden

The percentage of the population who live in households where total housing costs (net of housing allowances) represent more than 40% of disposable income (net of housing allowances) is the housing cost overburden rate.

The housing cost comprises monthly expenses incurred while living in a dwelling. It includes

- Utility costs (water, electricity, gas, heating)
- Structural insurance, mandatory services and charges
- Regular maintenance and repairs
- Taxes
- Mortgage interest payments net of any tax relief and gross of housing benefits (for homeowners)
- Rental payments gross of housing benefits (for tenants) ([Eurostat, ND](#))

Factors which contribute to unaffordable housing costs and increase in the number of unhoused people include:

- Rental and property prices outpacing income levels
- Inadequate provision of social housing
- Inadequate levels of construction of new housing stock
- Inadequate levels of renovation of existing stock
- Inadequate investment in social infrastructure

The 2024 Joint Employment Report found that housing costs were deemed to be a ‘financial burden’ by almost half of EU households in 2023. The housing cost overburden rate across the EU was 8.8% in 2023 but it was much higher for people already experiencing poverty risks – 33.5% ([EC, 2024a](#): 19; [Eurostat 2024](#)). The highest housing cost overburden rates in the EU ‘were in Greece (28.5%), Denmark (15.4%) and Germany (13.0%), the lowest in Cyprus (2.6%), Slovenia (3.7%) and Croatia (4.0%).’ ([Eurostat 2024](#)). Renting difficulties ‘during the last 12 months were reported by more than a quarter of people aged 30-54 experiencing poverty or social exclusion risks’ ([EC, 2024a](#): 19).

4.3.4 Housing - Unsustainable System Dynamics

[The EU Renovation Wave strategy](#) ‘sets out measures across the whole renovation chain that aim to increase the rate and depth of renovations.’ Measures include:

- strengthening information, legal certainty and incentives for public and private owners and tenants to undertake renovation:
- ensuring adequate and well-targeted funding
- increasing technical assistance to regional and local actors
- promoting renovation interventions for smart buildings and digital friendly renovations
- ensuring that the construction sector is fit to drive sustainable renovations and uses sustainable materials
- using renovation as a lever to address energy poverty
- promoting the decarbonisation of heating and cooling ([EC, NDb](#))

A shallow retrofit is a small scale change to enable energy saving. These can include measures such as:

- Insulation around pipes
- Upgrading the boiler
- New windows or doors
- Attic Insulation
- Wall insulation – includes cavity, internal and external wall insulation
- Heat pump

- Solar thermal
- Heating controls
- Solar PV

Ideally single measures should be taken as part of a long-term plan for a deep retrofit. If measures are taken in isolation, and in the wrong order, they may have to be redone at a later date to achieve further energy reductions. That said, small scale energy saving measures in rental properties can be very beneficial in reducing heating costs for tenants in the short term. Arguably, for tenants in particular, small-scale measures (that may have to be redone as part of a deep retrofit) are better than no measures, since landlords have limited incentives to do a deep retrofit at present.

A deep retrofit is a planned process which considers all aspects of the house (fabric, ventilation, heating and cooling systems) and constructs them so that they work in an integrated manner and CO₂ emissions are significantly reduced.

A deep retrofit generally entails three steps:

1. Building envelope improvements - reducing the amount of energy required to heat homes by adding insulation, reducing air leakage and upgrading windows and doors
2. Switch off fossil fuelled equipment – install of new heating technologies from a renewable source. Mechanical ventilation installed to ensure good indoor air quality.
3. If appropriate adding a renewable energy system such as solar PV panels to meet remaining energy needs. Storage batteries can also be included to store excess PV electricity for later use.

The benefits of a deep retrofit include:

- Greater comfort
- Improved indoor air quality
- Reduced health risks (good insulation protects inhabitants against extreme weather events, good ventilation against mold etc.)
- Lower operating costs
- Higher resale value
- Greater building durability
- Buyers can access ‘green’ mortgages offered with lower interest rates by lending institutions
- Switch to onsite energy production can free up electricity supply to replace fossil fuels elsewhere

Barriers to retrofit by tenure type

Homeowners

At the level of individual domestic dwelling owners there are capability, opportunity and motivation barriers to undertaking a deep retrofit of their property.

These barriers appear at various stages across the life cycle of a deep retrofit.

Table 28 - Barriers for domestic dwelling owners (SEAI, 2023: 19)

Barrier category	Barrier	Trigger	Research	Decide	Set up	Install	Use
Capability Barriers	Lack of awareness of what retrofits are/benefits	x	x	X			
	The number of options and decisions involved in the retrofitting process			X			
	Lack of financial literacy		X	x			

Opportunity Barriers	High costs of retrofitting	x	x	X			
	Split incentives to retrofit between homeowner landlord and tenants			X			
	Lack of skilled installers				x	X	
	Lack of social opportunity due to lack of visibility of retrofits	x	X				
Motivation barriers	Friction costs and hassle	x	x	x	x	x	X
	Moderate perceived benefits of retrofitting	x	x	X			
	Risk aversion – perceived risks related to financing the retrofit; finding competent installers etc			x			
	Low consumer trust towards messaging financing options and installer quality	x	x	x	x	x	x

Tenants

Although tenants and people living with family members may have the ability to install some energy saving measures in the home (e.g. energy efficient appliances and light bulbs), they cannot undertake a deep retrofit or single retrofit measure of the property they live in. When a domestic dwelling undergoes a deep retrofit tenants are at risk of rental price raise and renoeviction – eviction due to the need for major renovations, the aim to sell the property or having to leave a property due to a rent increase after the retrofit.

Unhoused people

Unhoused people have no access to a steady domestic dwelling.

Nomadic people

Nomadic people who live in mobile homes (e.g. Irish travellers) also have no access to retrofitting supports. There is limited capacity to retrofit existing mobile home due to space and structural limitations (e.g. the roof would not support solar panels, limited space on halting sites for solar panels, insulation and weatherproofing cannot be sufficiently applied to the mobile home, most Travellers do not own the land the mobile unit occupies). Unlike houses there is not scheme to rate the energy efficiency of mobile homes and newer mobile homes (since 2015) do 'not necessarily have energy efficient improvements in the home fabric and structures. They also did not contain appliances which would make the mobile any more efficient than the older surveyed homes' ([Ehimen et al, 2022: 13](#)).

Barriers to Travellers acquiring energy efficient mobile homes and integrating renewable energy systems (RES) on halting sites include:

- High upfront cost of energy efficient mobile homes (with starting prices at approximately 60000 euro)

- Low income
- High unemployment
- Financial exclusion (unable to get loans from banks)
- No rating scheme for mobile homes (purchasers don't have enough information about the energy efficiency of the unit)
- Mobile home roof may not be able to support solar panels
- Being ignored by RES marketers and manufacturers. Limited or no service or product-offering, focused on use in mobiles
- Lack of familiarity with RES operation, use and maintenance
- Limited space for ground-mounted solar or other RE systems
- Traveller owners of mobile homes do not usually own the land they reside on

There can be hesitancy to add RES in case there is a need to move the mobile home in the future ([Ehimen et al, 2022: 15-19](#)).

Although accessible and targeted funding is a central principle of the Renovation Wave national schemes still largely entail individualised risk and responsibility.

Individualised Responsibility and Risk in the Context of High Housing Costs

A key element of current retrofit schemes is that the onus to pay for retrofit works largely remains with individual property owners.

Where the owner is unable to pay the high upfront costs, and unable to access suitable financing, they are excluded from the energy savings and health benefits of climate resilient housing.

Private landlords, who are often primarily driven by perceived return on investment, have limited incentives to retrofit – this is known as the split incentive. As tenants pay the energy bills, and the return on investment in retrofit is perceived to be quite low, many landlords do not undertake retrofit.

Housing Market - Boom-Bust Cycle

During the 20th century the rental sector in many western European countries experienced long term decline. Post-war housing policies such as investment in social housing and supports for homeownership, followed by the neoliberal promotion of homeownership since the 1980s, increased homeownership rates, culminating in the global housing bubble of the 2000s. However, since 2008 access to homeownership has been undermined and a resurgence of renting has been triggered ([Byrne, 2019](#)).

Financialization can be defined as 'the increasing role of financial motives, financial markets, financial actors and financial institutions in the operation of the domestic and international economies' ([Epstein 2005](#) cited in [Epstein, 2015: 5](#)). The problems associated with financialization include 'increases in economic instability, increased short term orientation, reduced long-term investment in equipment, innovation, infrastructure and human capital and an increase in income and wealth inequality' ([Epstein, 2015: 6](#)).

In the housing market, one feature of financialization was that mortgage markets grew considerably during the late 1990s and early 2000s. This was facilitated largely through a weakening of credit standards (e.g. higher 'loan to value' and 'loan to income' ratios and longer maturities). After the crash of 2008 distressed or non-performing loans linked to real estate have been sold at significant discounts, particularly to hedge funds, private equity firms and Real Estate Investment Trusts ([Byrne, 2019: 3](#); [Farha, 2017: 9](#)).

Housing markets have long been understood as cyclical and financialization has arguably intensified the cyclical nature of property markets. As Byrne (2019) notes

Abundant credit heightens the 'up' phase of housing cycles, pushing house prices beyond wages and income and thus creating a growing affordability gap. In the 'down' phase of the cycle, house prices decline just as rapidly as they rose, but financialization creates a debt overhang which causes credit to 'crunch' and mortgage markets to freeze up [...]. As such, declining house prices do not ensure easier access to

homeownership once boom turns to bust, but rather further entrench the dynamics which exclude potential first-time buyers ([2019: 18](#))

In the post-war era, significant State intervention in the form of house building by the State, State provision of mortgage finance (especially to low and middle-income households), State subsidisation of mortgage providers and tax reliefs for home homeowners led to an expansion of home ownership.

With the onset of neoliberalism these State supports were largely withdrawn. Home ownership became accessible via the private market only. This has led to a concentration of property ownership by wealthier households who can 'draw on existing housing wealth and may acquire additional properties as landlords' ([Forrest and Hirayama, 2014](#); [Roland and Kadi, 2017](#) cited in [Byrne, 2015](#): 6)

A key element of enabling a just transition is to prioritise affordable, accessible and climate resilient housing for all. However, this is difficult, given the boom-bust cyclical nature of housing markets in high-income economies, and particularly in major cities.

Using London as a case study, [Dianati and Ryan-Collins \(2023\)](#) argue that two reinforcing feedback loops – the *Investment Loop* and *Housing-Finance Loops* – together 'establish a self-reinforcing engine of growth in house prices on the one hand and housing credit on the other' (2023: 25). This leads to the excessive accumulation of mortgage debt which – after some delay – feeds back to 'reverse the direction of the abovementioned reinforcing loops towards an eventual downturn' (2023: 25). They explain this using two balancing feedback loops – *Debt Burden* and *Leverage loops* – which only take effect after significant time delays. These time delays make the housing system in high income economies prone to recurring boom-and-bust cycles.

Although the feedback loops outlined below are based on the London housing market, the authors posit that they likely apply to many urban areas in high income economies. Further research would be required to assess the extent of their applicability to the case study countries – Portugal, Ireland and Germany.

Investment Loop

House prices are influenced by utility-based demand and investment demand. Drawing on existing research on the 'self-fulfilling' nature of house price appreciation, [Dianati and Ryan-Collins \(2023\)](#) demonstrate that in high income economies, as house prices rise, an expectation of an ongoing rise in future house prices is created. This instigates an increase in expected housing return on investment, which then leads to more investment in housing as a commodity, reinforcing an increase in house prices (R1 Investment Loop).

The rise in house prices results in more people being priced out of buying a house and either moving to, or staying in, the private rental sector. This increases both the privately rented tenure share and the cost of private rents. That rise in private rents further stimulates the expected housing return on investment which also reinforces an increase in house prices (R2 Priced Out Loop).

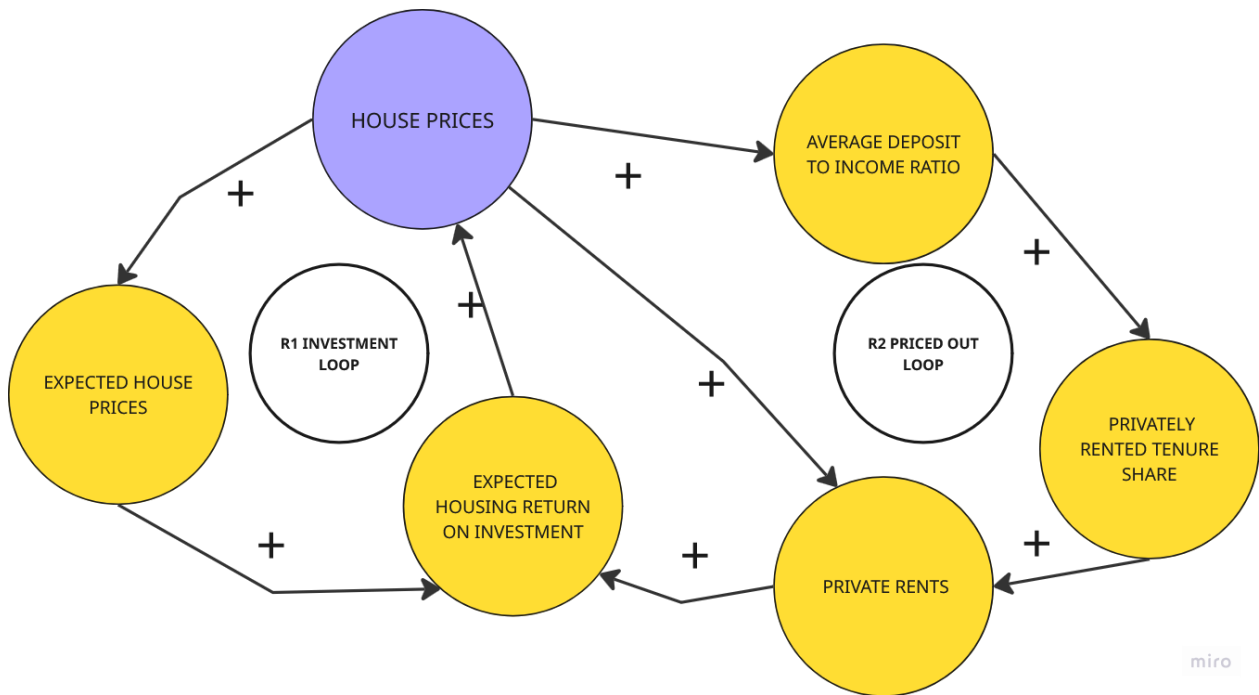


Figure 21 - Investment loop (adapted from [Dianati and Ryan-Collins, 2023](#))

Housing Finance Loops

[Dianati and Ryan-Collins \(2023\)](#) argue that as house prices rise more people need a mortgage to buy a house and the demand for new mortgages increases. An increase in new mortgage advances further inflates house prices which boosts demand and reinforces demand for mortgage advances (R3 Credit Creates its own Demand).

In dualist housing systems (dominant private market, public housing a tenure type of last resort) - governments are forced to support homeownership for marginal buyers. This further exacerbating boom-bust trends in these markets.

As house prices rise the value of the housing stock also rises. Homeowners have more collateral to raise credit against for re-mortgages or new home loans. This is the case in flexible mortgage markets in particular. This can lead to a further increase in house prices – as people invest in second houses (R4 Collateral loop).



As house prices increase, more mortgage advances are issued, and this leads to a growing stock of outstanding mortgage balances. There is a delay on this, as stock builds up gradually over time and continues to grow even after a peak of new mortgage advances. Substantially decreasing this stock of outstanding mortgage balances takes place over a long period of time.

The debt service coverage ratio is a measure of the degree to which the income of the property owner can cover their debt obligations. Among other factors, an increase in the mortgage interest rate can lead to a decrease in homeowners ability to cover their mortgage repayments. The debt service ratio (DSR) is ‘the amount of mortgage repayment (consisting of principal and interest) divided by household disposable income and is a measure of the burden of debt’ ([Dianati and Ryan-Collins, 2023: 14](#)). A high DSR is an early warning sign of mortgage defaults. Highly leveraged investors can ‘engage in fire sales of property with distressed debt’ ([Dianati and Ryan-Collins, 2023: 15](#)) leading to a collapse in house prices. A high number of mortgage defaults also contributes to a collapse in house prices, a debt overhang (lending institutions cannot fund many additional mortgages), and a shortage of new mortgage advances.

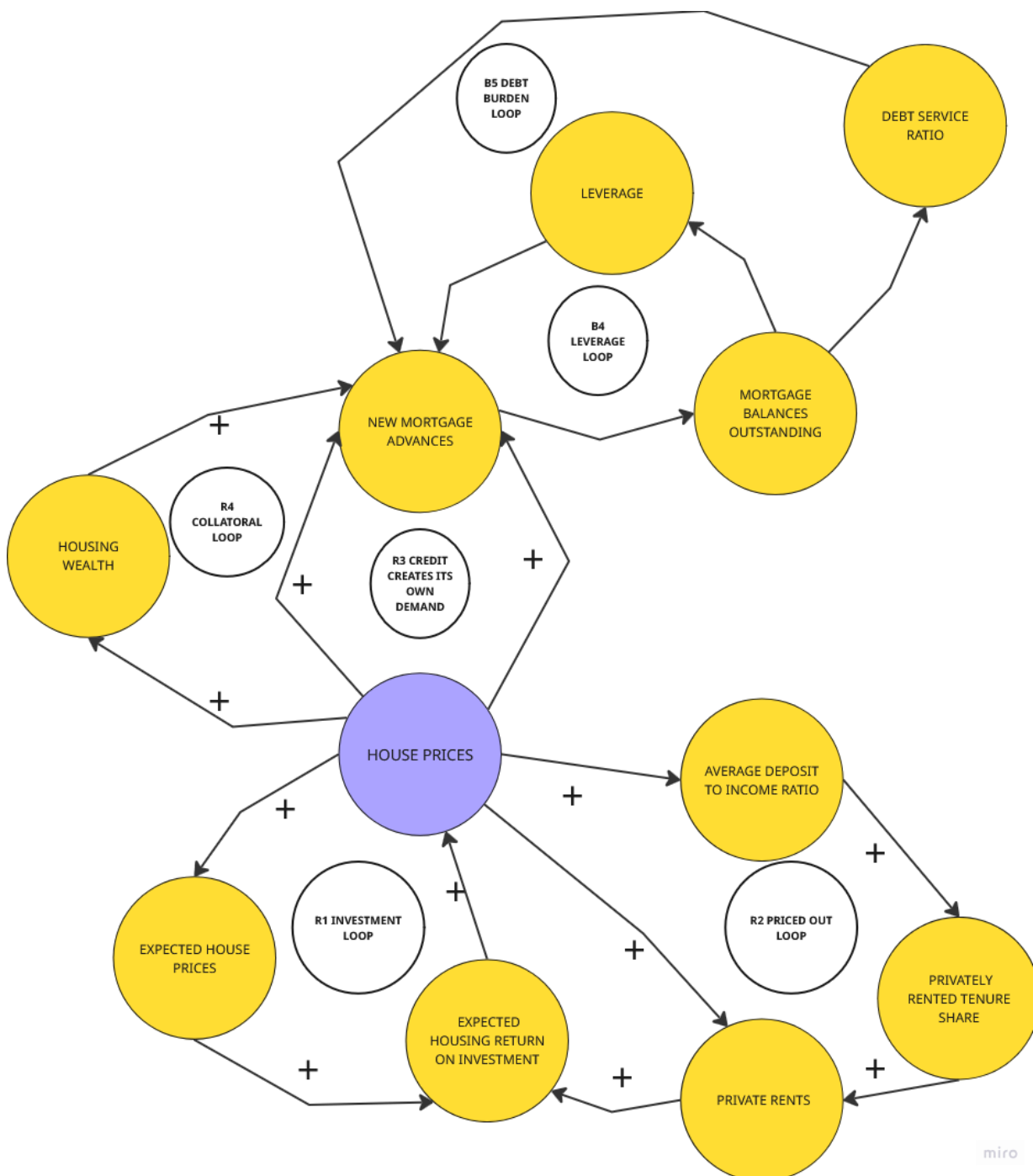


Figure 23 - Leverage and Debt Burden Loops B3 and B4 (adapted from [Dianati and Ryan-Collins, 2023](#))

Rising Housing Costs in the EU

Although there is considerable variation among countries, across the EU house prices and rents have increased since 2010.

From 2010 until Q1 2025, in the EU, rents increased by 27.8% and house prices by 57.9%. In the EU, Q1 2025, when compared with the same period in 2024, rents increased by 3.2% and house prices increased by 5.7% ([Eurostat, 2025b](#))

In Portugal and Ireland, housing cost overburden is experienced by renters at market price at a much higher rate than those with other tenure status. In Germany a significant proportion of owners with a mortgage or loan and renters at market price are experiencing housing cost overburden.

Table 29 - Housing Cost Overburden Rate

Type	Ireland (%)	Germany (%)	Portugal (%)	EU
Owner with mortgage or loan	1.3	14.5	3.8	5
Owner no outstanding mortgage or loan	1	6.8	2	4.4
Tenant, rent at market price	18.3	13.8	30.3	19.2
Tenant rent at reduced price or free	7.4	8.8	3.6	10

(Eurostat, 2025c)

The ‘priced out of home ownership’ dynamic is evident in Ireland where the private rental sector has grown from 8% in 1991 to 18% in 2023. This trend has a strong age and geographical dimension. The rise in private renting is more pronounced among younger households and urban areas have had a much steeper fall in home ownership relative to rural areas (NESC, 2023: 4-5).

Although housing unaffordability in the private rental sector is experienced across the EU some countries have had much higher rental inflation since 2010. Ireland and Hungary are notable in this regard with rental inflation at 115% and 124% respectively.

Table 30 - Rental Inflation Rate

	2010	2025-Q1	% Change
EU27_2020	93.19	119.14	27.85%
DE	93.8	116.9333	24.66%
ES	100.45	112.5333	12.03%
HU	90.55	202.5267	123.66%
IE	79.8	171.4667	114.87%
IT	95.2	111	16.60%
PT	90.86	134.04	47.52%

(Eurostat, 2025b)

Many renters in Ireland express a strong preference for home ownership. A telephone survey of renters conducted by the Irish Government Economic and Evaluation service found that 86.5% of respondents agreed that ‘owning a home makes more sense because you are protected against rent increases and owning is a good investment’ (Corrigan et al., 2019: 7). Other statements classified as ‘major reasons’ by a majority of respondents for a home ownership preference include:

- Owning gives you control over what you do with your living space, like renovations (78.8%)
- Owners have a physical structure where they and their family can feel safe (78%)
- Paying rent is not a good investment (75.5%)
- Owning means have a good place to raise children and provide them with a good education (73.3%)

Among respondents ‘who indicated that they expect to be renting privately in the medium to long-term, 11.6% indicated a preference for renting while 85.1% indicated that they would prefer to own a home but felt that this would be unlikely’ (Corrigan et al, 2019: 13). For those who judged themselves likely to be long-term renters the majority ‘did so on the basis that they would be unable to afford home-ownership’ (Corrigan et al, 2019: 18). Similarly, a survey conducted by Threshold of their former clients noted an increase from 49% in 2020 to 64% in 2022 of respondents who were renting because they were unable to buy a home. Only 14% of respondents said they were renting by choice (Threshold, 2022: 10).

A study of the rental housing market in the Lisbon metropolitan area (Área Metropolitana de Lisboa, AML) found that among surveyed renters, only about 5 % believe they will ever be able to buy their own home. Moreover, a vast majority (80%) said they do not expect to inherit a house. The survey also shows that many renters are in precarious situations: about 12 % live in shared housing (e.g. renting a room or sharing with others) and 37 % of these share-housing situations correspond to informal rental arrangements. The high “rent burden” (i.e. large share of income devoted to rent) is a major constraint (CES, 2025).

Another report focused on youth in Portugal indicates that in 2020, about two in three young renters (i.e. ~ 65 %) said they would prefer to own a home — but stated they lacked the financial conditions to do so (savings, down payment, credit access). Other reasons cited included seeing the purchase as a short-term investment not suited to their plans, concern about indebteding themselves, or preferring to rent due to flexibility (Idealista, 2023).

The first study, conducted in Lisbon in 2025, focuses on the metropolitan area but provides a valuable reflection of the broader Portuguese housing reality, as the capital’s market trends often mirror national patterns. Meanwhile, the 2022 study on young people across Portugal highlights similar challenges, showing that most young renters aspire to own a home but are prevented from doing so mainly due to financial constraints such as low income, lack of savings, and limited access to credit.

No similar studies of renter preferences were found for Germany.

Addressing systemic issues of financialization and commodification of housing

In 2017 the UN Special Rapporteur on adequate housing as a component of the right to an adequate standard of living, and on the right to non-discrimination in this context, defined the ‘financialization of housing’ as incorporating:

- the structural changes in housing and financial markets and global investment whereby housing is treated as a commodity, a means of accumulating wealth and often as security for financial instruments that are traded and sold on global markets
- the way capital investment in housing increasingly disconnects housing from its social function of providing a place to live in security and dignity and hence undermines the realization of housing as a human right
- the way housing and financial markets are oblivious to people and communities, and the role housing plays in their well-being (Farha, 2017: 3)

The financialisation of housing has led to money flowing out of communities to remote investors; incentives for foreclosure, particularly in desirable urban settings, rather than modifications of loan agreements to avoid evictions; and foreign and domestic investment in short-term rental properties which contribute to ‘escalating prices of housing and changes to the make-up of neighbourhoods, without creating affordable housing or other benefits for the local population’ (Farha, 2017: 9).

The UN Special Rapporteur noted that States are not addressing systemic issues of financialization and commodification of housing by reclaiming governance of housing systems from global credit markets. The Special Rapporteur recommended:

- a) The design of a strategy for engaging financial regulatory bodies in the realization of the goal of adequate housing for all by 2030
- b) Strategies to achieve SDG target 11.1 should ‘include a full range of taxation, regulatory and planning measures in order to re-establish housing as a social good, promote an inclusive housing system and prevent speculation and excessive accumulation of wealth;
- c) Trade and investment treaties should recognize the paramountcy of human rights, including the right to housing, and ensure that States are fully empowered to regulate private investment so as to ensure the realization of the right to housing;
- d) Business and human rights guidelines should, on a priority basis, be developed specifically for financial actors operating in the housing system;
- e) States should review all laws and policies related to foreclosure, indebtedness and housing, to ensure consistency with the right to adequate housing, including the obligation to prevent any eviction resulting in homelessness;
- f) States must ensure that courts, tribunals and human rights institutions recognize and apply the paramountcy of human rights and interpret and apply domestic laws and policies related to housing and housing finance consistently with the right to adequate housing;
- g) International, regional and national human rights bodies should devote more attention to the issue of financialization and clarify for States, through constructive dialogue during periodic reviews and in consideration of individual cases, their obligations in relation to the financialization of housing.’
([Farha, 2017: 21-22](#))

Climate resilient housing in the context of the financialization of housing

Given the current structure of most retrofit supports which require significant upfront investment and are targeted towards homeowners – low-income homeowners, and tenants in the private rental sector are unable to undertake retrofit works. People with high mortgage debt (homeowners), those unable to leave the private rental sector (due to high house and rent prices) and those being forced back into the private rental sector after defaulting on a mortgage are negatively impacted.

The unaffordability of deep retrofits for homeowners and the lack of incentives for landlords to retrofit properties can lead to higher energy costs and, in some cases, higher health costs. It also risks a potential raise in CO2 costs for energy.

The compounding affordability effects of not being able to access retrofits is shown in the reinforcing feedback loop below.

The low uptake of retrofit leads to worse housing conditions. This can lead to worse health conditions and higher healthcare costs which further exacerbates economic inequalities. Worse housing conditions also lead to higher energy use and costs, and this also exacerbates economic inequalities. For middle and low-income homeowners increased energy and health costs means they cannot afford to access retrofit schemes. For middle- and low-income tenants this means they cannot afford to rent high quality housing. This contributes to the lack of incentives for landlords to retrofit and contributes to the low uptake of retrofit schemes.

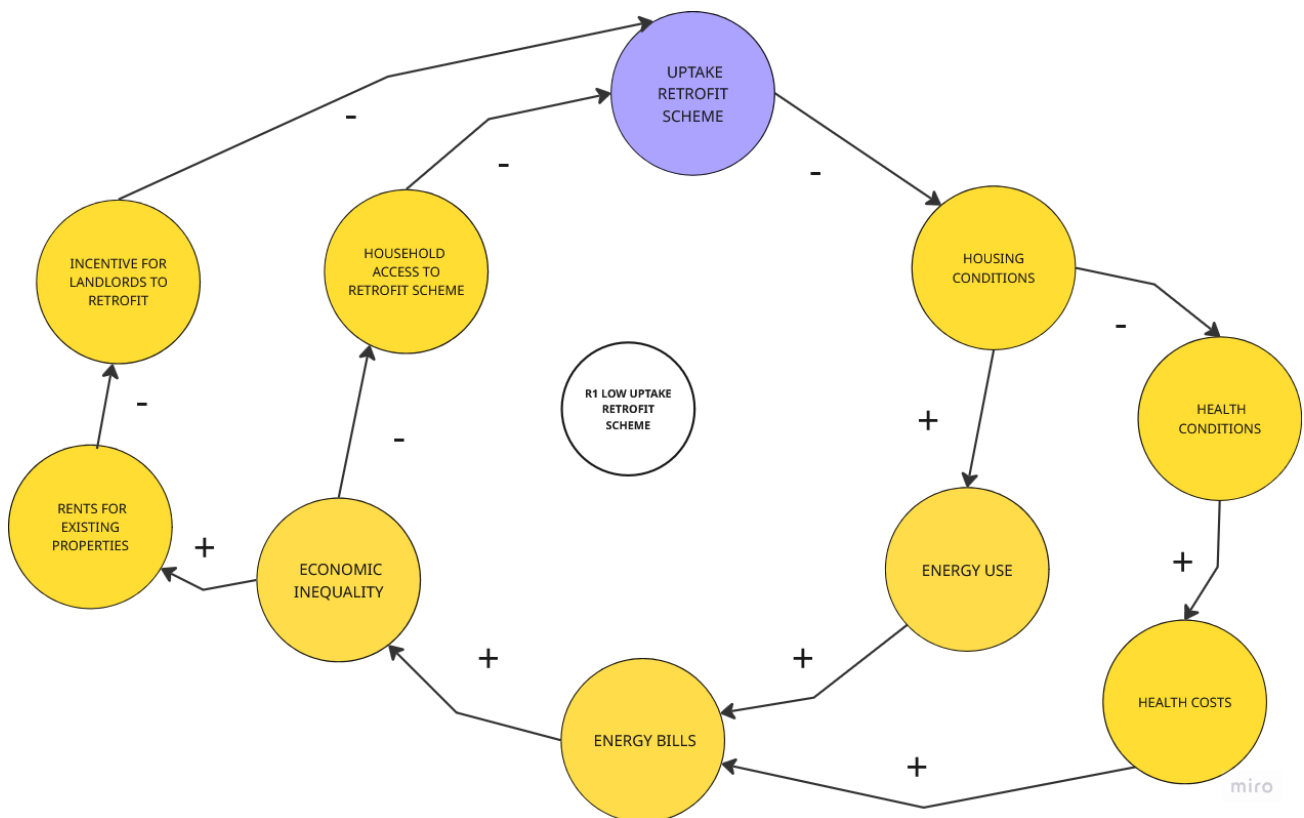


Figure 24 - Compounding Affordability Effects (Adapted from [Po et al, 2021](#))

Another significant barrier to retrofitting at scale is the underdeveloped market for retrofitting. This is due to the relatively low demand for retrofit in some countries which leads to some retrofitting work being limited in scope (e.g. only insulation, only replacing windows or doors) or to full retrofits being limited to single homes driven by motivated individual owner-occupiers, rather than at scale across a neighbourhood or housing block.

We can see this in the Low Demand High Cost loop (R1), within which the relatively low demand has led to limited financing options which leaves the cost of retrofit high. Given this high cost the return on investment is understood to be too low which leads to a low uptake of retrofit work, which reinforces the lack of retrofit suppliers, and retrofit works continue to be individualised and/or limited in scope.

In mixed tenure housing blocks and multi-apartment buildings more generally (local authority units, owner-occupier units and private rental sector units) it can be difficult to achieve consensus on the need for, and cost distribution of, retrofitting a housing block. Each stakeholder has different incentives and different degrees of agency with regard to the retrofit of their home. When a mix of incentives and capability constraints coexist in one building and are misaligned there are multiple barriers to retrofit. Property tenure can also limit neighbourhood scale retrofit solutions (e.g. district heating) ([Po et al, 2021](#)).

As noted above private landlords, who are often primarily driven by perceived return on investment, have no real incentive to retrofit. Even if the tenant is keen to undertake retrofit works, they have no agency to do so (R2 Low demand landlord split incentive). In multi occupier buildings, the unwillingness of landlords, and/or the unwillingness or inability of some owner occupiers to invest in retrofit works, can result in the whole block being unable to proceed with retrofit works. This also contributes to the low uptake of housing retrofit works (R3 Low Demand Mixed Tenure).

Another dynamic that requires further investigation is ‘renoviction’.

Other jurisdictions have greater tenant protections but rent increases after a renovation can render the property unaffordable for long term tenants with historically low rents or tenants already at the limit of their financial capabilities, particularly those on fixed or low incomes such as pensioners, disabled people and lone parent families.

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are decided in the landlords favour leading tenants to conclude that it is not worthwhile to refuse consent to the renovation. After renovation gradual rent increases are often introduced over a 2–5-year period. This is presented as a tenant-friendly principle allowing tenants to improve their financial situation to meet the rent increases, but it is particularly disadvantageous to those on fixed incomes. It can arguably work in the landlords favour as it leads to ‘prolonged displacement’ rather than a sudden increase in relocation statistics following renovation ([Baeten et al, 2017](#)).

A notable aspect of both large-scale renovation and neighbourhood displacement is that it is usually **stretched out over time**. This can lead to great anxiety and uncertainty for tenants over a long period of time, negatively impacting their health and wellbeing. As [Baeten et al \(2017\)](#) note this strategy of ‘temporal extension’ serves the displacement of tenants with historically low rents and their replacement with middle- and high-income earners in three ways. First the long time period obscures the displacement itself. During a long renovation process it is very difficult to identify the reasons behind tenants relocation, prior to, during and after the renovations. Second it allows for the removal of potential disagreements. Third ‘lengthy renovation projects exhaust tenants and produce the feeling that resistance is meaningless’ (2017: 646).

Further research is required to determine if, or the extent to which, this dynamic is applicable to the case study countries.

4.3.5 Stock Overview

Energy Consumption in the Residential Sector

In the EU in 2023, 62.5% of final energy consumption in the residential sector was for heating homes, followed by 15.1% for water heating.

Table 31 - Energy consumption in EU households, 2023 ([Eurostat, 2025a](#))

	Ireland (%)	Germany (%)	Portugal (%)	EU
Space heating	56.2	65.8	32.2	62.5
Water heating	21.5	16.6	16.8	15.1
Lighting and Electrical Appliances	19	9.8	20.4	14.5
Cooking	2.4	6.6	1.1	6.5
Other end use	1	0.9	0.0	0.8
Space cooling	0.0	0.2	1.1	0.6
Total				100

([Eurostat, 2025a](#))

In the case study countries there are also differences in the profile of energy consumption by fuel in the residential sector (2023).

Table 32 - Energy consumption by fuel type in the residential sector (2023)

Type	Ireland	Germany	Portugal	EU
Heating oil	41.72	18.8	11.05	10.3

Electricity	27.33	20.6	41.52	25.9
Natural gas	18.24	37.27	8.33	29.5
Renewables and biofuels	4.35	16.32	39.09	23.5
Solid fossil fuels	2.87	0.59	0.00	2.2
Derived heat?	0.00	6.42	0.00	8.5

(Eurostat 2025a)

Household types differ in terms of their heating requirements. For example, in Ireland ‘there are 45% more detached houses than semi-detached houses, but the total heating demand in detached houses is 129% higher than in semi-detached houses. A significant factor in this is the larger average floor area for detached homes compared to other properties’ (SEAI, 2022: 33-34).

Percentage breakdown of fuel used for space heating and hot water in the residential sector in Ireland, by building type

Table 33 - Fuel used for Space Heating and Hot Water, Ireland (2022)

	Electricity %	Natural gas %	Oil %	Solid Fuel %
Detached house	2	10	67	21
Semi-detached house	3	49	39	9
Terraced house	6	55	26	13
Apartment	38	47	11	4

(SEAI, 2022)

In Portugal, household heating behaviour and infrastructure vary considerably across dwelling types, though published data with full disaggregation by building category could not be sourced. National surveys indicate that 18.4% of dwellings have no heating equipment at all, and many rely on decentralized, low-efficiency systems—such as electric heaters and fireplaces—rather than centralized heating installations (Corrigan et al. 2019)

Among homes that do have heating systems, approximately 1.5 million households use gas boilers, and an increasing—but still modest—number rely on heat pumps, with around 277,000 units installed by 2022 (Corrigan et al. 2019). Because detached dwellings typically have larger envelope surfaces (roofs, walls) and more rooms, it is plausible that detached houses in Portugal have higher total heating demand than semi-detached houses or flats—despite being fewer in number. However, without disaggregated data by dwelling type, this remains a reasoned hypothesis rather than a confirmed fact.

Overall, the heating mix in Portugal is fragmented and decentralized compared with northern Europe: most homes rely on localized electric or biomass heaters rather than centralized systems.

Together, these data highlight a heterogeneous national energy landscape, where moderate climatic conditions, building typologies, and infrastructural diversity shape both heating demand and fuel choice across Portuguese households.

No national level data was found on fuels used by building type in Germany.

Energy Efficiency Ratings of Residential Dwellings

There are different energy efficiency ratings operative in the case study countries.

In Ireland the Building Energy Rating (BER) compares the energy performance of different dwellings on a like-for-like basis. It can be used as an indicator of the energy efficiency of a home and the running costs and carbon emissions associated with heating the home to a comfortable level. A-rated homes are the most energy-efficient and tend to have the lowest energy bills. G-rated homes are the least energy-efficient, generally require a lot of energy to heat, and have the highest energy bills.

The Programme for Government (PfG) commits to retrofitting 500,000 homes to a B2/Cost Optimal Equivalent BER standard by 2030. This is approximately 32% of the housing stock. However national level BER rating data weighted using Census of Population 2016 estimates that only **14%** of homes in Ireland have a BER rating of B2 or more, 39% are rated C3-B3, and 45% have a BER rating of D1 or lower ([CSO, 2023](#)).

Table 34 - Energy Efficiency Ratings - Weighted to National Level, Ireland (2009-2023)

Table 16 BERs Weighted to National Level (2009-2023)														
													% of row	
Dwelling Type	Energy Rating												Total	
	A	B1	B2	B3	C1	C2	C3	D1	D2	E1	E2	F		G
Apartment	13	2	5	9	11	10	10	10	9	6	4	4	7	166,823
Detached house	7	2	4	8	10	11	11	11	10	6	5	5	10	685,615
Semi-detached house	9	1	2	6	11	13	13	13	11	6	5	5	5	441,842
Terraced house	8	1	2	7	10	11	10	11	11	7	6	7	8	259,857
Total	9	2	3	7	10	11	11	11	10	6	5	5	8	1,554,137

([CSO, 2023](#)).

The rate for energy-efficient refurbishment in the German building stock was 0.69% in 2024, whilst it needs to be at 1,9% to reach climate targets (Bundesverband energieeffiziente Gebäudehülle, 2025).

In Portugal, of all houses that have been allocated an energy efficiency rating, 27.2% have a B rating or higher. There is no breakdown available by housing type.

Table 35 - Energy Efficiency Ratings Portugal (2014-2025) ([SCE, 2025](#))

Rating	% Residential Dwelling – 2014-2025 certifications	% Residential Dwelling – 2025 certification
A+	5.5	12.8
A	12.4	25.5
B	7.3	5.9
B-	8.1	6
C	24.4	18.5

D	21.3	15.6
E	13.1	9.8
F	7.8	5.9

5. The FITTER Platform: Overview and Structure

5.1 Introduction

The FITTER platform presents an innovative approach to supporting decision-making through policy appraisal, particularly through its Intelligent Predictive Decision Support System (IPDSS). By combining predictive modelling, quantitative risk assessment, systems thinking, stakeholder engagement, and gamified interfaces, it can support policy makers in making decisions regarding twin transition policies that promote eco-social justice and reduce structural inequalities. The platform is modular, scalable, and focused on transparency, interpretability, and inclusivity.

5.2 Methodological considerations and limitations

Before detailing the platform's functionalities, it is essential to acknowledge the methodological challenges inherent in applying machine learning (ML) and quantitative risk assessment to complex social systems:

Limits of predictive accuracy in social contexts and risks for policymaking:

ML models often struggle to predict outcomes in social settings shaped by complex, shifting, and context-sensitive factors. As [Buchholz and Grote \(2023\)](#) argue, ML models excel in domains with stable background conditions but fare poorly when applying to social systems. The Fragile Families Challenge ([Salganik et al., 2020](#)), a large-scale study that invited researchers to predict life outcomes for children using longitudinal social data, illustrated this limitation. Despite access to rich datasets, sophisticated algorithms, and rigorous validation methods, predictive accuracy remained low. More importantly, Salganik et al. warned that using such models in policymaking can be dangerous: inaccurate predictions may lead to misinformed decisions, misallocated resources, or unintended harm to vulnerable groups. These limitations must be explicitly acknowledged when deploying predictive tools like FITTER. Predictive modelling should serve as one input among many and must be complemented by qualitative insights, stakeholder engagement, and ethical safeguards. Platforms like FITTER, which aim to support inclusive and anticipatory governance, must be especially cautious in how they present and apply predictive outputs. Transparency about uncertainty, model limitations, and potential biases is essential to ensure that technology serves eco-social justice goals rather than undermining them.

Overreliance on Quantitative Risk Assessment (QRA):

FITTER's use of QRA assumes that social exclusion risks can be measured, ranked, and predicted with confidence. However, social systems – such as communities, institutions, and individual behaviour - are often unstable and reactive. This means that people may change how they behave simply because they know they're being assessed or because a policy or any unexpected event is introduced. In other words, these systems are non-stationary (they don't stay the same over time) and reflexive (they respond to predictions or interventions). As a result, patterns found in past data may not hold in the future, especially when new policies reshape the very conditions being modelled ([Northcott, 2020](#)). This makes it difficult to rely solely on past data to forecast future outcomes. This can be summarised saying that objectives and policy mechanisms, even perceptions of people facing the same objectives and policy mechanisms, change with the time.

How does FITTER seek to address these limitations?

- Module B of the FITTER platform empowers policymakers to actively redesign objectives and policy mechanisms in alignment with just transition requirements. It provides tools that enable them to revise and prioritize new objectives and mechanisms based on improved eco-social performance.
- FITTER adopts long-term objectives and policy mechanisms that are sufficiently general to ensure consistent public perception over time. The platform identifies and demonstrates the influencing factors and indicators both quantitatively and qualitatively. These are carefully selected and validated by social experts to ensure their stability and relevance as explanatory variables over time.

Data Limitations and Bias:

ML models rely on large, high-quality datasets to learn patterns and make predictions. In fields like social science, however, such datasets are often difficult to obtain. Data may be incomplete, outdated, inconsistently collected, or missing for certain groups, especially marginalised populations. For example, people with disabilities, ethnic minorities, or low-income communities may be underrepresented in official records or surveys. This lack of representation means that models trained on these datasets may not accurately reflect the experiences or needs of these groups. Moreover, training data may reflect existing social biases. For example, natural language processing systems have been shown to associate negative sentiment with terms related to race or disability ([Gebru, 2019](#); [Hutchinson et al., 2020](#)). Without careful bias mitigation, predictive models risk reinforcing structural inequalities.

How does FITTER seek to address these limitations?

- FITTER algorithms do not rely on natural language processing. Instead, they utilise the Naïve Bayes classifier and Bayesian Network analysis, both of which operate solely on structured data collected at the outset. This data is sampled according to Eurostat's socio-demographic proportions, ensuring representativeness. For instance, if 24% of the population is classified as disabled, then 24% of the dataset will consist of data from disabled individuals—preserving the population's structural integrity.
- These methods support both aggregate and targeted analysis. By filtering the dataset based on socio-demographic profiles, FITTER enables predictions for the entire population as a homogeneous group or for specific segments such as disabled individuals or older adults. This flexibility allows for nuanced insights while maintaining statistical robustness.

Opacity and Interpretability:

While FITTER incorporates Naïve Bayes classifier and Bayesian Networks methods, they can still function as “black boxes” where the internal logic of the model is difficult to understand or explain. This lack of transparency is problematic in policymaking, where decision-makers need to understand how conclusions are reached. As Buchholz and Grote (2023) note, the trade-off between predictive power and explanatory clarity remains a central challenge. This means that while they may produce accurate predictions, the reasoning behind those predictions is often difficult to trace or explain. For policymakers, this lack of transparency poses a serious challenge.

Evidence-based decision-making requires a clear understanding of how outputs were generated, as policymakers must be able to justify their decisions to stakeholders, including the public, oversight bodies, and other government departments. They need to know which variables influenced a prediction, how those variables interacted, and whether the model's assumptions align with real-world conditions. Without this clarity, it becomes difficult to assess the reliability of the model, identify potential biases, or adapt

policies based on new evidence. FITTER must navigate this tension carefully, ensuring that its outputs are not only technically sound but also understandable and actionable for decision-makers.

How does FITTER seek to address these limitations?

- FITTER prioritises interpretability and responsible use. The platform emphasises features such as visualisations and scenario-based explanations to make model outputs more accessible and understandable. It also provides guidance on the responsible use of these outputs, particularly in high-stakes contexts where decisions may impact structurally vulnerable populations. This approach supports transparent, accountable, and inclusive policymaking.
- A two-step analytical framework enhances traceability and clarity. First, a Naïve Bayes classifier identifies the key explanatory variables associated with each perceived hazard. Second, a Bayesian Network analysis simulates the effects of changes in hazard perception. Together, these methods ensure that the model's logic and outcomes are both transparent and easy to interpret.

Mismatch Between Risk and Impact Assessment:

Traditional impact assessments in policy contexts often involve qualitative analysis, stakeholder consultation, and consideration of broader social and environmental factors. In contrast, risk assessments focus on quantifying the likelihood and severity of specific outcomes. FITTER's reliance on QRA may not align with how policymakers typically evaluate social impacts.

How does FITTER seek to address these limitations?

- FITTER bridges the gap between risk and impact assessments by integrating qualitative components and offering guidance on how its outputs can complement conventional impact assessment frameworks. Moreover, one key output of the QRA conducted in Module A is a curated list of mitigation measures, which feeds into Module B.
- These mitigation measures have been developed through national co-creation labs using processes similar to traditional impact assessments.

Scalability and Replication Challenges:

Although FITTER is designed to be modular and scalable, applying it across different policy domains or Member States may be challenging. Social and infrastructural conditions vary widely, and models trained in one context may not perform well in another. This limits the generalisability of the platform's outputs ([Kyriazos & Poga, 2024](#)).

How does FITTER seek to address these limitations?

- FITTER will transparently communicate the boundaries of its applicability, ensuring that users understand the contextual limits of its insights and avoiding overstatements about their transferability.
- To enhance contextual relevance, FITTER gathers data from six countries across Europe (Ireland, Spain, Italy, Germany, Portugal and Hungary) spanning a broad West-to-East spectrum. This data is analysed on a country-by-country basis to identify significant differences in how citizens perceive risks associated with the twin transition under similar conditions.
- Where cross-national differences are identified, they will be explicitly incorporated into the development of predictive models. This approach ensures that FITTER's outputs are sensitive to regional variations and better suited for localised policy applications.

Gamification and Decision-Making:

Gamification elements can enhance user engagement and make the platform more accessible. However, there is a risk that these features may oversimplify complex policy decisions or distract from critical analysis. While interactive design can support onboarding and learning, it must not substitute for rigorous evaluation.

How does FITTER seek to address these limitations?

- FITTER ensures that gamified elements are used only complement (not replace) analytical depth and deliberative processes.

Nonlinear Complexity and Model Selection:

As Kyriazos and Poga (2024) emphasise, social phenomena often involve nonlinear relationships that traditional models overlook. In their view, ML models can capture these dynamics, but they introduce new challenges (such as overfitting, instability, and reduced interpretability).

How does FITTER seek to address these limitations?

- FITTER balances predictive power with transparency, particularly when modelling intricate interactions across socio-demographic groups. The platform employs two nonlinear methods (Naïve Bayes classifier and Bayesian Network analysis) which are well-suited to capturing these complex dynamics.
- FITTER contributes meaningfully to policy appraisal by fostering systems thinking and encouraging stakeholder engagement. However, it recognises the limitations of predictive modelling and promotes cautious use of its outputs.
- By openly acknowledging methodological constraints, integrating complementary appraisal approaches, and maintaining a commitment to transparency, FITTER supports more equitable and informed policymaking, without overstating the certainty of its predictions.

5.3 Platform structure: sections and modules (with the example of transport)

The FITTER platform is designed as a structured decision-support environment that helps policymakers navigate the complexities of the twin digital and green transitions. To achieve this, the platform is organised into sections and modules, each serving a distinct function: from providing information about societal challenges to guiding the redesign of policies through interactive tools.

This section illustrates the platform's structure using the transport sector as an example. By examining transport, we can see how the platform integrates:

- Information provision: framing accessibility and transport poverty as key policy concerns.
- Policy modules: assessing risks, identifying disadvantaged groups, and simulating mitigation measures.
- Decision-support tools: enabling policymakers to rethink objectives, explore systemic effects, and design transformative interventions.

The transport example serves as a template for how other sectors (energy, housing, etc.) can be analysed within the same structure.

SECTION 1: INFORMATION PROVISION

Transport policy literature suggests that transport systems' **contribution to human well-being** ought to lie in the **provision of accessibility**, meaning **easy access** to opportunities and places of interest (e.g. jobs, consumption, leisure or health services)

GOAL - Sustainable Accessibility - the provision of access via sustainable transport modes such as:

- active modes and micro-mobility,
- public transport
- other shared services

This goal supports present and future wellbeing.

This leads to policy action that:

- acknowledges the importance of **creating proximity**
- **values space-efficient transport modes** and focuses on facilitating access to places via such modes
- sees **high traffic volume as the problem** to solve and **road space reallocation (to active and public or shared transport modes) as the solution**

Transport Poverty

Overview - Transport poverty refers to a 'lack of adequate transport services necessary to access general services and work, or to the inability to pay for these transport services' (EPRS, 2022: 1). Elements that lead to transport poverty include:

- Lack of nearby services and opportunities (i.e. proximity not considered in spatial planning) – this is also linked to too much time spent travelling (time poverty)
- No or limited transport availability (underdeveloped active, shared and public transport infrastructure, lack of multimodal options, low frequency of available services)
- Inaccessible transport (for disabled people for example)
- Inability to meet the cost of transport
- Difficulty meeting the cost of transport e.g. forced car ownership – where no practical alternative to a private car exists but people struggle to pay the high costs of car ownership
- Dangerous or unsafe transport options

Other Equity Concerns

- Harassment – public or shared transport
- Information deficit – for those without digital skills
- Difficulty buying tickets – e.g. if no in person contact – impacts those with poor digital or literacy skills
- Safety – risk of accidents or unsafe conditions in poorly lit areas for example
- Inaccessible infrastructure (for disabled people, elders, children)

SECTION 2

Policy - Electric vehicle subsidy policy

MODULE A - Risk assessment

- Functionality 1.- Weighted hierarchy of Hazards for a specific region.
- Functionality 2. - Identifying Disadvantaged Groups. This feature lists cohorts most vulnerable to key hazards by analyzing socio-demographic variables with significant score differences on high-weighted risks. For example, if a major hazard in the transport sector is "My petrol/diesel car will lose value and be harder to sell in the future" and younger respondents rate it low while those

over 45 rate it high, then people aged 45+ would be classified as a disadvantaged group for the twin transition in this region.

- Functionality 3.- Mitigation plan for this region oriented towards transformative measures obtained from a systems thinking design process.
- Functionality 4.- Simulation of “What if any of these measures are implemented?”. How the weighted hierarchy of Hazards and the list of Disadvantaged groups change

MODULE B – Redesigning Objectives and Policies

- Functionality 5 - Gamified decision-support tool to help policy-makers redesign objectives and develop policies guided by systems thinking, through shifting mental models promoting transformative interventions.

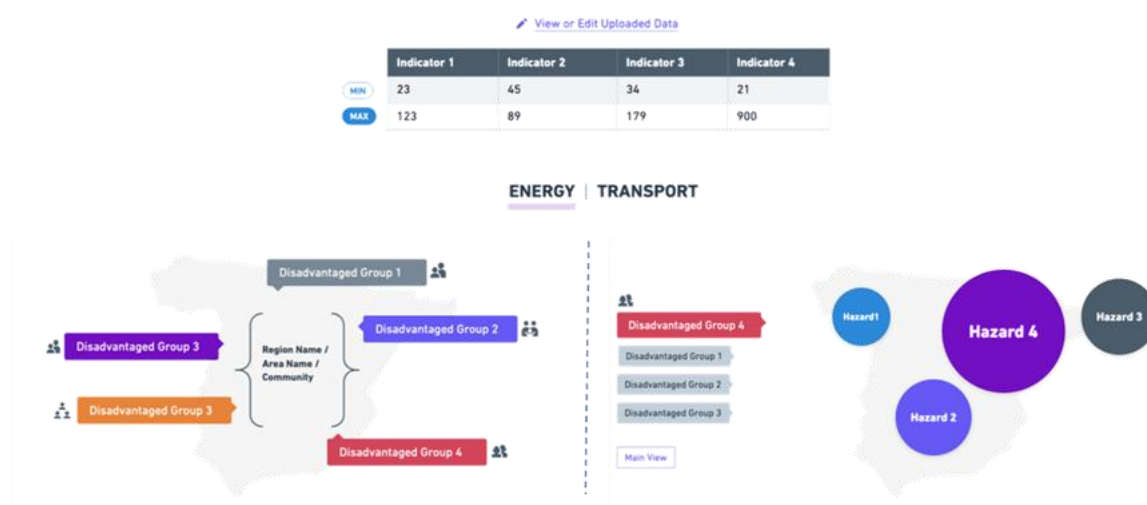
MODULE A

Process

The policy maker selects the region of interest, sector (transport, energy, housing), specific policy mechanism/intervention, and then uploads/pulls relevant and most recent metrics/indicators (e.g., Median salary, Arrears on utility bills, etc.) from publicly available datasets such as EUROSTAT. The platform utilises this information to estimate the most likely negatively affected groups a.k.a Disadvantaged Groups in that area and provides a list of hazards these groups may face if such measure is implemented. The hazards are ordered from highest to lowest based on the combination of their likelihood of occurrence and severity of negative impact as perceived by the Disadvantaged Groups

A **weighted hierarchy of hazards** and a **List of Disadvantaged Groups** will be offered (Functionalities 1 and 2).

The hazards associated with the implementation of specific policy interventions within each sector (e.g., the adoption of electric vehicles in the transport sector) and their respective reactive mitigation measures (e.g., subsidies for electric cars) were identified through desktop research and case studies conducted in six EU Member States. A survey was then designed to capture the perceptions of people and determine the most affected disadvantaged groups, assessing their views on the likelihood and potential negative impacts of these hazards.



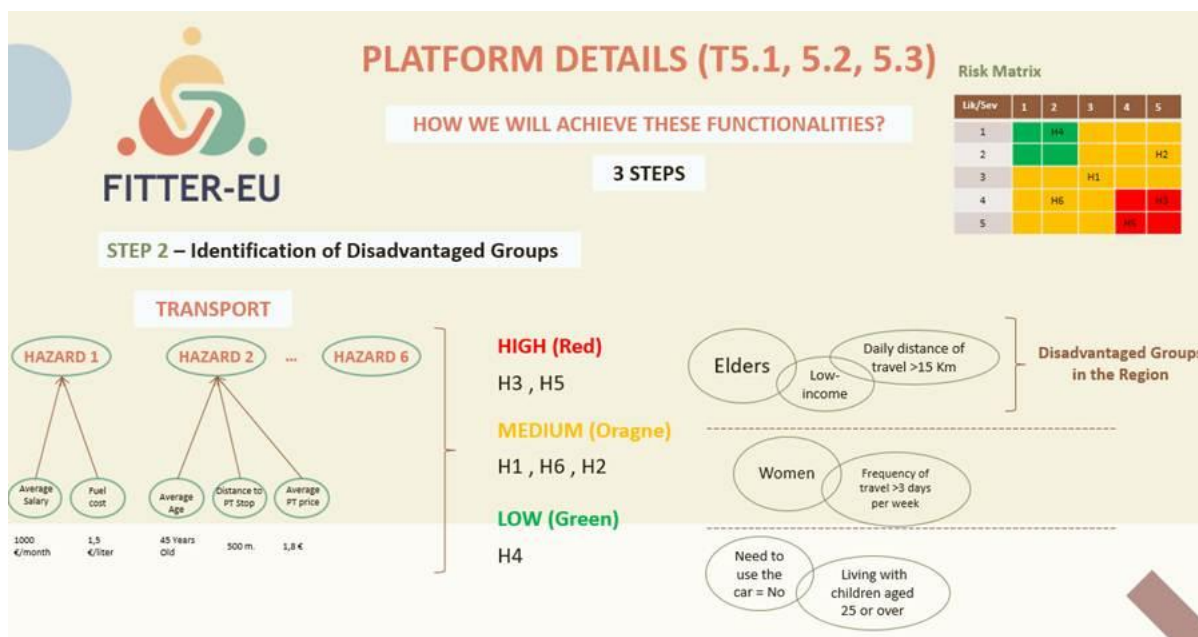


Figure 26 –Weighed hierarchy of hazards and identification of disadvantaged groups

Risk perception (by disadvantaged groups)

The indicators are combined in a model that will use algorithms to give an estimation of how people from a specific region (information coming from objective regional indicators) would perceive the hazards and negative impacts (coming from subjective perception by assessing in a Likert scale, through a survey launched in the Task 4.1, how citizens are perceiving the risk of the identified hazards) and which socio-demographic groups would be most negatively affected (subjective perceptions + objective regional indicators) by the policy mechanism

Algorithms of the predictive FITTER platform are oriented to predict perceptions on hazards and loss of rights due to the twin transitions, transform the system and, by achieving this, the perception of hazard by disadvantaged groups will be reduced. Some objective and metric-based indicators, as those indicated above (salary, fuel cost, distance to public transport stop, etc...) will be used as predictors of these perceptions with a specific confidence level.

MODULE B

Process

After visually mapping the identified hazards and affected disadvantaged groups, the platform will provide a range of engaging, systems-thinking-inspired interactive tools for policymakers. These tools and exercises will prompt users to analyse the root causes of problems, explore the systemic dynamics that generate them, and design transformative measures aligned with the long-term objectives of the Twin Transition.

Through a gamified and interactive environment, policymakers will be equipped with the skills to assess the transformative potential of specific policy mechanisms, recognise that reactive mitigation measures often lead to short-term fixes that fail, and be empowered to rethink objectives from a systemic perspective. Ultimately, the platform -conceived as a decision-support tool designed to foster critical thinking, reflection, and informed decision-making -will nurture the creation of transformative policies that

drive the digital and green transition, ensuring fairness and inclusion are embedded at the core of their design.

Policy Options

Goal - **Sustainable Accessibility**

(Outcome of T4.2)

12 policy proposals – 6 adjusted; 6 new

Analysis of transformative potential per policy

- Sectoral challenge – does it directly address a specific problem?
- Systemic effects – Does it generate wider impacts? In what domain?
- Rules of the game – Does it reshape institutions/regulations?
- Paradigms and attitudes – Can it shift political priorities
- Equity dimensions – Does it reduce inequalities in the green/digital transition?
- Barriers – Are there political, cultural or financial obstacles?
- Timing – Can it be implemented now or does it require preparation?

Categorisation

- Structural shift – Necessary for justice and long-term transition; Costly or politically challenging to implement
- Transformative bets – Potentially game-changing but uncertain risk or requires alignment/timing
- Quick wins – Easy to implement, cost-effective, broadly accepted; often low hanging fruit
- Short term stabiliser – Temporary fixes, subsidies or reactive measures; Useful politically but limited long term effect
- Non-starters – Initiatives, policies that are difficult to implement and would bring limited systemic change if realised



Figure 27 – Recommended mitigation plans per region

The policy maker (or those who support them) can play by simulating to implement a specific measure or recommendation and check how this would affect the list of Disadvantaged Groups and the perception of Hazards and loss of rights due to the twin transition (**Functionality 4**). This Functionality will allow the policy maker to select the intervention with a broader societal impacts and collective well-being.



Figure 28 – Impacts of mitigation plans on disadvantaged groups

6. Conclusion

Deliverable D2.3, has advanced the FITTER-EU project's commitment to embedding social justice at the core of Europe's twin green and digital transitions. By integrating systems inquiry, risk assessment, and similar methods to impact assessment, the FITTER approach recognises that achieving a just transition requires more than incremental policy adjustments - it demands transformative change in the structures, mindsets, and rules that shape our energy, transport, and housing systems.

The analysis demonstrates that policies focused solely on technological fixes or reactive measures are insufficient to address the root causes of inequality and environmental harm. Instead, a systems lens reveals the importance of redesigning infrastructures, reallocating resources, and shifting societal goals toward sustainable accessibility, energy democracy, and adequate housing for all. The FITTER methodology, grounded in a rights-based framework and informed by international and European legal standards, ensures that the benefits of transition are shared equitably and that the burdens do not fall disproportionately on already disadvantaged groups.

The document also highlights the limitations of predictive modelling and quantitative risk assessment in complex social contexts, emphasising the need for transparency, stakeholder engagement, and the integration of qualitative insights. The FITTER platform's modular and participatory design seeks to empower policymakers with tools that are both analytically robust and sensitive to the lived realities of diverse communities across Europe.

Ultimately, the FITTER-EU project calls for a shift from business-as-usual approaches to ambitious, justice-oriented transformation. By prioritising fairness, inclusivity, and ecological sustainability, FITTER aims to support policymakers in creating solutions that not only mitigate immediate risks but also reconfigure the underlying systems that perpetuate inequality and ecological overshoot. With the methodologies and insights developed in this deliverable, FITTER-EU provides a strong foundation for anticipatory governance and collective action toward a fairer, more sustainable future for all.